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JAMB Physics: The Complete Course

Principles, worked examples and exercises for the UTME

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Chapter 1: Measurements, Units, Dimensions and Errors

Physics is an experimental science. Every law in this book began as a pattern noticed in measurements, and every law is still tested by measurement today. A student who states a quantity without its unit, or who does not know how far a reading can be trusted, has not really said anything. This opening chapter therefore deals with the language in which the rest of physics is written: physical quantities, their units, the way units combine, the instruments used to measure length, mass and time, and the errors that unavoidably accompany every reading.

The chapter also introduces dimensional analysis, a powerful and quick method for checking whether an equation can possibly be correct. The UTME regularly asks for the dimensions of a quantity, for the reading on a vernier caliper or micrometer screw gauge, and for a percentage error, so each of these skills is developed with worked examples and practice.

1.1 Physical quantities and their units

A **physical quantity** is a property of a body or a process that can be measured. Length, mass, time, temperature, force and speed are all physical quantities. To measure a quantity means to compare it with an agreed standard of the same kind, called a **unit**. The result of a

measurement is therefore always a number multiplied by a unit. When we say that a table is 1.5 m long, the number is 1.5 and the unit is the metre. Without the unit the number carries no information.

Physical quantities fall into two groups. **Fundamental quantities** are those that are chosen as independent and are not defined in terms of any other quantity. **Derived quantities** are obtained by combining fundamental quantities through multiplication or division. Speed, for instance, is distance divided by time, so it is derived from length and time.

The internationally agreed system of units is the **Système International d'Unités**, abbreviated SI. It has seven base quantities, and the first four in the table below are those most needed at UTME level.

Base quantity	SI unit	Symbol	Dimension symbol
Length	metre	m	L
Mass	kilogram	kg	M
Time	second	s	T
Electric current	ampere	A	I
Thermodynamic temperature	kelvin	K	Θ
Amount of substance	mole	mol	N
Luminous intensity	candela	cd	J

Notice that the kilogram, not the gram, is the base unit of mass. This is a frequent source of mistakes: whenever you substitute into a formula, convert grams to kilograms first, otherwise the answer will not be in

standard SI units.

> **Key point.** The three quantities of mechanics are length, mass and time. Almost every mechanical quantity can be written in terms of metres, kilograms and seconds, which is why the system is sometimes called the m.k.s. system.

Prefixes

Quantities in nature range from the size of a nucleus to the distance between stars, so we attach prefixes to units to avoid very large or very small numbers.

Prefix	Symbol	Factor
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

Thus 1 km = 10^3 m, 1 mm = 10^{-3} m, and 1 nm = 10^{-9} m. Remember that squares and cubes of prefixes must be squared or cubed as well. Since 1 cm = 10^{-2} m, we have 1 cm^2 = 10^{-4} m^2 and 1 cm^3 = 10^{-6} m^3. A litre is 1000 cm^3, which equals 10^{-3} m^3.

1.2 Derived units and conversion of units

Derived units are products or quotients of base units. Some are given special names in honour of scientists. The most important are collected below, each with its expression in base units.

Quantity	Unit	Symbol	In base units
Area	square metre	m^2	m^2
Volume	cubic metre	m^3	m^3
Speed, velocity	metre per second	m/s	$m\ s^{-1}$
Acceleration	metre per second squared	m/s^2	$m\ s^{-2}$
Density	kilogram per cubic metre	kg/m^3	$kg\ m^{-3}$
Force	newton	N	$kg\ m\ s^{-2}$
Pressure	pascal	Pa	$kg\ m^{-1}\ s^{-2}$
Work, energy	joule	J	$kg\ m^2\ s^{-2}$
Power	watt	W	$kg\ m^2\ s^{-3}$
Frequency	hertz	Hz	s^{-1}
Electric charge	coulomb	C	A s

To convert from one unit to another, multiply by a conversion factor equal to one. For example, since 1 h = 3600 s and 1 km = 1000 m, a speed in km/h is converted to m/s by multiplying by 1000/3600, that is, dividing by 3.6.

WORKED EXAMPLE 1.1

A bus travelling from Lagos to Ibadan moves at 72 km/h. Express this speed in m/s.

Solution: 72 km/h = 72 × 1000 m ÷ 3600 s = 72 000/3600 m/s = 20 m/s. Equivalently, 72 ÷ 3.6 = 20 m/s.

WORKED EXAMPLE 1.2

The density of aluminium is 2.7 g/cm^3. Express it in kg/m^3.

Solution: 1 g = 10^{-3} kg and 1 cm^3 = 10^{-6} m^3. Therefore 2.7 g/cm^3 = 2.7 × 10^{-3} kg ÷ 10^{-6} m^3 = 2.7 × 10^3 kg/m^3 = 2700 kg/m^3.

Key point. To convert g/cm^3 to kg/m^3, multiply by 1000. To convert km/h to m/s, divide by 3.6. These two shortcuts save time in the examination.

Standard form and significant figures

Very large or very small numbers are written in **standard form** as $A \times 10^{n}$, where $1 \le A < 10$ and n is an integer. The speed of light, 300 000 000 m/s, becomes 3×10^{8} m/s, and the diameter of a hydrogen atom, about 0.000 000 000 1 m, becomes 1×10^{-10} m.

Significant figures are the digits in a number that carry meaningful information about the precision of a measurement. The rules are simple. All non-zero digits are significant. Zeros between non-zero digits are significant. Leading zeros, which only locate the decimal point, are not significant. Trailing zeros after a decimal point are significant. Thus 0.004050 has four significant figures (4, 0, 5, 0), while 3.20 has three. In multiplication and division the answer should be given to the same number of significant figures as the least precise quantity used. In addition and subtraction the answer is limited by the number of decimal places of the least precise quantity.

1.3 Dimensions and dimensional analysis

The **dimension** of a physical quantity tells us how it is built from the base quantities. We write the dimension of a quantity X in square brackets, as [X]. Length, mass and time have the dimensions L, M and T, so [length] = L, [mass] = M and [time] = T.

Dimensions multiply and divide just as the quantities do. Area is length times length, so [area] = L^{2}. Velocity is length divided by time, so [velocity] = LT^{-1}. Acceleration is velocity divided by time, giving LT^{-2}.

Force is mass times acceleration, so [force] = MLT^{-2}. Pure numbers, and ratios of like quantities such as angles in radians, strain and refractive index, have no dimensions; we say they are dimensionless. Constants such as ½ and 2π likewise have no dimension.

Quantity	Defining relation	Dimensions
Volume	length × length × length	L^3
Density	mass ÷ volume	ML^{-3}
Momentum	mass × velocity	MLT^{-1}
Force	mass × acceleration	MLT^{-2}
Pressure	force ÷ area	$ML^{-1}T^{-2}$
Work, energy	force × distance	ML^2T^{-2}
Power	work ÷ time	ML^2T^{-3}
Frequency	1 ÷ time	T^{-1}

WORKED EXAMPLE 1.3

Find the dimensions of pressure and state its SI unit in base units.

Solution: Pressure = force ÷ area. [force] = MLT^{-2} and [area] = L^2. Hence [pressure] = $MLT^{-2} \div L^2 = ML^{-1}T^{-2}$. The SI unit is therefore kg m^{-1} s^{-2}, which is the pascal (N/m^2).

Uses of dimensions

Dimensional analysis has three main uses.

1. **Checking the homogeneity of an equation.** Every term in a correct physical equation must have the same dimensions. You cannot add a length to a time. If the two sides of an equation differ in dimensions, the equation is certainly wrong. If they agree, the equation may be right, but this does not prove it, since numerical factors such as ½ are invisible to dimensions.
2. **Deriving the form of a relationship.** If we know which quantities a result depends on, we can often find how it depends on them.
3. **Converting units.** The dimensions show how the numerical value changes when the base units change.

WORKED EXAMPLE 1.4

Show that the equation $v^2 = u^2 + 2as$ is dimensionally consistent, where u and v are velocities, a is acceleration and s is distance.

Solution: $[v^2] = (LT^{-1})^2 = L^2T^{-2}$. $[u^2] = L^2T^{-2}$. For the last term, $[2as] = [a][s] = LT^{-2} \times L = L^2T^{-2}$, since the number 2 has no dimension. All three terms have dimensions L^2T^{-2}, so the equation is dimensionally consistent.

WORKED EXAMPLE 1.5

The period T of a simple pendulum is thought to depend on its length l, the mass m of the bob and the acceleration due to gravity g. Use dimensions to find how T depends on these quantities.

Solution: Assume $T = k\, m^a\, l^b\, g^c$, where k is a dimensionless constant. Writing dimensions: $T = M^a\, L^b\, (LT^{-2})^c = M^a\, L^{b+c}\, T^{-2c}$. Compare powers on both sides. For M: $a = 0$. For T: $-2c = 1$, so $c = -\tfrac{1}{2}$. For L: $b + c = 0$, so $b = \tfrac{1}{2}$. Hence $T = k\,\sqrt{(l/g)}$. The period does not depend on the mass. (A full analysis shows that $k = 2\pi$, which dimensions alone cannot supply.)

Key point. Dimensional analysis cannot determine dimensionless constants such as ½, 2 or π, and it fails when a quantity depends on a sum of terms, or on a function such as sine or logarithm of a dimensioned quantity. Arguments of trigonometric, exponential and logarithmic functions must be dimensionless.

1.4 Measuring instruments

Length: the metre rule, vernier caliper and micrometer screw gauge

A metre rule is graduated in millimetres, so its smallest division is 1 mm and readings can be made to about ±0.5 mm. To use it correctly, place your eye vertically above the mark being read to avoid **parallax error**, which is the apparent shift of a pointer against a scale when viewed from the side.

The **vernier caliper** measures lengths, internal diameters and depths to a fraction of a millimetre. It has a main scale graduated in millimetres (or tenths of a centimetre) and a sliding vernier scale. A typical vernier has 10 divisions covering 9 mm of the main scale, so each vernier division is 0.9 mm and the difference between one main-scale division and one vernier division is 0.1 mm. This difference is the **least count** of the instrument, equal to 0.1 mm or 0.01 cm. Some verniers have 50 divisions and a least count of 0.02 mm.

To read the vernier: (1) note the main-scale reading just before the zero of the vernier; (2) find the vernier division that lines up exactly with any main-scale division; (3) multiply that division number by the least count and add it to the main-scale reading.

Before use, close the jaws and check for a **zero error**. If the zero of the vernier is not aligned with the zero of the main scale, the instrument reads a fixed amount too high or too low. A positive zero error (the

reading is above zero when the jaws are closed) must be subtracted from every reading; a negative zero error must be added.

WORKED EXAMPLE 1.6

A vernier caliper with least count 0.01 cm is used to measure the diameter of a metal rod. The main scale reads 2.3 cm just before the vernier zero, and the 6th vernier division coincides with a main-scale division. When the jaws are closed the caliper has a zero error of +0.02 cm. Find the true diameter.

Solution: Observed reading = 2.3 + 6 × 0.01 = 2.36 cm. Zero error is +0.02 cm, so corrected reading = 2.36 − 0.02 = 2.34 cm.

The **micrometer screw gauge** is more precise and is used for thin wires, sheets and small spheres. A screw with a known **pitch** (the distance advanced in one complete turn) moves the measuring spindle. The thimble is divided into equal parts. If the pitch is 0.5 mm and the thimble has 50 divisions, then one division of the thimble corresponds to 0.5 ÷ 50 = 0.01 mm, which is the least count. The reading is the sleeve (main scale) reading, including any half-millimetre mark visible, plus the thimble division multiplied by the least count. A ratchet limits the force applied, so that the object is not squeezed.

WORKED EXAMPLE 1.7

A micrometer screw gauge has pitch 0.5 mm and 50 divisions on its thimble. In measuring the diameter of a wire the sleeve reads 5.5 mm and the thimble division 28 is in line with the datum line. The gauge has a zero error of −0.03 mm. Find the true diameter.

Solution: Least count = 0.5/50 = 0.01 mm. Observed reading = 5.5 + 28 × 0.01 = 5.5 + 0.28 = 5.78 mm. A negative zero error means the gauge reads too low, so we add 0.03 mm: true diameter = 5.78 + 0.03 = 5.81 mm.

Mass, time and volume

Mass is measured with a beam balance or an electronic balance, which compares the unknown mass with standard masses. A spring balance, in contrast, measures weight, and its reading changes with the local value of g.

Time is measured with a stopwatch or clock. In experiments such as the simple pendulum, the time for 20 oscillations is measured and divided by 20; this reduces the percentage error caused by reaction time, which is typically about 0.2 s at the start and end of timing. The oscillation count should begin when the bob passes the midpoint of its swing, because it moves fastest there and the reference mark is easier to judge.

The volume of a liquid is measured with a measuring cylinder or burette, reading the bottom of the meniscus with the eye level with the surface. The volume of an irregular solid that sinks is found by

displacement: the rise in the water level in a measuring cylinder equals the volume of the solid.

1.5 Accuracy, precision and errors

No measurement is perfectly exact. The difference between a measured value and the true value is the **error**. Two ideas are often confused. **Accuracy** describes how close a measurement is to the true value. **Precision** describes how closely repeated measurements agree with one another. A set of readings of 4.51 s, 4.52 s and 4.51 s is precise; if the true time is 4.20 s because the stopwatch runs fast, the readings are precise but not accurate.

Errors are classified into two main types.

- **Systematic errors** push all readings in the same direction by roughly the same amount. Examples are zero error of an instrument, a miscalibrated scale, and a faulty stopwatch. They cannot be reduced by averaging, but they can be removed by correcting the instrument or applying a known correction.
- **Random errors** vary unpredictably from one reading to the next, sometimes too high and sometimes too low. They arise from reaction time, judgement of a scale, draughts or fluctuating temperature. Taking many readings and averaging reduces their effect.

Parallax is a personal error that may be either. Mistakes such as misreading a scale or recording a wrong figure are blunders, not errors in the technical sense; they are removed by care.

Absolute, fractional and percentage error

If a quantity has measured value x and true value x_0, the **absolute error** is $|x - x_0|$. The **fractional error** is the absolute error divided by the true value, and the **percentage error** is the fractional error multiplied by 100%.

When the true value is not known, the measurement is quoted as $x \pm \Delta x$, where Δx is the estimated uncertainty, often half the smallest scale division for a single reading. For a set of readings the mean is taken as the best value, and the uncertainty is estimated from the spread.

WORKED EXAMPLE 1.8

A student times a trolley as 2.52 s over a distance for which the accepted time is 2.50 s. Find the absolute, fractional and percentage errors.

Solution: Absolute error = 2.52 − 2.50 = 0.02 s. Fractional error = 0.02/2.50 = 0.008. Percentage error = 0.008 × 100% = 0.8%.

Combining errors

When quantities with uncertainties are combined in a calculation, the uncertainties combine as well. At UTME level, use the following simple rules for the maximum possible error.

- **Addition or subtraction:** add the absolute errors. If $A = a \pm \Delta a$ and $B = b \pm \Delta b$, then $A + B$ and $A - B$ both have absolute error $\Delta a + \Delta b$.
- **Multiplication or division:** add the fractional (or percentage) errors.

- **Powers:** for a quantity raised to the power n, multiply its fractional error by n. A square, for instance, has double the fractional error.

WORKED EXAMPLE 1.9

The length and breadth of a rectangular plate are measured as (20.0 ± 0.1) cm and (5.0 ± 0.1) cm. Find the area and its percentage uncertainty, and state the area with its absolute uncertainty.

Solution: Area = 20.0 × 5.0 = 100 cm^2. Fractional error in length = 0.1/20.0 = 0.005, and in breadth = 0.1/5.0 = 0.02. Fractional error in area = 0.005 + 0.02 = 0.025, so the percentage uncertainty is 2.5%. Absolute uncertainty = 0.025 × 100 = 2.5 cm^2. The area is (100 ± 2.5) cm^2.

WORKED EXAMPLE 1.10

In an experiment with a simple pendulum a student finds l = 1.00 m and T = 2.01 s. Use $T = 2\pi\sqrt{(l/g)}$ to calculate g. If the length has a 1% error and the period a 0.5% error, what is the percentage error in g? (Take $\pi^2 = 9.87$.)

Solution: Squaring the formula, $T^2 = 4\pi^2 l/g$, so $g = 4\pi^2 l/T^2$. Then $g = 4 \times 9.87 \times 1.00 \div (2.01)^2 = 39.48 \div 4.0401 = 9.77$ m/s^2. For the error, g is proportional to l and to T^{-2}, so the fractional error in g = (1%) + 2 × (0.5%) = 2%. Thus g = 9.77 m/s^2 with an uncertainty of about 2%, that is, ± 0.2 m/s^2.

Key point. A quantity that is squared contributes twice its percentage error; one that is under a square root contributes half. Always add, never subtract, the percentage errors in the maximum-error method, even when a quantity is in the denominator.

1.6 Graphs and measurement

Plotting a graph averages out random errors across many readings and reveals systematic ones. When you plot a graph, choose scales so the plotted points fill at least half of each axis, label both axes with the quantity and its unit, and draw a smooth line or straight line of best fit so that points lie evenly on both sides of it. The **slope** of a straight line is the change in the vertical quantity divided by the change in the horizontal quantity, taken over a large triangle so that reading errors are small. The **intercept** is where the line cuts an axis.

For the pendulum, a graph of T^2 (vertical) against l (horizontal) is a straight line through the origin with slope $4\pi^2/g$. From the slope, $g = 4\pi^2/\text{slope}$. If the line has a positive intercept on the T^2 axis, it suggests a systematic error, for example, measuring the length to the top of the bob rather than to its centre.

To test whether two quantities are directly proportional, plot one against the other; a straight line through the origin confirms direct proportion. To test an inverse relationship, such as pressure and volume of a gas,

plot one quantity against the reciprocal of the other.

Estimating and checking answers

Examination questions in physics reward sensible estimation. A person walks at about 1.5 m/s, a car on the Lagos–Ibadan expressway about 25 m/s, sound travels at about 340 m/s and light at 3×10^8 m/s. A cup of water has a mass of about 0.25 kg, and the mass of an adult is about 70 kg. If your calculated answer departs wildly from such values, recheck the units and the powers of ten. An answer given with the wrong unit is usually marked as wrong even when the number is right, and an answer such as a speed of 5×10^9 m/s for a bicycle announces itself as a slip.

Exercise 1.1

1. Which of the following is a derived unit: kilogram, ampere, newton, kelvin?
2. A motorcyclist rides at 90 km/h. Express this speed in m/s.
3. The density of a mineral is 5 g/cm^3. Express it in kg/m^3.
4. Find the dimensions of work (energy).
5. Find the dimensions of power.
6. A vernier caliper has a least count of 0.01 cm. The main scale reads 4.1 cm and the 7th vernier division coincides with a main-scale division. There is no zero error. What is the reading?
7. A micrometer screw gauge has a pitch of 0.5 mm and 50 thimble divisions. The sleeve reads 3.5 mm and the thimble

reads 12. Find the diameter if the zero error is zero.

8. A balance reads 49.0 g for a standard mass whose true value is 50.0 g. Find the percentage error.
9. The mass of a block is (10.0 ± 0.1) g and its volume is (4.0 ± 0.1) cm^3. Find the density and its percentage uncertainty.
10. In the equation $s = ut + \tfrac{1}{2}at^2$, s is a length and t is a time. What are the dimensions of a?
11. Write 0.000 045 m in standard form and in micrometres.
12. How many significant figures are there in 0.004050?

Answers to Exercise 1.1

1. Newton ($kg\ m\ s^{-2}$); the others are base units.
2. $90 \div 3.6 = 25$ m/s.
3. $5 \times 1000 = 5000\ kg/m^3$.
4. Force × distance = $MLT^{-2} \times L = ML^2T^{-2}$.
5. Work ÷ time = $ML^2T^{-2} \div T = ML^2T^{-3}$.
6. $4.1 + 7 \times 0.01 = 4.17$ cm.
7. Least count 0.5/50 = 0.01 mm; $3.5 + 12 \times 0.01 = 3.62$ mm.
8. Error = 1.0 g; 1.0/50.0 × 100% = 2%.
9. Density = 10.0/4.0 = 2.5 g/cm^3; fractional errors 0.1/10.0 = 0.01 and 0.1/4.0 = 0.025; total 0.035, so 3.5% (about ± 0.09 g/cm^3).
10. $[a] = [s]/[t^2] = L/T^2 = LT^{-2}$.
11. 4.5×10^{-5} m; since 1 µm = 10^{-6} m, this is 45 µm.

12. Four (4, 0, 5, 0); the leading zeros are not significant.

Exercise 1.2

1. A student determines the length of a table as 1.52 m, 1.53 m, 1.51 m and 1.52 m. Find the mean length and the largest deviation from the mean.
2. Show that the quantity (force × time) has the same dimensions as momentum.
3. A formula for the speed v of waves on a stretched string is proposed as $v = \sqrt{(F/\mu)}$, where F is the tension and μ is the mass per unit length. Show that it is dimensionally consistent.
4. The side of a cube is measured as (5.0 ± 0.1) cm. Find the volume and its percentage uncertainty.
5. A pendulum of length 0.80 m takes 36 s for 20 oscillations. Find the period and calculate g using $g = 4\pi^2 l/T^2$ (take $\pi^2 = 9.87$, and give g to two decimal places).

Answers to Exercise 1.2

1. Sum = 6.08, mean = 1.52 m; largest deviation = 0.01 m.
2. $[F][t] = MLT^{-2} \times T = MLT^{-1}$, which equals $[mv] = M \times LT^{-1}$.
3. $[F]/[\mu] = MLT^{-2} \div (M\,L^{-1}) = L^2T^{-2}$; its square root is LT^{-1}, the dimension of speed.

4. $V = 5.0^3 = 125\ cm^3$; fractional error $3 \times (0.1/5.0) = 0.06$, so 6%, or $\pm 7.5\ cm^3$.
5. $T = 36/20 = 1.8$ s; $g = 4 \times 9.87 \times 0.80 \div 1.8^2 = 31.584 \div 3.24 = 9.75\ m/s^2$.

Chapter summary

- A measurement is a number times a unit. SI base units for mechanics are the metre, kilogram and second; the ampere, kelvin, mole and candela complete the system.
- Derived units come from base units; the newton is kg m s^{-2}, the joule kg m^2 s^{-2}, the watt kg m^2 s^{-3} and the pascal kg m^{-1} s^{-2}.
- Convert units with factors equal to one: km/h to m/s by dividing by 3.6, g/cm^3 to kg/m^3 by multiplying by 1000.
- Dimensions describe a quantity in terms of M, L and T. They are used to check equations, derive relationships (up to dimensionless constants) and convert units.
- Vernier least count is the difference between a main-scale division and a vernier division; a micrometer's least count is pitch divided by the number of thimble divisions. Always correct for zero error.
- Systematic errors shift all readings one way and cannot be removed by averaging; random errors scatter readings and are reduced by repeating and averaging.

- Percentage error = (absolute error ÷ true value) × 100%. For products and quotients add the percentage errors; for powers multiply by the exponent; for sums and differences add absolute errors.
- Give answers to a sensible number of significant figures, always with a unit.

Chapter 2: Scalars, Vectors and Motion in a Straight Line

Mechanics begins with the description of motion, a branch called **kinematics**. Before we ask why a body moves as it does, we must be able to say precisely where it is, how fast it is going and how its speed is changing. To do this we need two kinds of quantity: those that have size only, and those that also have direction. The first part of this chapter explains the difference and shows how vectors are added, resolved and subtracted. The second part develops the equations of uniformly accelerated motion, applies them to free fall, and shows how the graphs of motion can be read to give speed, acceleration and distance.

These topics are examined every year. Typical questions ask for the resultant of two forces, the distance covered from a velocity-time graph, the stopping distance of a vehicle, or the height reached by a stone thrown upwards. Each of these is built from a handful of ideas, and mastery of them is the foundation for every later chapter in mechanics.

2.1 Scalars and vectors

A **scalar quantity** is completely described by its magnitude (size) and unit. Mass, time, distance, speed, energy, temperature, density and work are scalars. A **vector quantity** has both magnitude and direction, and it obeys special rules of addition. Displacement, velocity, acceleration,

force, momentum and weight are vectors. To say that a Lagos-bound bus travelled 60 km tells us the distance, a scalar; to say that it moved 60 km on a bearing of 230° gives a displacement, a vector.

A vector is drawn as an arrow. The length of the arrow, to a chosen scale, represents the magnitude, and the direction of the arrow shows the direction of the quantity. In print a vector is often written in bold or with an arrow over the symbol, while its magnitude is written in ordinary type.

> **Key point.** Distance is the total length of path travelled and is a scalar. Displacement is the shortest straight-line distance from the starting point to the finishing point, in a stated direction, and is a vector. A runner who completes one lap of a 400 m track has travelled a distance of 400 m but has a displacement of zero.

2.2 Addition of vectors

Because vectors have direction, they cannot be added like ordinary numbers. A force of 3 N added to a force of 4 N may give 7 N, 1 N or anything between, depending on the directions. The single vector that has the same effect as the others acting together is called the **resultant**.

Vectors in the same or opposite directions

If two vectors act along the same line in the same direction, the resultant is their sum. If they act in opposite directions, the resultant is the difference of their magnitudes and points in the direction of the larger. This explains why two boys pushing a car from the same side add their efforts, while two teams in a tug of war nearly cancel each other.

The triangle and parallelogram laws

The **triangle law** states that if two vectors are represented in magnitude and direction by two sides of a triangle taken in order, their resultant is represented by the third side taken in the opposite order. In practice, draw the first vector, then draw the second starting at the tip of the first. The resultant runs from the tail of the first to the tip of the second.

The **parallelogram law** states that if two vectors acting at a point are represented by the adjacent sides of a parallelogram drawn from that point, their resultant is represented by the diagonal of the parallelogram drawn from the same point. The two laws are equivalent.

For two vectors P and Q with an angle θ between their directions (when drawn from the same point), the magnitude of the resultant R follows from the cosine rule:

$R^2 = P^2 + Q^2 + 2PQ\cos\theta$

and the angle α that R makes with P is given by $\tan\alpha = Q\sin\theta / (P + Q\cos\theta)$. For the very common special case of perpendicular vectors ($\theta = 90°$), $\cos\theta = 0$ and the result reduces to Pythagoras' theorem, $R = \sqrt{(P^2 +}$

Q^2), with $\tan \alpha = Q/P$.

WORKED EXAMPLE 2.1

Two forces of 3 N and 4 N act at a point at right angles to each other. Find the magnitude and direction of the resultant.

Solution: $R = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$ N. The angle with the 3 N force is $\alpha = \tan^{-1}(4/3) = 53.1°$. The resultant is 5 N, at 53.1° to the 3 N force.

WORKED EXAMPLE 2.2

Two forces of 10 N and 6 N act at a point with an angle of 60° between them. Calculate the resultant.

Solution: $R^2 = 10^2 + 6^2 + 2(10)(6) \cos 60° = 100 + 36 + 120 \times 0.5 = 100 + 36 + 60 = 196$. Therefore $R = \sqrt{196} = 14$ N.

Subtraction of vectors

To subtract a vector, add its negative, that is, a vector of the same size pointing the opposite way. The change in velocity of a body is therefore the final velocity plus the reverse of the initial velocity. A car moving at 20 m/s that turns a corner and still moves at 20 m/s has changed its velocity, because the direction has changed, even though its speed has not.

2.3 Resolution of vectors

Resolution is the opposite of addition. A single vector can be replaced by two **components** at right angles, which together have the same effect as the original. If a vector F makes an angle θ with the horizontal, then:

- the horizontal component is F cos θ;
- the vertical component is F sin θ.

Resolution is used whenever a force acts at a slant, for example a rope pulling a loaded cart, or gravity acting on a body on an inclined plane. When several vectors act at a point, the usual method is to resolve each one into horizontal and vertical components, add the components in each direction separately, and then combine the two totals using Pythagoras' theorem.

WORKED EXAMPLE 2.3

A woman pulls a loaded wheelbarrow with a rope that makes an angle of 30° with the ground. The tension in the rope is 50 N. Find the horizontal and vertical components of the pull.

Solution: Horizontal component = 50 cos 30° = 50 × 0.866 = 43.3 N. Vertical component = 50 sin 30° = 50 × 0.5 = 25 N. Only the 43.3 N moves the barrow forward; the 25 N slightly lifts it.

2.4 Speed, velocity and acceleration

Speed is the rate of change of distance with time. Average speed is the total distance divided by the total time taken. Its unit is metre per second (m/s); to change km/h to m/s, divide by 3.6, so 72 km/h is 20 m/s.

Velocity is the rate of change of displacement with time, or displacement per unit time in a stated direction. Average velocity = displacement ÷ time taken. Because displacement is a vector, so is velocity. A body can move at constant speed and yet have changing velocity, as we shall see in circular motion.

Acceleration is the rate of change of velocity with time:

$$a = (v - u)/t$$

where u is the initial velocity and v is the final velocity after a time t. Its unit is m/s^2, and it is a vector. When the velocity decreases the acceleration is negative, and it is often called **retardation** or **deceleration**. An acceleration is **uniform** if the velocity changes by equal amounts in equal intervals of time.

> **Key point.** Speed and velocity differ because velocity includes direction. Average speed over a round trip is not zero, but average velocity over a round trip is zero, because the displacement is zero.

Instantaneous quantities

The speed shown by a car's speedometer at a particular moment is the **instantaneous speed**. In mathematical terms, instantaneous velocity is the limit of the average velocity as the time interval is made extremely small. On a displacement-time graph it is the gradient of the tangent to the curve at that moment; on a velocity-time graph the gradient at a point gives the instantaneous acceleration.

2.5 Equations of uniformly accelerated motion

When a body moves in a straight line with constant acceleration a, its initial velocity u, final velocity v, displacement s and time t are connected by three equations:

1. $v = u + at$
2. $s = ut + \tfrac{1}{2}at^2$
3. $v^2 = u^2 + 2as$

A useful fourth relation follows from the average velocity of uniformly changing velocity: $s = \tfrac{1}{2}(u + v)t$.

The first equation is simply the definition of acceleration rearranged. For the second, the average velocity is $(u + v)/2 = u + \tfrac{1}{2}at$, and multiplying by t gives the displacement. For the third, eliminate t between the first two. These equations apply only when the acceleration

is constant, and they apply along a straight line, so a sign convention is needed: choose one direction as positive and give a negative sign to any quantity that points the opposite way.

WORKED EXAMPLE 2.4

A car moving at 5 m/s accelerates uniformly at 2 m/s² for 8 s. Find its final velocity and the distance covered in this time.

Solution: $v = u + at = 5 + 2 \times 8 = 21$ m/s. The distance is $s = ut + \tfrac{1}{2}at^2 = 5 \times 8 + \tfrac{1}{2} \times 2 \times 8^2 = 40 + 64 = 104$ m. Check: average velocity $= (5 + 21)/2 = 13$ m/s and $13 \times 8 = 104$ m.

WORKED EXAMPLE 2.5

A driver on the Ibadan expressway travelling at 20 m/s applies the brakes and the car decelerates uniformly at 4 m/s². Find the time taken to stop and the stopping distance.

Solution: Take the direction of motion as positive, so $u = 20$ m/s, $v = 0$, $a = -4$ m/s². Time: $t = (v - u)/a = (0 - 20)/(-4) = 5$ s. Distance: $v^2 = u^2 + 2as$ gives $0 = 400 + 2(-4)s = 400 - 8s$, so $s = 50$ m. The car stops in 5 s after covering 50 m.

Notice how the stopping distance depends on the square of the speed. If the speed in Example 2.5 were doubled to 40 m/s with the same deceleration, the stopping distance would be $1600 \div 8 = 200$ m, four times as large. This is the physical reason why speeding is so dangerous.

Distance in the nth second

The distance covered during the nth second alone equals the distance in n seconds minus the distance in (n − 1) seconds. Using $s = ut + \frac{1}{2}at^2$, this gives $s_n = u + a(n - \frac{1}{2})$. For a body starting from rest with a = 2 m/s², the distance in the fifth second is 2 × (5 − 0.5) = 9 m. Check: distance in 5 s is ½ × 2 × 25 = 25 m, in 4 s it is 16 m, and 25 − 16 = 9 m.

2.6 Free fall and motion under gravity

Near the Earth's surface, every body that falls freely, ignoring air resistance, gains velocity at the same rate. This is the **acceleration due to gravity**, g, which in this book is taken as 10 m/s² unless otherwise stated (the more precise value is about 9.8 m/s²). A feather and a stone fall at the same rate in a vacuum; in air the feather is slowed by air resistance.

Free fall problems use the same three equations with a = g. For a body released from rest and falling downwards (taking down as positive), u = 0, so v = gt, $h = \frac{1}{2}gt^2$ and $v^2 = 2gh$. For a body thrown upwards, take up as positive so that a = −g. At the highest point the velocity is momentarily zero. The time taken to rise equals the time taken to fall back to the starting level, and the body returns to its starting point with the same speed it was thrown with, but pointing downwards.

WORKED EXAMPLE 2.6

A mango falls from a branch 80 m above the ground. Taking g = 10 m/s², find the time of fall and the speed with which it strikes the ground.

Solution: From $h = \tfrac{1}{2}gt^2$, $80 = 5t^2$, so $t^2 = 16$ and $t = 4$ s. The speed is $v = gt = 10 \times 4 = 40$ m/s. Check with $v^2 = 2gh = 2 \times 10 \times 80 = 1600$, so $v = 40$ m/s.

WORKED EXAMPLE 2.7

A stone is thrown vertically upwards with a speed of 30 m/s. Find (a) the maximum height reached, (b) the total time of flight before it returns to the thrower's hand.

Solution: (a) At the top $v = 0$, so $0 = 30^2 - 2 \times 10 \times H$, giving $H = 900/20 = 45$ m. (b) Time to the top is $t = (30 - 0)/10 = 3$ s. The time to fall is the same, so the total time is 6 s. Check: $s = ut - \tfrac{1}{2}gt^2 = 30 \times 6 - 5 \times 36 = 180 - 180 = 0$, so the stone is back at its starting level.

2.7 Graphs of motion

Graphs give a picture of motion, and the UTME often asks for information to be read from them. There are three important rules.

Displacement-time graphs

The gradient of a displacement-time graph is the velocity. A straight line sloping upwards means constant velocity. A horizontal line means the body is at rest. A curve that steepens means increasing velocity, that is, acceleration. A line sloping downwards means motion back towards the starting point.

Velocity-time graphs

The **gradient** of a velocity-time graph is the acceleration. The **area under the graph** between two times is the displacement (the distance covered, if the velocity does not change sign). A horizontal line means uniform velocity, an upward straight line means uniform acceleration, and a downward straight line means uniform retardation. Areas below the time axis represent motion in the reverse direction.

> **Key point.** Gradient of s–t graph = velocity. Gradient of v–t graph = acceleration. Area under v–t graph = displacement. Area under a–t graph = change in velocity.

WORKED EXAMPLE 2.8

The velocity of a train leaving a station rises uniformly from 0 to 12 m/s in 4 s, stays constant at 12 m/s for the next 6 s, and then falls uniformly to rest in the following 4 s. Find (a) the acceleration in the first stage, (b) the retardation in the last stage, (c) the total distance and the average speed.

Solution: (a) $a = (12 - 0)/4 = 3\ \text{m/s}^2$. (b) $a = (0 - 12)/4 = -3\ \text{m/s}^2$, so the retardation is $3\ \text{m/s}^2$. (c) The graph is a trapezium. Distance in stage one = ½ × 4 × 12 = 24 m; stage two = 12 × 6 = 72 m; stage three = ½ × 4 × 12 = 24 m. Total distance = 24 + 72 + 24 = 120 m. The total time is 14 s, so the average speed = 120/14 = 8.57 m/s. As a check, the area of the trapezium is ½ × (parallel sides 14 and 6) × 12 = ½ × 20 × 12 = 120 m.

2.8 Relative velocity

The velocity of a body A *relative to* a body B is the velocity A appears to have to an observer moving with B. It is found by vector subtraction: $v_{AB} = v_A - v_B$. For two bodies moving along the same straight line in the same direction, the relative speed is the difference of the speeds; for bodies moving towards each other it is the sum. When a boat crosses a river, the velocity of the boat relative to the ground is the vector sum of its velocity relative to the water and the velocity of the water relative to the ground.

WORKED EXAMPLE 2.9

Car A travelling at 30 m/s is 100 m behind car B travelling at 20 m/s along a straight road in the same direction. How long does A take to catch B, and how far does A travel in this time?

Solution: The speed of A relative to B is 30 − 20 = 10 m/s. A must close a gap of 100 m, so t = 100/10 = 10 s. Distance travelled by A = 30 × 10 = 300 m. Check: B travels 20 × 10 = 200 m, and 200 + 100 = 300 m, so they meet.

WORKED EXAMPLE 2.10

A boatman rows at 4 m/s relative to the water, heading directly across a river 200 m wide. The river flows at 3 m/s. Find (a) the resultant speed of the boat relative to the bank, (b) the time to cross, (c) how far downstream the boat lands.

Solution: (a) The two velocities are perpendicular, so $v = \sqrt{(4^2 + 3^2)} = 5$ m/s. (b) Only the component across the river carries the boat over, so t = 200/4 = 50 s. (c) During this time the current carries the boat 3 × 50 = 150 m downstream. Check: 5 × 50 = 250 m along the path, and $\sqrt{(200^2 + 150^2)} = \sqrt{(40000 + 22500)} = \sqrt{62500} = 250$ m.

2.9 Practical hints for the examination

- Convert all quantities to SI units before substituting. Change km/h to m/s by dividing by 3.6.
- List the known quantities u, v, a, s and t first, mark the unknown, then choose the equation that contains the four relevant letters and not the one you do not need.
- For vertical motion, decide at the start whether up or down is positive and keep to it.
- When reading a velocity-time graph, look for gradient and area; when reading a displacement-time graph, look for gradient only.
- If a problem has two stages, treat each stage separately and note that the final velocity of one stage is the initial velocity of the next.

Exercise 2.1

1. Which of the following is a scalar quantity: velocity, momentum, work, force?
2. Two forces of 6 N and 8 N act at a point at right angles. Find the magnitude of the resultant.
3. Find the horizontal and vertical components of a force of 100 N acting at 60° above the horizontal.
4. A car accelerates uniformly from rest to 30 m/s in 10 s. Find the acceleration and the distance covered.
5. A ball is dropped from the top of a building 125 m high. Taking g = 10 m/s², find the time of fall and the speed on

reaching the ground.

6. A stone is thrown upwards at 20 m/s. Find the maximum height reached and the time it takes to return to the thrower.
7. A train slows uniformly from 40 m/s to 10 m/s in 6 s. Find the retardation and the distance covered in this time.
8. Two forces of 10 N each act at a point with an angle of 120° between them. Find the resultant.
9. A vehicle travelling at 72 km/h is brought to rest by a uniform retardation of 5 m/s². Find the stopping distance.
10. A body starts from rest and moves with a uniform acceleration of 2 m/s². Find the distance it covers in the fifth second.
11. Forces of 5 N and 12 N act at right angles. Find the resultant and the angle it makes with the 5 N force.
12. A runner completes two full laps of a 400 m track in 160 s. Find the average speed and the average velocity.

Answers to Exercise 2.1

1. Work, because it has magnitude only. The other three are vectors.
2. $R = \sqrt{(36 + 64)} = \sqrt{100} = 10$ N.
3. Horizontal $= 100 \cos 60° = 50$ N; vertical $= 100 \sin 60° = 86.6$ N.
4. $a = 30/10 = 3$ m/s²; $s = \tfrac{1}{2} \times 3 \times 100 = 150$ m.
5. $125 = 5t^2$, so $t^2 = 25$, $t = 5$ s; $v = 10 \times 5 = 50$ m/s.
6. $H = 400/20 = 20$ m; time up $= 2$ s, so the total is 4 s.

7. $a = (10 - 40)/6 = -5$ m/s², retardation 5 m/s²; $s = \tfrac{1}{2}(40 + 10) \times 6 = 150$ m.
8. $R^2 = 100 + 100 + 2(10)(10)\cos 120° = 200 - 100 = 100$, so $R = 10$ N.
9. 72 km/h = 20 m/s; $0 = 400 - 10s$, so $s = 40$ m.
10. $s_5 = \tfrac{1}{2} \times 2 \times 25 = 25$ m and $s_4 = 16$ m, so the fifth second gives 9 m.
11. $R = \sqrt{(25 + 144)} = 13$ N; angle $= \tan^{-1}(12/5) = 67.4°$ to the 5 N force.
12. Distance 800 m, so speed = 800/160 = 5 m/s; displacement is zero, so the average velocity is zero.

Exercise 2.2

1. A body moving with uniform acceleration has velocity 4 m/s at one instant and 16 m/s after 6 s. Find the acceleration and the distance moved in the 6 s.
2. A car A at 25 m/s follows a lorry B at 15 m/s on a straight road and is initially 80 m behind. After how long does A draw level with B?
3. A velocity-time graph of a body is a straight line from (0 s, 0 m/s) to (5 s, 20 m/s), then a horizontal line to 15 s. Find the total distance covered in 15 s.
4. A stone dropped into a well is heard to strike the water after 3 s. Ignoring the time for the sound to return, and taking $g = 10$ m/s², find the depth of the well.

5. A boy throws a ball vertically upwards and catches it after 4 s. With what speed was it thrown and how high did it rise?
6. A boat heads straight across a river at 6 m/s relative to the water. The current is 8 m/s. Find the speed of the boat relative to the bank.
7. A body moves from rest with uniform acceleration and covers 36 m in the first 6 s. Find the acceleration and the velocity at the end of 6 s.
8. The displacement-time graph of a cyclist is a straight line passing through (0 s, 0 m) and (20 s, 100 m). What is the velocity?
9. A particle starts from rest with $a = 3\ m/s^2$ for 4 s, then moves at constant velocity for 5 s. Find the total displacement.

Answers to Exercise 2.2

1. $a = (16 - 4)/6 = 2\ m/s^2$; $s = \tfrac{1}{2}(4 + 16) \times 6 = 60$ m.
2. Relative speed = 10 m/s; $t = 80/10 = 8$ s.
3. Area $= \tfrac{1}{2} \times 5 \times 20 + 20 \times 10 = 50 + 200 = 250$ m.
4. $h = \tfrac{1}{2} \times 10 \times 9 = 45$ m.
5. Time up = 2 s, so $u = 10 \times 2 = 20$ m/s; $H = 400/20 = 20$ m.
6. $v = \sqrt{(36 + 64)} = 10$ m/s.
7. $36 = \tfrac{1}{2} \times a \times 36$, so $a = 2\ m/s^2$; $v = 2 \times 6 = 12$ m/s.
8. Gradient $= 100/20 = 5$ m/s.

9. After 4 s: $v = 12$ m/s and $s = \frac{1}{2} \times 3 \times 16 = 24$ m; then $12 \times 5 = 60$ m; total = 84 m.

Chapter summary

- Scalars have magnitude only (distance, speed, mass, time, energy); vectors have magnitude and direction (displacement, velocity, acceleration, force, momentum).
- Vectors are added by the triangle or parallelogram law: $R^2 = P^2 + Q^2 + 2PQ\cos\theta$, and for perpendicular vectors $R = \sqrt{(P^2 + Q^2)}$.
- A vector F at angle θ to the horizontal has components $F\cos\theta$ (horizontal) and $F\sin\theta$ (vertical).
- Speed = distance/time; velocity = displacement/time; acceleration = change in velocity/time. Convert km/h to m/s by dividing by 3.6.
- For constant acceleration: $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s = \frac{1}{2}(u + v)t$.
- Free fall has $a = g = 10$ m/s² downwards. Upward throws reach $H = u^2/2g$, and the time of flight is $2u/g$.
- On a v–t graph the gradient is acceleration and the area is displacement; on an s–t graph the gradient is velocity.
- Relative velocity is found by vector subtraction; for a boat on a river, add the velocities as vectors.

Chapter 3: Projectile Motion, Circular Motion and Newton's Laws

Chapter 2 described how bodies move. This chapter explains why. The central ideas are the three laws of motion stated by Isaac Newton, which connect force, mass and acceleration. From them follow the conservation of linear momentum, the analysis of friction, and the conditions under which forces and turning effects balance. We also extend kinematics to motion in two dimensions: the flight of a projectile, and the motion of a body on a circular path.

Questions from this chapter are among the most frequent in the UTME. They include the range and time of flight of a projectile, the tension in a string whirled in a circle, the apparent weight of a person in a lift, the recoil of a gun, the acceleration of a block on a rough slope, and the reactions at the supports of a loaded beam. Each is treated below with full worked examples.

3.1 Projectile motion

A **projectile** is a body thrown or launched into the air and moving under the influence of gravity alone. A football kicked at Teslim Balogun Stadium, a stone thrown across a stream and water from a fountain all

follow curved paths. The key to analysing them is the **independence of horizontal and vertical motions**. If air resistance is ignored, the only force on the projectile is its weight, which acts vertically. Therefore:

- the horizontal component of velocity stays constant (no horizontal force, so no horizontal acceleration);
- the vertical motion has the constant acceleration g downwards.

The two motions are treated separately and are linked only by the time of flight.

Horizontal projection

Suppose a body leaves a point at a height h with a horizontal speed u. Its initial vertical velocity is zero, so the time to reach the ground is found from $h = \tfrac{1}{2}gt^2$, that is, $t = \sqrt{(2h/g)}$. The horizontal distance covered, the **range**, is $R = ut$. At landing, the vertical velocity is $v_y = gt$, and the speed is $\sqrt{(u^2 + v_y^2)}$.

WORKED EXAMPLE 3.1

A ball rolls off the edge of a platform 45 m high with a horizontal speed of 12 m/s. Find the time of flight, the horizontal range and the speed on reaching the ground.

Solution: $t = \sqrt{(2 \times 45/10)} = \sqrt{9} = 3$ s. Range $= 12 \times 3 = 36$ m. Vertical velocity at landing $= 10 \times 3 = 30$ m/s. Speed $= \sqrt{(12^2 + 30^2)} = \sqrt{(144 + 900)} = \sqrt{1044} = 32.3$ m/s.

Projection at an angle

When a body is projected with speed u at an angle θ above the horizontal, the initial velocity is resolved into u cos θ (horizontal) and u sin θ (vertical). The vertical motion is that of a stone thrown upwards, as in Chapter 2. The results are:

- time of flight: $T = 2u \sin\theta / g$;
- maximum height: $H = u^2 \sin^2\theta / (2g)$;
- horizontal range: $R = u^2 \sin 2\theta / g$.

The time to reach the highest point is T/2. At the top, the vertical velocity is zero, but the horizontal velocity u cos θ remains, so the speed there is not zero. The range is greatest when $\sin 2\theta = 1$, that is, $\theta = 45°$, giving $R_{max} = u^2/g$. Two angles that add up to 90° (for example 30° and 60°) give the same range, but the larger angle gives a greater height and longer time.

WORKED EXAMPLE 3.2

A stone is thrown with a speed of 50 m/s at 30° above the horizontal. Find (a) the time of flight, (b) the maximum height, (c) the range. Take $g = 10\ m/s^2$.

Solution: Horizontal component = 50 cos 30° = 43.3 m/s; vertical component = 50 sin 30° = 25 m/s. (a) T = 2 × 25/10 = 5 s. (b) H = 25²/(2 × 10) = 625/20 = 31.25 m. (c) R = 43.3 × 5 = 216.5 m. Check with the formula: R = 2500 × sin 60°/10 = 250 × 0.866 = 216.5 m.

WORKED EXAMPLE 3.3

What is the greatest horizontal distance that a footballer can kick a ball if the greatest speed he can give it is 20 m/s?

Solution: The maximum range occurs at 45°: $R_{max} = u^2/g = 400/10 = 40$ m.

3.2 Uniform circular motion

A body moving in a circle at constant speed has a velocity that is always changing in direction, so it is accelerating even though its speed is constant. Circular motion is described by:

- the **period** T, the time for one complete revolution;
- the **frequency** f = 1/T, the number of revolutions per second, in hertz (Hz);
- the **angular velocity** ω, the angle swept per second, in radians per second.

One complete revolution is 2π radians (360°), so ω = 2π/T = 2πf. The linear speed v of a body moving on a circle of radius r is v = ωr. To convert from revolutions per minute, divide by 60 to get revolutions per second.

Centripetal acceleration and force

The velocity of the body is tangent to the circle, and its change is directed towards the centre. The acceleration towards the centre is the **centripetal acceleration**:

$$a = v^2/r = \omega^2 r$$

By Newton's second law, a net inward force is required to produce this acceleration. This is the **centripetal force**:

$$F = mv^2/r = m\omega^2 r$$

The centripetal force is not a new kind of force; it is whatever real force supplies the inward pull. For a stone whirled on a string it is the tension; for a car on a bend it is friction between the tyres and the road; for the Moon it is the Earth's gravitational pull. Because the force is perpendicular to the velocity, it does no work and does not change the speed.

> **Key point.** If the string breaks, the stone does not fly outwards along the radius. It moves off along the tangent at the point of release, with the speed it had, because no force then acts on it (Newton's first law).

WORKED EXAMPLE 3.4

A stone of mass 0.5 kg is whirled in a horizontal circle of radius 2 m with a speed of 4 m/s. Find the centripetal acceleration, the tension in the string, the angular velocity and the period.

Solution: $a = v^2/r = 16/2 = 8$ m/s². Tension $= ma = 0.5 \times 8 = 4$ N. $\omega = v/r = 4/2 = 2$ rad/s. $T = 2\pi/\omega = \pi = 3.14$ s.

WORKED EXAMPLE 3.5

A car of mass 1000 kg rounds a flat bend of radius 50 m at 15 m/s. (a) What inward force is needed? (b) If the coefficient of friction between the tyres and the road is 0.5, what is the greatest speed at which the bend can be taken without skidding?

Solution: (a) $F = mv^2/r = 1000 \times 225/50 = 4500$ N. (b) The maximum friction is $\mu mg = 0.5 \times 1000 \times 10 = 5000$ N. At the limit, $mv^2/r = \mu mg$, so $v = \sqrt{(\mu g r)} = \sqrt{(0.5 \times 10 \times 50)} = \sqrt{250} = 15.8$ m/s. The mass cancels, so the limiting speed is the same for any car.

3.3 Newton's laws of motion

The first law: inertia

Newton's first law states that a body remains at rest, or continues in uniform motion in a straight line, unless acted upon by an external resultant force. The property of a body by which it resists a change in its state of rest or motion is its **inertia**, and the measure of inertia is the mass. This is why passengers in a bus lurch forward when the driver brakes suddenly: their bodies tend to continue at the old velocity.

The second law

Newton's second law states that the rate of change of momentum of a body is proportional to the resultant force acting on it and takes place in the direction of the force. For a body of constant mass this reduces to

$$F = ma$$

where F is the resultant force in newtons, m the mass in kilograms and a the acceleration in m/s². One **newton** is the force that gives a mass of 1 kg an acceleration of 1 m/s². Note that F is the *resultant* force; all forces on the body must be added as vectors before F = ma is applied.

Weight is the force of gravity on a body, W = mg. It is a vector, measured in newtons, and varies slightly from place to place, while mass does not. A student of mass 60 kg has weight 600 N on Earth.

WORKED EXAMPLE 3.6

A man of mass 60 kg stands on a weighing scale in a lift. Find the reading of the scale when the lift (a) accelerates upwards at 2 m/s², (b) accelerates downwards at 2 m/s², (c) moves at constant velocity, (d) falls freely.

Solution: The scale reading is the normal reaction R on the man. (a) Upward acceleration: R − mg = ma, so R = m(g + a) = 60 × 12 = 720 N. (b) Downward acceleration: mg − R = ma, so R = m(g − a) = 60 × 8 = 480 N. (c) a = 0, so R = mg = 600 N. (d) In free fall a = g, so R = 0; the man is apparently weightless.

The third law

Newton's third law states that to every action there is an equal and opposite reaction. More precisely, if body A exerts a force on body B, then B exerts a force of equal magnitude and opposite direction on A. The two forces act on *different* bodies, so they never cancel each other. A rocket moves forward because it pushes hot gases backwards, and the gases push it forwards; a swimmer pushes water backwards and is pushed forwards.

WORKED EXAMPLE 3.7

Two blocks, of masses 3 kg and 2 kg, are connected by a light string passing over a smooth pulley. Find the acceleration and the tension when they are released.

Solution: The heavier block accelerates downwards, the lighter upwards, with the same acceleration a. For the 3 kg block: $30 - T = 3a$. For the 2 kg block: $T - 20 = 2a$. Adding the equations: $10 = 5a$, so $a = 2\ \text{m/s}^2$. Then $T = 20 + 2 \times 2 = 24$ N. Check with the first equation: $30 - 24 = 6 = 3 \times 2$.

3.4 Linear momentum and impulse

Linear momentum is the product of mass and velocity, $p = mv$, with unit kg m/s (or N s). It is a vector. Newton's second law in its original form says that the net force equals the rate of change of momentum: $F = (mv - mu)/t$.

The product of a force and the time for which it acts is the **impulse**: impulse $= Ft = mv - mu =$ change of momentum. Impulse explains why a cricketer draws his hands back when catching a fast ball: lengthening the time of the stopping reduces the force for the same change of momentum. It is also the reason for crumple zones in cars and for padding in helmets.

WORKED EXAMPLE 3.8

A ball of mass 0.4 kg strikes a wall at 10 m/s and rebounds along the same line at 6 m/s. The contact lasts 0.02 s. Find the impulse and the average force.

Solution: Take the initial direction as positive. Change of momentum = $0.4 \times (-6) - 0.4 \times 10 = -2.4 - 4 = -6.4$ N s. The impulse has magnitude 6.4 N s, directed away from the wall. Average force = 6.4/0.02 = 320 N.

Conservation of linear momentum

The principle of conservation of linear momentum states that when bodies in a system interact, the total momentum of the system remains constant, provided no external resultant force acts on it. It follows from Newton's second and third laws, because the forces between two colliding bodies are equal and opposite and act for the same time, so the impulses cancel.

In an **elastic collision**, both momentum and kinetic energy are conserved. In an **inelastic collision**, momentum is conserved but some kinetic energy is changed to heat, sound and deformation. The most inelastic case is one where the bodies stick together. Momentum is conserved in both cases.

WORKED EXAMPLE 3.9

A 2 kg trolley moving at 3 m/s collides with a stationary 1 kg trolley and the two stick together. Find their common velocity and the kinetic energy lost.

Solution: Momentum before = 2 × 3 = 6 kg m/s. After, (2 + 1)v = 6, so v = 2 m/s. Kinetic energy before = ½ × 2 × 9 = 9 J. After = ½ × 3 × 4 = 6 J. Loss = 3 J.

WORKED EXAMPLE 3.10

A rifle of mass 4 kg fires a bullet of mass 0.02 kg with a velocity of 400 m/s. Find the recoil velocity of the rifle.

Solution: Initial momentum is zero, so 0 = 0.02 × 400 + 4V. Hence V = −8/4 = −2 m/s. The rifle recoils at 2 m/s opposite to the bullet.

3.5 Friction

Friction is the force that opposes the relative motion, or attempted motion, of two surfaces in contact. It acts along the surfaces and is caused by the roughness of the surfaces at microscopic scale. **Static friction** acts between surfaces at rest and increases to match the applied force up to a maximum value, the **limiting friction**. Once sliding starts, **kinetic (dynamic) friction** acts, and is usually slightly less than the limiting friction.

The laws of friction, verified by experiment, state that: (1) friction opposes motion; (2) the limiting friction is directly proportional to the normal reaction between the surfaces; (3) it is independent of the area of contact, provided the normal reaction is unchanged; and (4) kinetic friction is nearly independent of the speed. The **coefficient of friction** is

$$\mu = F/R$$

where F is the friction force and R the normal reaction. It is a ratio, and has no unit. On a horizontal surface R = mg, so F = μmg.

The inclined plane

A block of mass m rests on a plane inclined at angle θ to the horizontal. The weight mg is resolved into a component mg sin θ along the plane, pulling the block down the slope, and a component mg cos θ perpendicular to the plane, which equals the normal reaction. The maximum friction is μmg cos θ. If the plane is smooth, the acceleration down the slope is g sin θ. If it is rough and the block slides, the acceleration is a = g(sin θ − μ cos θ).

As the plane is slowly raised, the block just begins to slide when mg sin θ = μmg cos θ, that is, tan θ = μ. The angle θ at which this happens is the **angle of friction** (or angle of repose), and measuring it provides a simple experiment for finding μ.

WORKED EXAMPLE 3.11

A block of mass 20 kg rests on a rough horizontal floor where the coefficient of friction is 0.4. A horizontal force of 100 N is applied. Find the acceleration.

Solution: Normal reaction = $20 \times 10 = 200$ N. Friction = $0.4 \times 200 = 80$ N. Net force = $100 - 80 = 20$ N. $a = 20/20 = 1$ m/s².

WORKED EXAMPLE 3.12

A 10 kg block slides down a plane inclined at 30° to the horizontal. Find the acceleration if (a) the plane is smooth, (b) the coefficient of friction is 0.2.

Solution: (a) $a = g \sin 30° = 10 \times 0.5 = 5$ m/s². (b) $a = g(\sin\theta - \mu\cos\theta) = 10(0.5 - 0.2 \times 0.866) = 10(0.5 - 0.1732) = 3.27$ m/s².

3.6 Equilibrium of forces

A body is in **equilibrium** when it has no tendency to change its state of motion. Two conditions must be satisfied. First, the resultant of all the forces acting on it must be zero (**translational equilibrium**). Second, the sum of the turning effects about any point must be zero (**rotational equilibrium**).

Concurrent forces

For forces meeting at a point, the first condition alone is enough. The forces, drawn head to tail, form a closed polygon; for three forces, a closed triangle. A convenient method is to resolve all forces into horizontal and vertical components and require that the sum of each set is zero. Three forces in equilibrium also obey Lami's theorem: each force is proportional to the sine of the angle between the other two.

WORKED EXAMPLE 3.13

A load of weight 100 N is suspended from the midpoint of a rope whose two halves each make an angle of 30° with the horizontal. Find the tension in the rope.

Solution: By symmetry the tension T is the same in each half. The vertical components support the weight: $2T\sin 30° = 100$, so $T = 100/(2 \times 0.5) = 100$ N. The horizontal components are equal and opposite, so they cancel.

Moment of a force, couples and parallel forces

The **moment** of a force about a point is the product of the force and the perpendicular distance from the point to the line of action of the force. Its unit is the newton metre (N m). The **principle of moments** states that when a body is in equilibrium, the sum of the clockwise moments about any point equals the sum of the anticlockwise moments about that point.

A **couple** is a pair of equal, parallel and opposite forces whose lines of action do not coincide; it produces rotation without translation. The moment of a couple is one of the forces times the perpendicular distance between them, as when a driver turns a steering wheel with two hands.

The **centre of gravity** of a body is the point through which the whole weight of the body appears to act. For a uniform regular body it is at the geometric centre. A body is in **stable** equilibrium if, when slightly displaced, it returns to its original position (its centre of gravity rises); **unstable** if it moves further away (the centre of gravity falls); and **neutral** if it stays in its new position (the centre of gravity stays at the same height, like a ball on a table). A wide base and a low centre of gravity improve stability, which is why a fully loaded lorry carrying cement at the bottom is safer than one with a high load.

WORKED EXAMPLE 3.14

A uniform plank 6 m long weighing 100 N rests on supports at its two ends. A load of 60 N is placed 2 m from the left end. Find the reaction at each support.

Solution: Let the reactions be R_1 (left) and R_2 (right). Take moments about the left end: $R_2 \times 6 = 100 \times 3 + 60 \times 2 = 300 + 120 = 420$, so $R_2 = 70$ N. Vertical equilibrium: $R_1 + R_2 = 160$, so $R_1 = 90$ N.

Key point. When solving a moments problem, choose the pivot at the point where an unknown force acts. That force then has zero moment and drops out of the equation, leaving one unknown at a time.

Exercise 3.1

1. A projectile is launched at 40 m/s at 30° above the horizontal. Find the time of flight, the maximum height and the range.
2. A ball is thrown horizontally at 20 m/s from the top of a cliff 80 m high. How far from the foot of the cliff does it land?
3. A body of mass 2 kg moves in a circle of radius 0.5 m at 5 m/s. Find the centripetal force.
4. A wheel of radius 0.25 m rotates at 120 revolutions per minute. Find its angular velocity and the speed of a point on its rim.
5. A force of 20 N acts on a 5 kg block on a rough surface where friction is 5 N. Find the acceleration.
6. A man of mass 80 kg stands in a lift which accelerates downwards at 1.5 m/s². Find the normal reaction on him.
7. A 3 kg trolley moving at 4 m/s hits a stationary 5 kg trolley and they stick together. Find the common velocity.
8. A truck of mass 2000 kg moving at 15 m/s is brought to rest in 5 s. Find the average braking force.

9. Find the friction acting on a 10 kg block on a horizontal surface if $\mu = 0.3$.
10. A 30 kg child sits 2 m from the pivot of a see-saw. At what distance from the pivot must a 20 kg child sit to balance it?
11. A bullet of mass 0.05 kg is fired at 300 m/s from a gun of mass 3 kg. Find the recoil speed of the gun.

Answers to Exercise 3.1

1. $T = 2 \times 20/10 = 4$ s; $H = 400/20 = 20$ m; $R = 40 \cos 30° \times 4 = 34.64 \times 4 = 138.6$ m.
2. $t = \sqrt{(160/10)} = 4$ s; range $= 20 \times 4 = 80$ m.
3. $F = mv^2/r = 2 \times 25/0.5 = 100$ N.
4. $f = 2$ Hz, $\omega = 2\pi \times 2 = 12.6$ rad/s; $v = \omega r = 4\pi \times 0.25 = \pi = 3.14$ m/s.
5. Net force $= 15$ N; $a = 15/5 = 3$ m/s².
6. $R = m(g - a) = 80 \times 8.5 = 680$ N.
7. $12 = 8v$, so $v = 1.5$ m/s.
8. $F = m(v - u)/t = 2000 \times 15/5 = 6000$ N.
9. $F = \mu mg = 0.3 \times 10 \times 10 = 30$ N.
10. $30 \times 2 = 20 \times d$, so $d = 3$ m.
11. $0.05 \times 300 = 3V$, so $V = 5$ m/s.

Exercise 3.2

1. What is the maximum range of a ball thrown with a speed of 30 m/s?
2. A projectile launched at 30° has the same range as another launched at the same speed. At what angle was the second launched?
3. A car of mass 800 kg moves round a bend of radius 40 m at 20 m/s. Find the required centripetal force.
4. A block just starts to slide on a plank when the plank is tilted to the angle whose tangent is 0.75. Find μ.
5. A force of 50 N acts for 0.2 s on a 0.5 kg ball initially at rest. Find the impulse and the final speed.
6. A 50 N weight hangs from a string. It is pulled aside by a horizontal force until the string makes 45° with the vertical. Find the horizontal force and the tension.
7. A 1 kg ball moving at 6 m/s makes a perfectly elastic head-on collision with an identical stationary ball. Find the velocities after impact.
8. A 5 kg block is pulled along a horizontal surface by a 40 N horizontal force. The coefficient of friction is 0.5. Find the acceleration.
9. A 10 kg block slides down a smooth slope 30° to the horizontal. What force acts down the slope, and what is the normal reaction?

10. A stone of mass 0.2 kg is whirled on a string 1 m long at 3 revolutions per second. Find the tension (take $\pi^2 = 10$).

Answers to Exercise 3.2

1. $R_{max} = u^2/g = 900/10 = 90$ m.
2. Complementary angles give equal ranges: $90° - 30° = 60°$.
3. $F = 800 \times 400/40 = 8000$ N.
4. $\tan\theta = \mu$, so $\mu = 0.75$.
5. Impulse $= Ft = 50 \times 0.2 = 10$ N s; $v = 10/0.5 = 20$ m/s.
6. $F = W \tan 45° = 50$ N; $T = W/\cos 45° = 50/0.7071 = 70.7$ N.
7. Equal masses exchange velocities: the first stops, and the second moves at 6 m/s.
8. Friction $= 0.5 \times 50 = 25$ N; $a = (40 - 25)/5 = 3$ m/s^2.
9. Down-slope force $= 100 \sin 30° = 50$ N; normal reaction $= 100 \cos 30° = 86.6$ N.
10. $\omega = 2\pi \times 3 = 6\pi$; $T = m\omega^2 r = 0.2 \times 36\pi^2 \times 1 = 0.2 \times 36 \times 10 = 72$ N.

Chapter summary

- In projectile motion the horizontal velocity is constant and the vertical acceleration is g. For projection at θ: $T = 2u \sin\theta/g$, $H = u^2 \sin^2\theta/2g$, $R = u^2 \sin 2\theta/g$, with maximum range at 45°.
- Uniform circular motion has $\omega = 2\pi/T = 2\pi f$ and $v = \omega r$. The centripetal acceleration is v^2/r and the centripetal force is

mv^2/r, directed to the centre.

- Newton's first law defines inertia; the second gives $F = ma$ (resultant force); the third states that action and reaction are equal, opposite and act on different bodies.
- Weight $W = mg$. In a lift, the reading is $m(g + a)$ when accelerating up and $m(g - a)$ when accelerating down.
- Momentum $p = mv$; impulse Ft = change of momentum. Momentum is conserved in all collisions when no external force acts; kinetic energy is conserved only in elastic collisions.
- Friction: $F = \mu R$, with μ = tan of the angle of friction. On a slope, $a = g(\sin\theta - \mu\cos\theta)$.
- Equilibrium requires zero resultant force and zero resultant moment. Choose the pivot at an unknown force to simplify the moments equation.
- Stable, unstable and neutral equilibrium depend on whether the centre of gravity rises, falls or stays level when the body is displaced.

Chapter 4: Work, Energy, Power and Simple Machines

Forces make things move, but forces also make things happen. A generator turns water power into electricity, a car engine turns the chemical energy of petrol into motion, and a labourer on a building site in Abuja uses a pulley to raise blocks to the third floor. In every case the questions are the same: how much work is done, what energy is transferred, how fast is it done, and how efficiently? This chapter develops the concepts of work, energy and power, states the principle of conservation of energy, and applies these ideas to the simple machines that the UTME syllabus requires: the lever, pulley, inclined plane, wheel and axle, and screw.

The chapter relies on the mechanics of Chapters 2 and 3. Throughout, $g = 10\ \text{m/s}^2$ unless another value is stated.

4.1 Work

In physics, **work** is done when a force moves its point of application in the direction of the force, or in part of that direction. The work done by a constant force F whose point of application moves through a displacement s is

$$W = F\,s\cos\theta$$

where θ is the angle between the force and the displacement. When the force and displacement are in the same direction, θ = 0° and W = Fs. The SI unit of work is the **joule** (J): one joule is the work done when a force of one newton moves its point of application through one metre in the direction of the force. Work is a scalar quantity, even though it is calculated from two vectors.

Key point. No work is done when (a) the force is zero, (b) the displacement is zero, or (c) the force is perpendicular to the displacement (cos 90° = 0). A porter carrying a load at constant height along a level road does no work on the load against gravity, because the supporting force is vertical while the displacement is horizontal. A satellite in a circular orbit has a gravitational force perpendicular to its motion, so gravity does no work on it.

Work can be negative. If the force opposes the displacement (θ = 180°), then W = −Fs. Friction on a sliding block does negative work: it removes energy from the block.

When the force varies, the work is the area under the graph of force against displacement. For a force that rises uniformly from zero to F over a distance x (as in a stretched spring), the area is a triangle and W = ½Fx.

WORKED EXAMPLE 4.1

Sulaiman pulls a loaded sledge along level ground with a rope inclined at 60° to the horizontal. The tension in the rope is 200 N and the sledge moves 15 m. Calculate the work done by the tension. How much work is done by the weight of the sledge?

Solution: $W = F\,s\cos\theta = 200 \times 15 \times \cos 60° = 200 \times 15 \times 0.5 = 1{,}500$ J. The weight acts vertically downwards and the displacement is horizontal, so the angle is 90° and the work done by the weight is zero.

4.2 Energy

Energy is the capacity to do work. It is measured in joules, the same unit as work. Energy exists in many forms: mechanical (kinetic and potential), heat, light, sound, electrical, chemical and nuclear. When work is done on a body, energy is transferred to it; when a body does work, it loses energy.

Kinetic energy

Kinetic energy (K.E.) is the energy a body possesses because of its motion. A body of mass m moving with speed v has

$$\text{K.E.} = \tfrac{1}{2}mv^2$$

This follows from the equations of motion. A resultant force F acting on a mass m over a distance s produces $v^2 = u^2 + 2as$, and since F = ma, the work Fs = mas = $\frac{1}{2}m(v^2 - u^2)$. The work done by the resultant force equals the change in kinetic energy. This is the **work–energy theorem**:

Work done by the resultant force = $\frac{1}{2}mv^2 - \frac{1}{2}mu^2$

Because the speed is squared, doubling the speed of a vehicle multiplies its kinetic energy, and therefore its braking distance for a given braking force, by four. This is the physical reason why speed limits matter on the Lagos–Ibadan expressway.

Potential energy

Potential energy (P.E.) is the energy a body has because of its position or condition. *Gravitational* potential energy is the work done in raising a body of mass m through a height h against gravity:

P.E. = mgh

The height is measured from any convenient reference level, because only changes in potential energy have physical meaning. *Elastic* potential energy is stored in a stretched or compressed spring and is treated in Chapter 6. Chemical energy stored in fuel and food is another form of potential energy.

Conservation of energy

Key point. The **principle of conservation of energy** states that energy can neither be created nor destroyed; it can only be changed from one form to another. The total energy of an isolated system is constant.

In mechanical problems with no friction or air resistance, the sum of kinetic and potential energy is constant. A ball thrown upwards loses kinetic energy as it rises and gains an equal amount of potential energy; at the top, all its energy is potential. When friction is present, some mechanical energy changes into heat, but the total energy of all forms is still conserved.

WORKED EXAMPLE 4.2

A car of mass 1,200 kg increases its speed from 10 m/s to 20 m/s over a distance of 150 m on a level road. Find the increase in kinetic energy and the average resultant force on the car.

Solution: Initial K.E. = ½ × 1,200 × 10^2 = 60,000 J. Final K.E. = ½ × 1,200 × 20^2 = 240,000 J. Increase = 240,000 − 60,000 = 180,000 J. By the work–energy theorem, F × 150 = 180,000, so F = 1,200 N.

WORKED EXAMPLE 4.3

A ball of mass 0.4 kg is dropped from a height of 20 m. Neglecting air resistance, find (a) its speed on hitting the ground, (b) its kinetic and potential energies when it is 5 m above the ground.

Solution: (a) Energy at the start is purely potential: mgh = 0.4 × 10 × 20 = 80 J. This becomes kinetic energy at the ground: ½ × 0.4 × v^2 = 80, so v^2 = 400 and v = 20 m/s. (b) At 5 m, P.E. = 0.4 × 10 × 5 = 20 J. The ball has fallen 15 m, so K.E. = 0.4 × 10 × 15 = 60 J. The sum is 20 + 60 = 80 J, as conservation requires.

A pendulum and a roller coaster

Where a body moves along a curved smooth path, only the vertical drop matters, not the shape of the path. A pendulum bob released from a height h above its lowest point reaches the lowest point with v = √(2gh). If the bob is released from a point 0.8 m above the bottom, v = √(2 × 10 × 0.8) = √16 = 4 m/s. The tension in the string is always perpendicular to the motion, so it does no work.

4.3 Power

Power is the rate of doing work, or the rate at which energy is transferred:

P = W/t = E/t

The SI unit is the **watt** (W), equal to one joule per second. Larger units are the kilowatt (1 kW = 1,000 W) and the megawatt (1 MW = 10^6 W). An older unit, the horse power, is equal to about 746 W.

If a force F acts on a body moving with constant velocity v in the direction of the force, the work done in time t is Fvt, so

$$P = Fv$$

This relation explains why a car slows down on a steep hill if the driver does not press the accelerator: for a given engine power, a larger driving force means a lower speed.

WORKED EXAMPLE 4.4

A pump raises 1,200 kg of water every minute from a well to a tank 15 m above the water level. The pump is 60% efficient. Calculate the useful power output and the power supplied to the pump.

Solution: Work done per minute = mgh = 1,200 × 10 × 15 = 180,000 J. Output power = 180,000/60 = 3,000 W = 3 kW. Efficiency = output/input, so input = 3,000/0.60 = 5,000 W = 5 kW.

WORKED EXAMPLE 4.5

A car of mass 1,000 kg climbs a road that rises 1 m for every 20 m of road length. It moves at a constant speed of 15 m/s against a total frictional resistance of 500 N. Find the power developed by the engine.

Solution: Component of weight down the slope = mg sin θ = 1,000 × 10 × (1/20) = 500 N. At constant speed the driving force equals the sum of the opposing forces: F = 500 + 500 = 1,000 N. Power = Fv = 1,000 × 15 = 15,000 W = 15 kW.

The kilowatt-hour

Electrical energy is sold in **kilowatt-hours** (kWh). One kilowatt-hour is the energy transferred by a device of power 1 kW working for 1 hour: 1 kWh = 1,000 W × 3,600 s = 3.6×10^6 J. A 2 kW electric heater used for 3 hours consumes 6 kWh. If electricity costs ₦80 per kWh, the cost is 6 × 80 = ₦480.

Efficiency

No real machine delivers as much useful work as the energy supplied to it, because some energy is dissipated as heat, sound or friction. The **efficiency** is

efficiency = (useful energy output / total energy input) × 100% = (useful power output / power input) × 100%

Efficiency cannot exceed 100%. A petrol engine converts only about 25–30% of the chemical energy of the fuel into useful work, the rest being lost as exhaust heat and friction.

4.4 Sources of energy

Energy sources are grouped as **renewable** and **non-renewable**. Renewable sources are replaced naturally on a human time-scale: solar energy, hydroelectric power, wind, tides, biomass and geothermal energy. Nigeria's Kainji, Jebba and Shiroro dams are hydroelectric stations in which water falling from a height drives turbines; the gravitational potential energy of the water becomes kinetic energy, then rotational energy of the turbine, then electrical energy in the generator. Non-renewable sources, such as coal, crude oil, natural gas and nuclear fuels, exist in finite amounts and take millions of years to form.

The power available from a hydroelectric scheme is P = (mass flowing per second) × g × h × efficiency. If 500 kg of water falls each second through 40 m and the overall efficiency is 70%, the electrical power is 500 × 10 × 40 × 0.70 = 140,000 W = 140 kW.

4.5 Simple machines

A **machine** is a device that enables a force applied at one point (the **effort**, E) to overcome a force at another point (the **load**, L). Machines do not create energy. They make work easier by changing the size or

direction of a force, or by changing the distance moved. Three quantities describe every machine.

- **Mechanical advantage** (M.A.) = load/effort = L/E. It has no unit.
- **Velocity ratio** (V.R.) = distance moved by the effort / distance moved by the load. It depends only on the geometry of the machine.
- **Efficiency** = work output/work input × 100% = (L × load distance)/(E × effort distance) × 100% = (M.A./V.R.) × 100%.

> **Key point.** Efficiency = (M.A./V.R.) × 100%. For an ideal machine without friction, M.A. = V.R. and the efficiency is 100%. For a real machine M.A. is always less than V.R. A machine can have M.A. greater than 1 and still be less than 100% efficient; a machine with M.A. less than 1 may be useful if it gains speed or convenience of direction.

The lever

A lever is a rigid bar turning about a fixed point called the **fulcrum**. By the principle of moments, in equilibrium E × (effort arm) = L × (load arm), so the ideal M.A. = effort arm/load arm, which is also the V.R. Levers are classified by the position of the fulcrum, load and effort.

- **First class:** fulcrum between effort and load (crowbar, seesaw, scissors, pliers).
- **Second class:** load between fulcrum and effort (wheelbarrow, nutcracker, bottle opener). M.A. is always greater than 1.

- **Third class:** effort between fulcrum and load (a pair of tongs, the human forearm, a fishing rod). M.A. is always less than 1.

Pulley systems

A single fixed pulley changes only the direction of the force: V.R. = 1. A single movable pulley supported by two rope segments has V.R. = 2. In a **block and tackle** the V.R. equals the number of rope segments supporting the movable block, that is, the number of rope sections that carry the load. If the effort must be pulled 4 m to raise the load by 1 m, the V.R. is 4. Real pulleys have friction and the movable block has weight, so the efficiency is below 100%.

WORKED EXAMPLE 4.6

A pulley system of velocity ratio 4 is used to raise a load of 1,000 N through 2 m by an effort of 300 N. Calculate the mechanical advantage, the efficiency, the work done by the effort and the work done on the load.

Solution: M.A. = 1,000/300 = 3.33. Efficiency = (3.33/4) × 100% = 83.3%. The effort moves 4 × 2 = 8 m, so work by effort = 300 × 8 = 2,400 J. Work on load = 1,000 × 2 = 2,000 J. Check: 2,000/2,400 = 0.833, which is 83.3%. The 400 J difference is the energy wasted in friction and in lifting the movable pulley.

The inclined plane

Raising a load through a vertical height h by pushing it up a slope of length l needs a smaller force than lifting it vertically, but over a longer distance. The effort moves l while the load rises h, so

V.R. = l/h = 1/sin θ

For a smooth slope the effort is E = L sin θ = Lh/l. Loading a truck at the market by rolling a drum up a ramp is a familiar application.

WORKED EXAMPLE 4.7

A load of 600 N is pulled up a plank 5 m long to a platform 1.5 m high by a force of 250 N parallel to the plank. Find the M.A., V.R. and efficiency, and the work done against friction.

Solution: M.A. = 600/250 = 2.4. V.R. = 5/1.5 = 3.33. Efficiency = 2.4/3.33 × 100% = 72%. Work input = 250 × 5 = 1,250 J. Useful work = 600 × 1.5 = 900 J. Work done against friction = 1,250 − 900 = 350 J.

The wheel and axle

A wheel and axle is a large wheel of radius R fixed rigidly to a smaller axle of radius r. An effort applied to a rope on the rim of the wheel raises a load on a rope wound round the axle. In one turn the effort moves 2πR and the load moves 2πr, so V.R. = R/r. A steering wheel, a door knob, a

screwdriver and the windlass of a well are examples. The same principle applies to a pair of gear wheels, where V.R. is the ratio of the numbers of teeth.

The screw

A screw is an inclined plane wrapped round a cylinder. The distance advanced in one complete turn is the **pitch**, p. In a screw jack, a handle of length (radius) r turns the screw. In one turn the effort moves 2πr while the load rises p, so V.R. = 2πr/p. A car jack has a very large V.R., so a small effort lifts a heavy car, but the efficiency is low (often below 40%) because of friction in the thread. This large friction is useful: it prevents the load from running back down when the effort is released. A machine in which the load stays in place when the effort is removed is called **self-locking**, and this requires an efficiency below 50%.

WORKED EXAMPLE 4.8

A screw jack has a pitch of 5 mm and a handle 35 cm long. Find its velocity ratio. If its efficiency is 30%, find the effort required to lift a load of 6,000 N.

Solution: V.R. = 2πr/p = 2 × 3.14 × 0.35/0.005 = 2.198/0.005 = 439.6, approximately 440. Efficiency 30% gives M.A. = 0.30 × 439.6 = 131.9. Effort = load/M.A. = 6,000/131.9 = 45.5 N.

4.6 Reading the examination question

Questions on this chapter fall into four families. The first asks for work or power from forces and distances; check the angle and the units (convert grams to kilograms, minutes to seconds, kilometres per hour to metres per second by dividing by 3.6). The second uses the conservation of energy to find a speed or a height, which is quicker than using the equations of motion. The third concerns efficiency, where you must keep clear whether a quoted percentage refers to energy, power, or M.A./V.R. The fourth is the machine question, where V.R. is read from the geometry and M.A. from the forces. Always find V.R. from distances and M.A. from forces; do not mix them.

Exercise 4.1

1. A force of 50 N acts at 60° to the horizontal on a trolley and moves it 8 m along level ground. Calculate the work done by the force.
2. How much work is done in lifting a 2 kg bag of rice through 3 m?
3. Find the kinetic energy of a 0.5 kg ball moving at 12 m/s.
4. A car of mass 800 kg travelling at 15 m/s is brought to rest in 45 m. Find the average braking force.
5. A stone of mass 0.2 kg is thrown vertically upwards at 15 m/s. Use energy to find the maximum height reached and the potential energy there.

6. A motor raises a 300 kg load through 12 m in 20 s. Find the useful power. If the motor is 75% efficient, what is its input power?
7. An engine delivers 40 kW to propel a vehicle at a steady 25 m/s. Find the driving force.
8. A machine of velocity ratio 5 and efficiency 80% is operated by an effort of 100 N. What load can it raise?
9. A 200 kg drum is rolled up a smooth ramp 10 m long onto a truck bed 1.5 m high. What effort parallel to the ramp is needed? If the effort needed is actually 400 N because of friction, find the efficiency.
10. A 2 kg block slides from rest down a smooth curved track and falls through a vertical height of 1.8 m. It then slides on a rough horizontal surface with coefficient of friction 0.3. Find its speed at the bottom and the distance it travels before stopping.
11. A ball of mass 0.5 kg is dropped from 4 m and rebounds to 2.5 m. How much mechanical energy is lost in the impact?

Answers to Exercise 4.1

1. $W = 50 \times 8 \times \cos 60^\circ = 50 \times 8 \times 0.5 = 200$ J.
2. $W = mgh = 2 \times 10 \times 3 = 60$ J.
3. K.E. $= \tfrac{1}{2} \times 0.5 \times 12^2 = 0.25 \times 144 = 36$ J.
4. K.E. $= \tfrac{1}{2} \times 800 \times 225 = 90{,}000$ J; $F = 90{,}000/45 = 2{,}000$ N.

5. ½ × 0.2 × 15^2 = 22.5 J; mgh = 22.5 gives 0.2 × 10 × h = 22.5, so h = 11.25 m; P.E. at top = 22.5 J.
6. Work = 300 × 10 × 12 = 36,000 J; P = 36,000/20 = 1,800 W; input = 1,800/0.75 = 2,400 W.
7. F = P/v = 40,000/25 = 1,600 N.
8. M.A. = 0.80 × 5 = 4; load = 4 × 100 = 400 N.
9. Smooth: E = 2,000 × 1.5/10 = 300 N. With friction: M.A. = 2,000/400 = 5; V.R. = 10/1.5 = 6.67; efficiency = 5/6.67 = 75%.
10. v = √(2 × 10 × 1.8) = √36 = 6 m/s. Friction deceleration = 0.3 × 10 = 3 m/s^2; distance = v^2/(2a) = 36/6 = 6 m.
11. Energy lost = mg(4 − 2.5) = 0.5 × 10 × 1.5 = 7.5 J.

Chapter summary

- Work W = F s cos θ, measured in joules; no work is done when the force is perpendicular to the displacement. The area under a force–displacement graph is the work.
- Kinetic energy = ½mv^2; gravitational potential energy = mgh. The work done by the resultant force equals the change in kinetic energy.
- Energy is conserved: it changes form but is never created or destroyed. In frictionless motion K.E. + P.E. is constant.
- Power P = W/t = Fv, in watts. One kilowatt-hour = 3.6 × 10^6 J. Efficiency = useful output/input × 100%, always below 100% for real devices.

- Energy sources are renewable (solar, hydro, wind, tides, biomass) or non-renewable (coal, oil, gas, nuclear fuel).
- For machines: M.A. = load/effort; V.R. = effort distance/load distance; efficiency = M.A./V.R. × 100%.
- V.R. values: lever, arm ratio; pulley system, number of supporting rope segments; inclined plane, l/h; wheel and axle, R/r; screw jack, $2\pi r/p$.

Chapter 5: Gravitation and Simple Harmonic Motion

Every object released near the Earth falls; the Moon circles the Earth; the Earth circles the Sun. These observations are all consequences of a single law of attraction between masses. This chapter studies Newton's law of universal gravitation, the field strength of the Earth, the laws of planetary motion stated by Kepler, the motion of artificial satellites and the escape of a rocket from the Earth. It then turns to a different but equally important kind of motion, the to-and-fro oscillation of a pendulum, a spring or a vibrating string, which is called simple harmonic motion and which underlies the study of waves in later chapters.

Where numerical values are needed, take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}, the radius of the Earth $R = 6.4 \times 10^6$ m, and $g = 10$ m/s^2 at the surface of the Earth unless stated otherwise.

5.1 Newton's law of universal gravitation

Newton proposed that every particle in the universe attracts every other particle with a force that acts along the line joining them. The law is stated as follows.

Key point. The gravitational force between two point masses is directly proportional to the product of the masses and inversely proportional to the square of the distance between them: $F = G m_1 m_2 / r^2$.

G is the **universal gravitational constant**. It has the same value everywhere, and was first measured in the laboratory by Henry Cavendish using a torsion balance. Its small value, 6.67×10^{-11} N m^2 kg^{-2}, explains why we do not notice the attraction between ordinary objects: only when at least one of the masses is as large as a planet does the force become large. The forces on the two masses are equal and opposite, as Newton's third law requires, although the smaller mass has the greater acceleration.

Because the force depends on $1/r^2$, doubling the separation reduces the force to one quarter, and tripling it reduces it to one ninth. This is called an **inverse-square law**. For extended spherical bodies of uniform density, the force is calculated as though all the mass were concentrated at the centre, so r is the distance between centres.

WORKED EXAMPLE 5.1

Find the gravitational force between two students of masses 50 kg and 80 kg whose centres are 2 m apart.

Solution: $F = G\,m_1m_2/r^2 = 6.67 \times 10^{-11} \times 50 \times 80/2^2 = 6.67 \times 10^{-11} \times 4{,}000/4 = 6.67 \times 10^{-11} \times 1{,}000 = 6.67 \times 10^{-8}$ N. This is far too small to feel.

5.2 Gravitational field strength and the acceleration due to gravity

The region around a mass in which another mass experiences a gravitational force is a **gravitational field**. The **gravitational field strength** g at a point is the force per unit mass placed at that point: $g = F/m$. Its unit is N/kg, which is equivalent to m/s^2. For a body of mass m at the surface of the Earth, whose mass is M and radius R, the weight mg equals the gravitational force:

$mg = GMm/R^2$, so $g = GM/R^2$

The mass m cancels, so all bodies fall with the same acceleration in the absence of air. At a distance r from the centre of the Earth ($r \geq R$), $g_r = GM/r^2 = g(R/r)^2$.

Using the given numbers, the mass of the Earth follows from $M = gR^2/G = 10 \times (6.4 \times 10^6)^2/(6.67 \times 10^{-11}) = 10 \times 4.096 \times 10^{13}/(6.67 \times 10^{-11}) = 6.1 \times 10^{24}$ kg. This agrees with the accepted figure of about 6.0×10^{24}

kg.

Variation of g

- **With height:** g decreases as one moves away from the Earth, according to the inverse-square law. At a height equal to the Earth's radius (r = 2R), g falls to a quarter of its surface value.
- **With depth:** inside the Earth, only the mass lying nearer the centre than the point attracts it in the manner above, and g decreases, becoming zero at the centre.
- **With latitude:** the Earth is slightly flattened at the poles and rotates, so g is about 9.83 m/s² at the poles and 9.78 m/s² at the equator.

Body	Surface g (m/s²)
Earth	9.8
Moon	1.6
Mars	3.7
Jupiter	24.8

Mass and weight. The **mass** of a body is the quantity of matter it contains; it is a scalar measured in kilograms and does not change from place to place. The **weight** is the gravitational force on the body, a vector measured in newtons, and it depends on the local value of g. A student of mass 60 kg weighs 600 N on Earth (taking g = 10 m/s²) but only 60 × 1.6 = 96 N on the Moon. Her mass is the same on both.

WORKED EXAMPLE 5.2

A satellite is at a height of 3.2×10^6 m above the surface of the Earth. Calculate the acceleration due to gravity there. ($R = 6.4 \times 10^6$ m, $g = 10$ m/s² at the surface.)

Solution: $r = R + h = 6.4 \times 10^6 + 3.2 \times 10^6 = 9.6 \times 10^6$ m. So $r/R = 1.5$. $g_r = g(R/r)^2 = 10/(1.5)^2 = 10/2.25 = 4.44$ m/s².

WORKED EXAMPLE 5.3

A planet has a mass 8 times that of the Earth and a radius twice that of the Earth. What is the acceleration due to gravity on its surface, in terms of the Earth's value?

Solution: $g = GM/R^2$. Then $g_{planet}/g_{Earth} = (M_p/M_E) \times (R_E/R_p)^2 = 8 \times (1/2)^2 = 8/4 = 2$. So $g_{planet} = 2 \times 10 = 20$ m/s².

5.3 Gravitational potential energy and escape velocity

Near the surface of the Earth, the potential energy of a body is mgh. This formula assumes that g is constant, which is true only for small heights. A more general definition takes the zero of potential energy at infinite distance, where the force is zero. The potential energy of a mass m at a distance r from the centre of a body of mass M is then

$$E_p = -GMm/r$$

The negative sign shows that work must be done to take the mass to infinity: the mass is bound. The **gravitational potential** at the point is $V = -GM/r$, the potential energy per unit mass.

The **escape velocity** is the minimum speed with which a body must be projected from the surface of a planet so that it escapes the planet's gravitational field and never returns. At escape, the kinetic energy provided just cancels the binding energy:

$\tfrac{1}{2}mv^2 = GMm/R$, so $v = \sqrt{(2GM/R)} = \sqrt{(2gR)}$

For the Earth, $v = \sqrt{(2 \times 10 \times 6.4 \times 10^6)} = \sqrt{(1.28 \times 10^8)} = 1.13 \times 10^4$ m/s, that is 11.3 km/s (the accepted value is 11.2 km/s, using $g = 9.8$). The escape velocity does not depend on the mass of the projectile or the direction of projection, provided there is no air resistance. For the Moon, with $g = 1.6$ m/s² and radius 1.74×10^6 m, $v = \sqrt{(2 \times 1.6 \times 1.74 \times 10^6)} = \sqrt{(5.57 \times 10^6)} = 2{,}360$ m/s, which is why the Moon cannot retain an atmosphere.

5.4 Kepler's laws of planetary motion

From the careful observations made by Tycho Brahe, Johannes Kepler deduced three laws describing the motion of planets around the Sun.

1. **First law (law of orbits):** each planet moves in an ellipse with the Sun at one focus.
2. **Second law (law of areas):** the line joining a planet to the Sun sweeps out equal areas in equal intervals of time. A planet therefore moves faster when it is nearer the Sun (perihelion) and slower when

it is farthest (aphelion). This expresses the conservation of angular momentum.

3. **Third law (law of periods):** the square of the period T of revolution of a planet is proportional to the cube of its mean distance r from the Sun: $T^2 \propto r^3$, or $T^2/r^3 = \text{constant}$.

The third law follows from Newton's law for a circular orbit. The gravitational force supplies the centripetal force: $GMm/r^2 = m\omega^2 r = m(4\pi^2/T^2)r$. Hence $T^2 = (4\pi^2/GM)r^3$, in which the constant $4\pi^2/GM$ depends only on the central mass.

WORKED EXAMPLE 5.4

A planet orbits a star at a mean distance four times the Earth–Sun distance, and the star has the same mass as the Sun. Find the period of the planet in years. A second planet has a period of 125 years; find its distance in terms of the Earth–Sun distance.

Solution: $T^2 \propto r^3$, so $T = (r/r_E)^{(3/2)}$ years. For $r = 4$, $T = 4^{(3/2)} = 8$ years. For $T = 125$, $r^3 = T^2 = 15{,}625$, so $r = 25$ (since $25^3 = 15{,}625$). The second planet is 25 times the Earth–Sun distance away.

5.5 Satellites

An artificial satellite stays in orbit because its speed carries it round the Earth just as quickly as gravity pulls it towards the centre; it is in continuous free fall. For a circular orbit of radius r, gravity provides the

centripetal force:

$GMm/r^2 = mv^2/r$, so $v = \sqrt{(GM/r)} = \sqrt{(gR^2/r)}$

The period is $T = 2\pi r/v$. A satellite in an orbit just above the surface ($r \approx R$) has $v = \sqrt{(gR)} = \sqrt{(10 \times 6.4 \times 10^6)} = \sqrt{(6.4 \times 10^7)} = 8{,}000$ m/s, and period $T = 2\pi R/v = 2 \times 3.14 \times 6.4 \times 10^6/8{,}000 = 5{,}027$ s, about 84 minutes. Higher orbits have lower speeds and longer periods.

A **geostationary** (or parking) orbit has a period of exactly 24 hours, a circular path in the equatorial plane, and the same sense of rotation as the Earth. The satellite then remains above the same point on the ground, which is why it is used for television broadcasting and telecommunication, including the signals received by dish antennas in Nigeria. The radius of this orbit is about 4.2×10^7 m from the centre of the Earth, that is about 3.6×10^7 m (36,000 km) above the surface. Satellites in low polar orbits, with periods of about 100 minutes, are used for weather observation, mapping and surveillance.

Weightlessness

Astronauts in an orbiting spacecraft seem to float. Gravity has not vanished (at the height of the International Space Station g is still about 8.7 m/s²), but both the astronaut and the spacecraft are in free fall with the same acceleration, so there is no contact force between them. The apparent weight, which is the reaction a supporting surface exerts, is zero. This is **apparent weightlessness**. A person in a freely falling lift is in the same condition.

5.6 Simple harmonic motion

An oscillation is a repeated to-and-fro motion about a fixed position. A body performs **simple harmonic motion** (S.H.M.) if its acceleration is directly proportional to its displacement from a fixed point and is always directed towards that point:

$a = -\omega^2 x$

The negative sign shows that the acceleration opposes the displacement. Because F = ma, the restoring force is proportional to the displacement, $F = -kx$.

Terms used

- **Amplitude** (A): the maximum displacement from the equilibrium position.
- **Period** (T): the time for one complete oscillation, in seconds.
- **Frequency** (f): the number of oscillations per second, in hertz; $f = 1/T$.
- **Angular frequency** (ω): $\omega = 2\pi f = 2\pi/T$, in rad/s.
- **Phase:** the stage of the oscillation reached; two points in the cycle differing by 2π are in phase.

Equations of S.H.M.

If the motion is timed from the equilibrium position, the displacement is $x = A \sin \omega t$. The corresponding results are:

- velocity at displacement x: $v = \omega\sqrt{(A^2 - x^2)}$;

- maximum speed (at the centre): $v_{max} = \omega A$;
- maximum acceleration (at the ends): $a_{max} = \omega^2 A$.

The velocity is greatest where the acceleration is zero (the centre) and zero where the acceleration is greatest (the extremes). This is the same as a ball thrown upwards, which is momentarily at rest at the top but has acceleration g there.

WORKED EXAMPLE 5.5

A particle executes S.H.M. with amplitude 0.05 m and frequency 2 Hz. Calculate its angular frequency, maximum speed, maximum acceleration, and speed when 0.03 m from the centre.

Solution: $\omega = 2\pi f = 2 \times 3.14 \times 2 = 12.57$ rad/s. $v_{max} = \omega A = 12.57 \times 0.05 = 0.628$ m/s. $a_{max} = \omega^2 A = (12.57)^2 \times 0.05 = 157.9 \times 0.05 = 7.90$ m/s². At $x = 0.03$ m: $v = \omega\sqrt{(A^2 - x^2)} = 12.57 \times \sqrt{(0.0025 - 0.0009)} = 12.57 \times \sqrt{0.0016} = 12.57 \times 0.04 = 0.503$ m/s.

The mass–spring system

A mass m on a spring of force constant k (see Chapter 6) experiences a restoring force $F = -kx$, so $a = -(k/m)x$, which is S.H.M. with $\omega^2 = k/m$. The period is

$$T = 2\pi\sqrt{(m/k)}$$

A heavier mass or a softer spring gives a longer period. The result is the same whether the spring is horizontal or hangs vertically, because the equilibrium position simply moves to include the weight.

WORKED EXAMPLE 5.6

A mass of 0.4 kg is attached to a spring of force constant 40 N/m and set oscillating. Find the period and frequency.

Solution: $T = 2\pi\sqrt{(m/k)} = 2 \times 3.14 \times \sqrt{(0.4/40)} = 6.28 \times \sqrt{0.01} = 6.28 \times 0.1 = 0.628$ s. Frequency $= 1/0.628 = 1.59$ Hz.

The simple pendulum

A **simple pendulum** is a small heavy bob hung on a light inextensible string of length l. For small angles of swing (below about 10°) the restoring force is approximately proportional to the displacement, and the motion is S.H.M. with

$$T = 2\pi\sqrt{(l/g)}$$

The period does not depend on the mass of the bob or on the amplitude (for small swings). It increases with the square root of the length: quadrupling the length doubles the period. A pendulum with a period of exactly 2 s is a *seconds pendulum*; on Earth its length is $l = gT^2/4\pi^2 = 10 \times 4/39.5 = 1.01$ m, which is very nearly one metre.

WORKED EXAMPLE 5.7

A seconds pendulum is carried to the Moon, where g = 1.6 m/s². What is its period there, and how long would a pendulum have to be to keep a period of 2 s on the Moon?

Solution: $T \propto 1/\sqrt{g}$, so $T_{Moon}/T_{Earth} = \sqrt{(10/1.6)} = \sqrt{6.25} = 2.5$. $T_{Moon} = 2.5 \times 2 = 5$ s. For T = 2 s on the Moon, $l = gT^2/4\pi^2 = 1.6 \times 4/39.5 = 0.162$ m.

Energy in S.H.M.

In an oscillator, energy is exchanged between kinetic and potential forms. At the extreme positions, the speed is zero and the energy is all potential; at the centre, the potential energy is least and the speed greatest. For a spring system the total energy is constant and equal to the maximum potential energy:

$$E = \tfrac{1}{2}kA^2 = \tfrac{1}{2}m\omega^2A^2$$

For the data of Worked example 5.5, with m = 0.2 kg, $\omega^2 = 157.9$, $k = m\omega^2 = 31.6$ N/m and $E = \tfrac{1}{2} \times 31.6 \times (0.05)^2 = 0.0395$ J.

Damping and resonance

In practice an oscillator loses energy to friction and air resistance, and its amplitude decays; the oscillation is **damped**. The shock absorbers of a vehicle apply damping deliberately so that the car body does not bounce for long after hitting a pothole. If a periodic force is applied to an oscillator, it performs **forced oscillations** at the frequency of the

driving force. When the driving frequency is equal to the natural frequency of the oscillator, the amplitude becomes very large. This is **resonance**. A child on a swing pushed at the right moment in each cycle, a radio receiver tuned to a station, and the shattering of a glass by a singer's note are all examples. Soldiers break step when crossing a bridge, because the regular frequency of their feet might resonate with the bridge.

Exercise 5.1

1. Find the gravitational force between two masses of 10 kg each whose centres are 0.5 m apart.
2. What is the acceleration due to gravity at a distance of 3R from the centre of the Earth, where R is the radius of the Earth? (Surface g = 10 m/s².)
3. A planet has a mass 8 times that of the Earth and a radius twice that of the Earth. What is the escape velocity from it, compared to that from the Earth?
4. Calculate the escape velocity from the Moon, given that g = 1.6 m/s² and the radius is 1.74×10^6 m.
5. A planet takes 125 years to orbit the Sun. How many times farther from the Sun is it than the Earth?
6. Find the orbital speed of a satellite skimming the surface of the Earth ($R = 6.4 \times 10^6$ m, g = 10 m/s²).
7. A simple pendulum is 0.4 m long. Find its period, taking g = 10 m/s² and $\pi = 3.14$.

8. A body of mass 0.25 kg suspended from a spring oscillates with period 0.5 s. Find the force constant of the spring.
9. A particle executes S.H.M. described by $x = 0.1 \sin 5t$ (metres). Find the period, maximum velocity, maximum acceleration, and speed when $x = 0.06$ m.
10. A pendulum has a period of 2 s on the Earth. What is its period on the surface of the Moon, where $g = 1.6$ m/s²?

Answers to Exercise 5.1

1. $F = 6.67 \times 10^{-11} \times 10 \times 10/(0.5)^2 = 6.67 \times 10^{-9}/0.25 = 2.67 \times 10^{-8}$ N.
2. $g_r = g/9 = 10/9 = 1.11$ m/s².
3. $g_p = 2g$ from Example 5.3. $v \propto \sqrt{(gR)}$, so $v_p/v_E = \sqrt{(2 \times 2)} = 2$. It is twice the Earth's value, about 22.6 km/s.
4. $v = \sqrt{(2 \times 1.6 \times 1.74 \times 10^6)} = \sqrt{(5.568 \times 10^6)} = 2{,}360$ m/s.
5. $r^3 = T^2 = 15{,}625$, so r = 25 times the Earth–Sun distance.
6. $v = \sqrt{(gR)} = \sqrt{(6.4 \times 10^7)} = 8{,}000$ m/s.
7. $T = 2\pi\sqrt{(0.4/10)} = 6.28 \times 0.2 = 1.26$ s.
8. $k = 4\pi^2 m/T^2 = 39.5 \times 0.25/0.25 = 39.5$ N/m.
9. $A = 0.1$ m, $\omega = 5$ rad/s. $T = 2\pi/5 = 1.26$ s. $v_{max} = 5 \times 0.1 = 0.5$ m/s. $a_{max} = 25 \times 0.1 = 2.5$ m/s². At $x = 0.06$: $v = 5\sqrt{(0.01 - 0.0036)} = 5 \times 0.08 = 0.4$ m/s.
10. $T = 2 \times \sqrt{(10/1.6)} = 2 \times 2.5 = 5$ s.

Chapter summary

- Newton's law: $F = Gm_1m_2/r^2$, an inverse-square force. $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.
- Field strength $g = GM/r^2$; g falls with height, falls inside the Earth towards the centre, and varies with latitude. Mass is constant; weight = mg depends on place.
- Gravitational potential energy is $-GMm/r$. The escape velocity is $\sqrt{(2GM/R)} = \sqrt{(2gR)}$, about 11.2 km/s for the Earth.
- Kepler's laws: elliptical orbits; equal areas in equal times; $T^2 \propto r^3$.
- Circular orbit speed $v = \sqrt{(GM/r)}$. A geostationary satellite has a 24 hour period, orbits above the equator and appears fixed in the sky. Orbiting bodies are in free fall and apparently weightless.
- S.H.M.: $a = -\omega^2x$; $T = 2\pi/\omega$; $v = \omega\sqrt{(A^2 - x^2)}$; $v_{max} = \omega A$; $a_{max} = \omega^2 A$; total energy $\frac{1}{2}kA^2$.
- Spring: $T = 2\pi\sqrt{(m/k)}$. Simple pendulum: $T = 2\pi\sqrt{(l/g)}$, independent of mass and (for small swings) amplitude.
- Damping reduces amplitude; resonance occurs when driving frequency equals natural frequency.

Chapter 6: Elasticity, Density, Pressure and Fluids at Rest and in Motion

Solid objects stretch when they are pulled and regain their shape when released; liquids exert a push on the walls of every vessel that holds them; a ship made of steel floats, while a small steel nail sinks; and a hydraulic jack in a mechanic's workshop in Ibadan lifts a lorry with a gentle push of the hand. All of these familiar facts belong to one branch of physics, the mechanical properties of matter. In this chapter we study how solids respond to forces (elasticity), how the compactness of a substance is measured (density), how forces spread over areas (pressure), how liquids and gases behave when they are at rest, and finally how they behave when they flow.

Unless a question states otherwise, take $g = 10\ \text{m/s}^2$, the density of water as $1{,}000\ \text{kg/m}^3$ ($1\ \text{g/cm}^3$), the density of mercury as $13{,}600\ \text{kg/m}^3$ and standard atmospheric pressure as $1.0 \times 10^5\ \text{Pa}$ (equivalent to 76 cm of mercury).

6.1 Elasticity and Hooke's law

A body is **elastic** if it returns to its original size and shape after the forces that deformed it have been removed. A steel spring, a rubber band, a bow and a diving board are all elastic within certain limits.

The simplest quantitative law of elasticity was discovered by Robert Hooke in the seventeenth century. It describes a spring or a wire that is being stretched by a steadily increasing load.

> **Key point. Hooke's law.** Provided the elastic limit is not exceeded, the extension of a spring or wire is directly proportional to the force (load) producing it: F = ke, where k is the force constant (spring constant) in N/m and e is the extension in metres.

The **force constant** k measures the stiffness of the spring. A large k means that a large force is needed for each metre of extension. If a graph of the force against the extension is plotted, it is a straight line through the origin up to a certain point, and the gradient of the line is k. Beyond a point called the **limit of proportionality** the graph begins to curve, because the extension grows faster than the force. A little further on is the **elastic limit**, the largest force for which the material still returns to its original length when the force is removed. Beyond it the material suffers permanent deformation. Beyond it the material is permanently deformed.

Springs in series have a smaller combined constant $k = k_1k_2/(k_1 + k_2)$; springs in parallel share the load and their constants add, $k = k_1 + k_2$.

Energy stored in a stretched spring

Work must be done to stretch a spring, and this work is stored as **elastic potential energy**. Because the force rises steadily from zero to the final value F = ke, the average force is half the final force, and the work done is the average force multiplied by the extension:

$E = \tfrac{1}{2}Fe = \tfrac{1}{2}ke^2$

On a force–extension graph, this energy is the area under the line. Notice that doubling the extension quadruples the stored energy.

WORKED EXAMPLE 6.1

A spring extends by 3.0 cm when a force of 12 N is applied. Calculate (a) the force constant, (b) the extension produced by a force of 20 N, (c) the energy stored in the spring in case (b).

Solution: (a) e = 3.0 cm = 0.030 m, so k = F/e = 12/0.030 = 400 N/m. (b) e = F/k = 20/400 = 0.050 m = 5.0 cm. (c) $E = \tfrac{1}{2}ke^2 = \tfrac{1}{2} \times 400 \times (0.050)^2 = 200 \times 0.0025 = 0.50$ J. As a check, $\tfrac{1}{2}Fe = \tfrac{1}{2} \times 20 \times 0.050 = 0.50$ J, which agrees.

Stress, strain and the Young modulus

The force constant k depends on the shape of the particular spring or wire, so it cannot be used to compare materials. To describe the material itself, we divide out the effect of the dimensions. **Tensile stress** is the force applied per unit cross-sectional area of the wire, stress = F/A,

measured in pascals (Pa, which is N/m^2). **Tensile strain** is the fractional increase in length, strain = e/L, where L is the original length. Strain is a ratio of two lengths and therefore has no unit.

Within the limit of proportionality, stress is proportional to strain, and the ratio is a constant for a given material, called the **Young modulus** E:

$E = \text{stress/strain} = (F/A)/(e/L) = FL/(Ae)$

The Young modulus of steel is about 2×10^{11} Pa, that of copper about 1.2×10^{11} Pa and that of rubber only about 10^7 to 10^8 Pa. A material with a large Young modulus is stiff, because a large stress is needed to produce a small strain. A wire that is twice as long extends twice as much under the same load; a wire of twice the cross-sectional area extends half as much. These facts are summarised by $e = FL/(AE)$.

WORKED EXAMPLE 6.2

A steel wire 2.0 m long and of cross-sectional area 1.0×10^{-6} m² carries a load of 200 N and extends by 2.0 mm. Find the stress, the strain and the Young modulus of the steel. What would be the extension if the wire were replaced by one of the same steel, 3.0 m long and with twice the area, under the same load?

Solution: Stress = F/A = $200/(1.0 \times 10^{-6}) = 2.0 \times 10^{8}$ Pa. Strain = e/L = $(2.0 \times 10^{-3})/2.0 = 1.0 \times 10^{-3}$. E = stress/strain = $2.0 \times 10^{8}/1.0 \times 10^{-3} = 2.0 \times 10^{11}$ Pa. For the new wire, e = FL/(AE) = $200 \times 3.0/(2.0 \times 10^{-6} \times 2.0 \times 10^{11}) = 600/(4.0 \times 10^{5}) = 1.5 \times 10^{-3}$ m = 1.5 mm. This agrees with the argument that the extension is multiplied by 3.0/2.0 for the length and by 1/2 for the area: 2.0 mm × 1.5 × 0.5 = 1.5 mm.

6.2 Density and relative density

The **density** of a substance is its mass per unit volume: ρ = m/V. The SI unit is kg/m³, though g/cm³ is often more convenient. To convert, remember that 1 g/cm³ = 1,000 kg/m³. Density characterises a substance, whether the sample is large or small: a drop of water and a tanker of water have the same density.

The **relative density** (formerly called specific gravity) of a substance is the ratio of its density to the density of water: relative density = density of substance/density of water. It can also be written as the ratio of the mass of any volume of the substance to the mass of an equal volume of

water. Relative density is a pure number. Since the density of water is 1 g/cm³, the relative density of a substance equals its density in g/cm³; mercury has a relative density of 13.6.

WORKED EXAMPLE 6.3

A rectangular block of metal measures 20 cm by 10 cm by 5.0 cm and has a mass of 2.7 kg. Find its density in kg/m³ and its relative density. A density bottle has a mass of 25 g when empty, 75 g when full of water and 65 g when full of palm oil. Find the relative density and the density of the palm oil.

Solution: The volume is 20 × 10 × 5.0 = 1,000 cm³ = 1.0 × 10⁻³ m³. The density is 2.7/(1.0 × 10⁻³) = 2,700 kg/m³, and the relative density is 2,700/1,000 = 2.7 (the metal is probably aluminium). For the oil, the mass of water = 75 − 25 = 50 g and the mass of oil = 65 − 25 = 40 g, so the relative density is 40/50 = 0.80 and the density of the oil is 0.80 × 1,000 = 800 kg/m³.

6.3 Pressure and the pressure in a liquid

When a force acts at right angles to a surface, its effect depends on the area over which it is spread. The **pressure** is the normal force per unit area: P = F/A. The SI unit is the pascal (Pa), equal to one newton per square metre.

To derive the formula, imagine a vertical column of liquid of height h and cross-sectional area A. Its volume is Ah, its mass is ρAh and its weight is ρAhg. This weight presses on an area A at the base, so the pressure due to the liquid is

$$P = \rho g h$$

Note carefully that the pressure does not depend on the area of the base or on the total volume of liquid; a narrow tube of water 10 m high gives the same pressure at its bottom as a wide lake of the same depth. This surprising fact is sometimes called the hydrostatic paradox. The total (absolute) pressure at a depth h below the surface of a liquid open to the atmosphere is $P = P_{atm} + \rho g h$. The part ρgh is the **gauge pressure**.

WORKED EXAMPLE 6.4

A diver in a lake is 12 m below the surface. Calculate (a) the pressure due to the water, (b) the total pressure, and (c) the force on a flat window of area 0.020 m² on the diver's equipment, assuming it faces horizontally and the pressure inside is negligible.

Solution: (a) $P = \rho g h = 1{,}000 \times 10 \times 12 = 1.2 \times 10^5$ Pa. (b) Total pressure $= 1.0 \times 10^5 + 1.2 \times 10^5 = 2.2 \times 10^5$ Pa. (c) $F = PA = 2.2 \times 10^5 \times 0.020 = 4{,}400$ N. The pressure at 12 m is more than double the pressure at the surface.

6.4 Pascal's principle and the hydraulic machine

Liquids are almost incompressible, and they transmit pressure very well. This is expressed in **Pascal's principle**: pressure applied to an enclosed fluid is transmitted undiminished to every part of the fluid and to the walls of the container. The hydraulic press, the hydraulic jack, the brake system of a car and the lift of a mechanical workshop all use this principle.

Suppose a small piston of area a and a large piston of area A are joined by a pipe full of oil. A force f applied to the small piston produces a pressure f/a, which is transmitted to the large piston, where it produces a force F = (f/a) × A. Therefore

F/f = A/a

The force is multiplied in the ratio of the areas, so the **mechanical advantage** of an ideal hydraulic machine is A/a. Energy is not created, however. Since liquids are incompressible, the volume displaced at each piston is equal, so the small piston must move through a distance greater than the large piston in the same ratio, and the work done by each piston is the same. In a real machine, friction and leakage mean that the work obtained is less than the work put in, and the efficiency is below 100%.

WORKED EXAMPLE 6.5

In a hydraulic press the small piston has an area of 5.0 cm² and the large piston has an area of 400 cm². A force of 100 N is applied to the small piston. Find (a) the force on the large piston, (b) the distance the small piston must move to raise the large piston by 1.0 cm, assuming the machine is ideal.

Solution: (a) The ratio of areas is 400/5.0 = 80, so F = 80 × 100 = 8,000 N. (b) The volumes displaced are equal, so $a \times d_1 = A \times d_2$, and $d_1 = (A/a) \times d_2 = 80 \times 1.0 = 80$ cm. Check by work: input work = 100 N × 0.80 m = 80 J; output work = 8,000 N × 0.010 m = 80 J, which agree for an ideal machine.

6.5 Atmospheric pressure and its measurement

The Earth is surrounded by a layer of air several tens of kilometres thick, and the weight of this air exerts the **atmospheric pressure**. At sea level it averages about 1.01×10^5 Pa. Atmospheric pressure decreases with altitude, because there is less air above; at the top of a high mountain or on the Jos Plateau it is a little lower than at the coast in Lagos.

The **simple mercury barometer** is made by filling a glass tube about 1 m long, closed at one end, with mercury, and inverting it with the open end under the surface of mercury in a dish. The mercury falls until the

column is about 76 cm high above the level in the dish, and the space above the column is a vacuum (the Torricellian vacuum). The pressure at the level of the dish surface is atmospheric on the open surface and is balanced by the pressure of the column inside the tube, so $P_{atm} = \rho g h$. With $h = 0.76$ m, $P = 13{,}600 \times 10 \times 0.76 = 1.03 \times 10^5$ Pa (using $g = 10$ m/s²; with $g = 9.8$ m/s² the value is 1.013×10^5 Pa). Note that the height of the column does not change if the tube is tilted (the vertical height stays the same and the column becomes longer), or if a tube of a different width is used, but it does fall if a little air gets into the vacuum, and it changes with altitude.

Manometers and the siphon

An open-tube **manometer** is a U-tube partly filled with a liquid, one limb of which is connected to a gas supply while the other is open to the atmosphere. If the liquid level is higher in the open limb, the gas pressure exceeds atmospheric pressure by $\rho g h$, where h is the difference in levels; if lower, the gas pressure is smaller by $\rho g h$. Pressures are often quoted directly in centimetres of mercury; a gas pressure that is 20 cm of mercury above an atmosphere of 76 cm of mercury is simply 96 cmHg.

6.6 Archimedes' principle and flotation

When a body is partly or fully immersed in a fluid, the fluid pushes it upwards. This force is called **upthrust**. It arises because the pressure at the bottom of the body is greater than the pressure at the top, so the net

pressure force on the body is upward. The magnitude of the upthrust is given by the principle that Archimedes is said to have discovered while taking a bath.

> **Key point. Archimedes' principle.** When a body is wholly or partly immersed in a fluid, it experiences an upthrust equal to the weight of the fluid it displaces. Upthrust = $\rho_{\text{fluid}} \times V_{\text{immersed}} \times g$.

The apparent weight of a body in a fluid is its true weight minus the upthrust. Therefore, **loss in weight = upthrust = weight of fluid displaced**. This leads to a useful method of finding the relative density of a solid that sinks:

relative density of solid = weight in air/(weight in air − weight in water)

and of a liquid, using the same solid immersed in the liquid and in water:

relative density of liquid = (weight in air − weight in liquid)/(weight in air − weight in water).

WORKED EXAMPLE 6.6

A stone has a mass of 0.60 kg. It weighs 4.0 N when completely immersed in water. Find (a) the upthrust, (b) the volume of the stone, (c) its density and relative density.

Solution: The weight in air is mg = 0.60 × 10 = 6.0 N. (a) Upthrust = 6.0 − 4.0 = 2.0 N. (b) The upthrust equals $\rho_w V g$, so V = 2.0/(1,000 × 10) = 2.0×10^{-4} m^3 (200 cm^3). (c) Density = m/V = 0.60/(2.0×10^{-4}) = 3,000 kg/m^3. The relative density is 3,000/1,000 = 3.0, which agrees with 6.0/(6.0 − 4.0) = 3.0.

The law of flotation

A body floats when the upthrust on it can equal its weight before it is completely submerged. In equilibrium the upthrust equals the weight of the body, and since the upthrust equals the weight of fluid displaced, we obtain the **law of flotation**: a floating body displaces its own weight of the fluid in which it floats. For a uniform body floating in a liquid, with the body's density ρ_b and the liquid's density ρ_l, the fraction of the volume submerged is ρ_b/ρ_l. A body sinks if its average density is greater than that of the liquid, floats fully submerged (it is in neutral buoyancy) if the densities are equal, and floats with part of its volume above the surface if it is less dense.

WORKED EXAMPLE 6.7

A uniform block of wood of density 600 kg/m³ floats in water. What fraction of its volume is submerged? If the block is 0.50 m long, 0.20 m wide and 0.10 m high and it floats with the 0.50 m × 0.20 m face horizontal, find the depth to which it sinks and the largest mass that can be placed on its top before it is just fully submerged.

Solution: The fraction submerged is 600/1,000 = 0.60. The volume is 0.50 × 0.20 × 0.10 = 0.010 m³, so the mass of the block is 600 × 0.010 = 6.0 kg. The depth submerged is 0.60 × 0.10 m = 0.060 m = 6.0 cm. When the block is just fully submerged, the upthrust is 1,000 × 0.010 × 10 = 100 N, which supports a total weight of 100 N, i.e. a total mass of 10 kg. The extra mass is 10 − 6.0 = 4.0 kg.

6.7 Fluids in motion, viscosity and surface tension

In steady (streamline) flow of an incompressible fluid through a pipe of changing cross-section, the volume passing each point per second is the same, so $A_1v_1 = A_2v_2$: the fluid moves faster where the pipe is narrower. **Bernoulli's principle** adds that where the speed of a fluid is greater, the pressure is lower. This explains the lift on an aeroplane wing and the working of a paint spray. **Viscosity** is the internal friction of a fluid; a sphere falling through a viscous liquid reaches a constant *terminal*

velocity when weight equals upthrust plus viscous drag. **Surface tension** is the force per unit length in a liquid surface; it makes drops spherical, lets a needle float, and, together with the adhesion between water and glass, causes *capillary rise* in narrow tubes, the rise being greater in a narrower tube.

Exercise 6.1

1. A spring has a force constant of 250 N/m. What force is needed to extend it by 8.0 cm?
2. A spring extends by 5.0 cm under a load of 10 N. Assuming the elastic limit is not exceeded, what is the extension under a load of 16 N?
3. Calculate the energy stored in a spring of force constant 200 N/m when it is extended by 10 cm.
4. Two identical springs, each of force constant 300 N/m, are placed side by side (in parallel) and carry a total load of 12 N. Find the extension of the combination.
5. A wire of length 1.5 m and cross-sectional area 2.0×10^{-6} m² is made of a material whose Young modulus is 2.0×10^{11} Pa. Find its extension when it carries a load of 400 N.
6. A tank of base 2.0 m by 1.5 m is filled with water to a depth of 2.0 m. Find the pressure on the base due to the water and the force on the base.
7. In a hydraulic jack the diameter of the large piston is ten times that of the small piston. What force on the small piston is

needed to support a load of 5,000 N on the large piston (ideal machine)?

8. A piece of metal weighs 8.0 N in air, 5.0 N in water and 5.6 N in a liquid X. Find the relative density of the metal and of the liquid X.
9. A block of ice of density 900 kg/m³ floats in water. What percentage of its volume is above the water surface?
10. An open manometer containing mercury is joined to a gas supply. The mercury level in the open limb is 20 cm higher than in the limb joined to the gas supply. What is the gas pressure in pascals if atmospheric pressure is 76 cm of mercury? (Density of mercury 13,600 kg/m³.)

Answers to Exercise 6.1

1. 20 N. $F = ke = 250 \times 0.080 = 20$ N.
2. 8.0 cm. The extension is proportional to the load: $5.0 \times 16/10 = 8.0$ cm.
3. 1.0 J. $E = \tfrac{1}{2}ke^2 = \tfrac{1}{2} \times 200 \times (0.10)^2 = 100 \times 0.010 = 1.0$ J.
4. 2.0 cm. For springs in parallel $k = 300 + 300 = 600$ N/m, so $e = 12/600 = 0.020$ m.
5. 1.5 mm. $e = FL/(AE) = (400 \times 1.5)/(2.0 \times 10^{-6} \times 2.0 \times 10^{11}) = 600/(4.0 \times 10^{5}) = 1.5 \times 10^{-3}$ m.
6. 2.0×10^4 Pa and 6.0×10^4 N. $P = \rho gh = 1{,}000 \times 10 \times 2.0 = 2.0 \times 10^4$ Pa; area $= 3.0$ m²; $F = PA = 6.0 \times 10^4$ N.

7. 50 N. The ratio of areas is the square of the ratio of diameters, $10^2 = 100$, so f = 5,000/100 = 50 N.
8. Metal: 8.0/(8.0 − 5.0) = 2.67 (approximately). Liquid X: (8.0 − 5.6)/(8.0 − 5.0) = 2.4/3.0 = 0.80.
9. 10%. The fraction submerged is 900/1,000 = 0.90, so 10% is above the surface.
10. 1.3×10^5 Pa (approximately). The gas pressure is the atmosphere plus 20 cm of mercury, which is 96 cmHg, so $P = 13{,}600 \times 10 \times 0.96 = 1.31 \times 10^5$ Pa.

Chapter summary

- Hooke's law states that within the elastic limit the extension is proportional to the applied force, F = ke; the energy stored in a stretched spring is $\tfrac{1}{2}ke^2$. Springs in series have $k = k_1k_2/(k_1 + k_2)$; springs in parallel have $k = k_1 + k_2$.
- Stress = F/A, strain = e/L, and the Young modulus E = stress/strain = FL/(Ae) is a property of the material, measured in pascals.
- Density ρ = m/V in kg/m³; relative density = density of substance/density of water, with no unit.
- Pressure P = F/A; the pressure in a liquid at depth h is ρgh, acts in all directions, and is the same at all points at the same depth. Total pressure = atmospheric pressure + ρgh.
- Pascal's principle: pressure applied to an enclosed fluid is transmitted undiminished; a hydraulic machine has F/f = A/a,

and the work on each piston is the same in an ideal machine.

- Atmospheric pressure is measured by a mercury barometer: $P = \rho g h$, about 76 cm Hg or 1.01×10^5 Pa at sea level; it falls with altitude.
- Archimedes' principle: the upthrust equals the weight of fluid displaced. A floating body displaces its own weight of fluid; the submerged fraction is $\rho_{body}/\rho_{liquid}$.
- For steady flow of an incompressible fluid, $A_1 v_1 = A_2 v_2$; by Bernoulli's principle, pressure is lower where the speed is greater.
- Viscosity is internal fluid friction and gives rise to terminal velocity; surface tension is the force per unit length in a liquid surface and explains drops, meniscus shape and capillarity.

Chapter 7: Temperature, Heat, Gases and the Transfer of Thermal Energy

On a hot afternoon in Kano a metal roof is too hot to touch, a bucket of water left in the sun warms slowly, and a tin of palm oil left near a fire runs freely. Heat is part of daily life, and the physics of heat explains why a railway line has gaps, why a kettle takes minutes to boil, why a sealed can of aerosol must never be thrown into a fire, and why a flask keeps tea hot. In this chapter we distinguish temperature from heat, study thermal expansion, calculate the energy needed to warm or melt a substance, state the gas laws, and describe the three ways in which thermal energy travels.

Where numbers are needed, use: specific heat capacity of water 4,200 J/kg K; specific latent heat of fusion of ice 3.36×10^5 J/kg; specific latent heat of vaporisation of water 2.26×10^6 J/kg; absolute zero = −273 °C.

7.1 Temperature and its measurement

Heat is energy that flows from one body to another because of a difference in temperature; it is measured in joules. **Temperature** is the degree of hotness of a body, which determines the direction in which heat will flow: heat flows from the body at the higher temperature to the

one at the lower temperature, until they reach thermal equilibrium. A small cup of boiling water has a higher temperature than a lake, but the lake contains far more heat energy. In terms of the kinetic theory, temperature is a measure of the average kinetic energy of the particles of a substance, whereas the heat content depends also on how many particles there are.

A thermometer uses a physical property that varies regularly with temperature, such as the length of a mercury thread, the resistance of a wire or the e.m.f. of a thermocouple. Two **fixed points** are chosen: the **lower fixed point** is the temperature of pure melting ice (0 °C) and the **upper fixed point** is the temperature of steam above water boiling at standard pressure (100 °C). The interval between them is divided into 100 equal parts for the Celsius scale. If a mercury-in-glass thermometer has thread lengths l_0 at the ice point, l_{100} at the steam point and l at an unknown temperature, then

$$\theta = [(l - l_0)/(l_{100} - l_0)] \times 100\ ^\circ\text{C}.$$

The SI scale is the **Kelvin (absolute) scale**, on which the lowest possible temperature, absolute zero, is 0 K. A temperature interval of 1 K is equal to 1 °C, and $T(\text{K}) = \theta(^\circ\text{C}) + 273$. A **clinical thermometer** has a narrow constriction that prevents the mercury from returning to the bulb, so the maximum reading is retained; its range is only about 35 °C to 42 °C. Mercury is a good thermometric liquid because it is opaque, expands uniformly and regularly, has a low specific heat capacity, conducts heat well and does not wet glass; alcohol is used for very low temperatures because it freezes at about −115 °C.

WORKED EXAMPLE 7.1

The mercury thread of a thermometer is 20 mm long in melting ice and 220 mm long in steam. What is the temperature when the thread is 50 mm long? Express the answer in kelvin.

Solution: $\theta = [(50 - 20)/(220 - 20)] \times 100 = (30/200) \times 100 = 15$ °C. In kelvin, $T = 15 + 273 = 288$ K.

7.2 Thermal expansion

When a solid is heated, its particles vibrate with greater amplitude and the average separation between them increases, so the body expands. The **linear expansivity** α of a material is the fractional increase in length per unit rise in temperature: $\alpha = \Delta L/(L_0 \Delta\theta)$, where L_0 is the original length and $\Delta\theta$ is the temperature rise; its unit is K^{-1}. Hence the increase in length is

$$\Delta L = \alpha L_0 \Delta\theta.$$

For a given solid the **area expansivity** is approximately 2α and the **cubic (volume) expansivity** is approximately 3α. Typical values of α are $1.1 \times 10^{-5}\ K^{-1}$ for steel, $1.2 \times 10^{-5}\ K^{-1}$ for concrete, $1.7 \times 10^{-5}\ K^{-1}$ for copper and $2.3 \times 10^{-5}\ K^{-1}$ for aluminium. Liquids expand more than solids for the same temperature rise, and gases expand much more than liquids. When a liquid is heated in a container, the container expands too, so the observed (apparent) expansion of the liquid is less than the real expansion.

The expansion of solids is a practical matter for engineers. Railway lines are laid with small gaps, and long steel bridges are fitted with rollers or expansion joints, so that they will not buckle in the dry-season heat. Concrete roads are laid in slabs separated by tar-filled gaps. A glass bottle may crack when hot water is poured in because the inside expands before the outside. A **bimetallic strip**, made of two metals such as brass and iron riveted together, bends on heating because one metal expands more than the other; it is used in thermostats, in electric irons, in fire alarms and in the flasher units of vehicle indicators.

Water behaves unusually. If water at 0 °C is heated, it first contracts until it reaches 4 °C, at which temperature its density is a maximum (1,000 kg/m³), and then it expands as usual. This is the **anomalous expansion of water**. In cold countries it means that the water at the bottom of a lake stays at about 4 °C while ice forms at the top, and so fish survive in winter.

WORKED EXAMPLE 7.2

A steel rail 20 m long at 25 °C is heated by the sun to 65 °C. If the linear expansivity of steel is $1.2 \times 10^{-5}\ \text{K}^{-1}$, calculate the increase in length.

Solution: $\Delta\theta = 65 - 25 = 40$ K. $\Delta L = \alpha L_0 \Delta\theta = 1.2 \times 10^{-5} \times 20 \times 40 = 9.6 \times 10^{-3}$ m = 9.6 mm. The gap left between rails must therefore be at least about 10 mm.

7.3 Heat capacity and specific heat capacity

The heat capacity of a body is the quantity of heat required to raise its temperature by one kelvin; its unit is J/K. Since it depends on the mass, the more useful quantity is the **specific heat capacity** c, the quantity of heat required to raise the temperature of 1 kg of a substance by 1 K, measured in J/(kg K). The heat gained or lost by a body whose temperature changes by Δθ, with no change of state, is

Q = mcΔθ.

Water has an unusually large specific heat capacity, 4,200 J/(kg K), which is why it is used for cooling engines and for hot-water bottles, and why coastal places such as Lagos have a milder temperature range than inland places. Copper has c ≈ 400 J/(kg K), aluminium 900 J/(kg K) and iron about 450 J/(kg K), so a metal pot heats far faster than the water in it.

When a hot body is placed in contact with a cold body in a well-insulated container, the heat lost by the hot body equals the heat gained by the cold body. This **principle of heat exchange** is the basis of the *method of mixtures* for finding specific heat capacities, in which a heated solid is dropped into water in a lagged calorimeter. For an electrical method, a heater of power P supplies energy Pt, so Pt = mcΔθ if there are no heat losses.

WORKED EXAMPLE 7.3

An electric heater of power 2.0 kW is used to heat 2.0 kg of water from 20 °C to 70 °C. Calculate the energy required and the time taken, assuming no heat is lost.

Solution: $Q = mc\Delta\theta = 2.0 \times 4{,}200 \times 50 = 4.2 \times 10^5$ J. Since $Q = Pt$, $t = Q/P = 4.2 \times 10^5 / 2{,}000 = 210$ s, which is 3.5 minutes.

WORKED EXAMPLE 7.4

A copper block of mass 0.40 kg at 100 °C is dropped into 0.50 kg of water at 20 °C. If the specific heat capacity of copper is 400 J/(kg K), find the final temperature, ignoring heat lost to the surroundings and to the container.

Solution: Let the final temperature be θ. Heat lost by copper $= 0.40 \times 400 \times (100 - \theta) = 160(100 - \theta)$. Heat gained by water $= 0.50 \times 4{,}200 \times (\theta - 20) = 2{,}100(\theta - 20)$. Equating: $16{,}000 - 160\theta = 2{,}100\theta - 42{,}000$, so $58{,}000 = 2{,}260\theta$ and $\theta = 25.7$ °C (to one decimal place).

7.4 Change of state and latent heat

When a solid is heated, its temperature rises until the melting point is reached; during melting the temperature stays constant even though heat is still being supplied. The energy supplied is used to overcome the forces holding the particles in fixed positions, not to increase their

kinetic energy. This energy is called **latent heat**. The **specific latent heat of fusion** is the quantity of heat required to change 1 kg of a solid to a liquid without a change in temperature; the **specific latent heat of vaporisation** is the heat required to change 1 kg of a liquid to vapour at its boiling point. In each case,

$Q = mL$.

The latent heat of vaporisation is much larger than that of fusion because, in turning from liquid to vapour, the particles must be separated completely and must also push back the atmosphere. This explains why steam at 100 °C causes a more severe burn than boiling water at 100 °C: each kilogram of steam releases 2.26×10^6 J on condensing, before it begins to cool.

Evaporation is the change of liquid to vapour at any temperature, taking place only at the surface; it is faster when the temperature is higher, the surface area larger, the air drier or moving, and it causes cooling because the more energetic molecules escape. This is how sweat cools the body, and how an earthen pot (a Nigerian water pot) keeps its contents cool. **Boiling** happens at one definite temperature, the boiling point, at which bubbles of vapour form throughout the liquid. The boiling point rises with increase of pressure (as in a pressure cooker, which cooks food faster at about 120 °C) and falls with decrease of pressure (water boils below 100 °C on a mountain). The melting point of ice is lowered slightly by an increase in pressure, and impurities such as salt lower the freezing point of water and raise its boiling point. A

refrigerator works by a volatile liquid that absorbs latent heat as it evaporates inside the cabinet and releases it as it condenses in the coils at the back.

The air always contains some water vapour. The **dew point** is the temperature at which the air becomes saturated with water vapour and dew begins to form. The **relative humidity** is the ratio of the mass of water vapour present in a given volume of air to the mass needed to saturate it at the same temperature, expressed as a percentage. It is high on a rainy day in Port Harcourt and low during the harmattan.

WORKED EXAMPLE 7.5

Calculate the heat required to change 0.20 kg of ice at 0 °C to water at 30 °C. Then find the final temperature when 0.050 kg of steam at 100 °C is passed into 1.0 kg of water at 20 °C and condensed completely, ignoring heat losses.

Solution: First part: melting needs $0.20 \times 3.36 \times 10^5 = 67{,}200$ J; warming the water needs $0.20 \times 4{,}200 \times 30 = 25{,}200$ J; the total is 92,400 J, or 9.24×10^4 J. Second part: let the final temperature be θ. Heat given out by the steam $= 0.050 \times 2.26 \times 10^6 + 0.050 \times 4{,}200 \times (100 - \theta) = 113{,}000 + 210(100 - \theta)$. Heat gained by cold water $= 1.0 \times 4{,}200 \times (\theta - 20)$. Equating: $113{,}000 + 21{,}000 - 210\theta = 4{,}200\theta - 84{,}000$, so $218{,}000 = 4{,}410\theta$ and $\theta = 49.4$ °C.

7.5 The gas laws

The behaviour of a fixed mass of gas is described by its pressure P, volume V and absolute temperature T. Experiments by Boyle, Charles and others gave three laws, which are valid for real gases at moderate pressure and temperatures well above their liquefaction points.

> **Key point. The gas laws.** Boyle's law: at constant temperature, PV = constant ($P_1V_1 = P_2V_2$). Charles's law: at constant pressure, V/T = constant. Pressure law: at constant volume, P/T = constant. Temperature T must always be in kelvin.

Boyle's law says that if a gas is compressed to half its volume without a change in temperature, its pressure doubles. Charles's law implies that the volume of a gas at constant pressure would shrink to zero at −273 °C, which defines absolute zero; in practice all gases liquefy before reaching it. The three laws combine into the **general gas equation** for a fixed mass of gas:

$$P_1V_1/T_1 = P_2V_2/T_2,$$

and for any quantity of an **ideal gas** the equation of state is PV = nRT, where n is the number of moles and R = 8.31 J/(mol K). **Standard temperature and pressure (s.t.p.)** means 273 K (0 °C) and 1.01×10^5 Pa (760 mmHg); at s.t.p. one mole of any gas occupies about 22.4 litres.

The **kinetic theory of gases** explains the laws. A gas consists of a very large number of tiny molecules in continuous random motion, colliding elastically with each other and with the walls. The pressure of the gas is the result of the countless collisions of the molecules with the walls; when the volume is reduced, collisions become more frequent and the pressure rises (Boyle's law). When the temperature rises, the molecules move faster, so collisions are more frequent and more forceful (the pressure law). The same theory explains *Brownian motion*, the random zigzag motion of smoke particles in air, which is caused by uneven bombardment by the invisible molecules, and *diffusion*, the spreading of a gas through another.

WORKED EXAMPLE 7.6

A fixed mass of gas occupies 2.0 litres at 27 °C and a pressure of 1.0 atmosphere. Calculate its pressure when its volume is 4.0 litres and its temperature is 127 °C. A sealed rigid cylinder holds air at 27 °C and a pressure of 2.0×10^5 Pa; what is the pressure when it is heated to 87 °C?

Solution: First part: $T_1 = 27 + 273 = 300$ K and $T_2 = 127 + 273 = 400$ K. From $P_1V_1/T_1 = P_2V_2/T_2$, $P_2 = P_1V_1T_2/(T_1V_2) = (1.0 \times 2.0 \times 400)/(300 \times 4.0) = 800/1{,}200 = 0.67$ atmosphere. Second part: the volume is constant, so $P_2 = P_1 \times T_2/T_1 = 2.0 \times 10^5 \times 360/300 = 2.4 \times 10^5$ Pa.

7.6 Transfer of thermal energy

Heat is transferred from one place to another in three ways: conduction, convection and radiation.

Conduction is the transfer of heat through a material from particle to particle without any movement of the material as a whole. In metals, free electrons that move through the lattice carry energy rapidly, so metals are good conductors; non-metals such as wood, plastic, glass, air and water are poor conductors (insulators). Substances such as wool, polystyrene foam and trapped air are used for insulation. The rate of conduction through a uniform slab or rod, in the steady state, is directly proportional to the area A and the temperature difference $\Delta\theta$, and inversely proportional to the thickness L:

$Q/t = kA\Delta\theta/L$,

where k is the **thermal conductivity** of the material in W/(m K). Metals have values of k of the order of 100 to 400 W/(m K), while for glass it is about 0.8 and for still air only 0.026.

Convection is the transfer of heat in a fluid by the movement of the fluid itself. When part of a fluid is heated, it expands, becomes less dense and rises, while cooler, denser fluid sinks to take its place, setting up a convection current. Convection explains the heating of water in a pot, sea breezes along the Lagos coast (land warms faster than the sea during the day, so air rises over land and cooler air blows in from the

sea), the draught up a chimney, and the ventilation of a room using an opening near the ceiling. A hot-water system installed with the tank above the boiler works because of convection.

Radiation is the transfer of heat by electromagnetic waves (mostly infrared) and needs no material medium; this is how the energy of the Sun reaches the Earth through space. Dull black surfaces are good absorbers and good emitters of radiation, while shiny, light-coloured surfaces are poor absorbers and poor emitters but good reflectors. This is why houses in hot regions are painted white, and why the car of a person who parks in the sun is hotter inside if the seats are dark. The **vacuum flask** reduces all three methods: the double-walled glass container with a vacuum between the walls prevents conduction and convection, the silvered walls reduce radiation, and the cork or plastic stopper reduces conduction and convection at the top.

WORKED EXAMPLE 7.7

A glass window of area 2.0 m² and thickness 5.0 mm has a temperature of 25 °C on the outside and 15 °C on the inside. Find the rate of heat flow through the window, if the thermal conductivity of the glass is 0.80 W/(m K).

Solution: $Q/t = kA\Delta\theta/L = 0.80 \times 2.0 \times 10/(5.0 \times 10^{-3}) = 16/(5.0 \times 10^{-3}) = 3{,}200$ W. The window therefore transmits energy at a rate of 3.2 kW.

Exercise 7.1

1. A steel bar is 5.0 m long at 20 °C. Find its increase in length when its temperature is raised to 120 °C ($\alpha = 1.1 \times 10^{-5}\ \text{K}^{-1}$).
2. How much heat is needed to raise the temperature of 3.0 kg of water from 25 °C to 65 °C?
3. An immersion heater rated 3.0 kW heats 5.0 kg of water from 20 °C to 80 °C. How long does it take, assuming no heat loss?
4. What mass of ice at 0 °C can be melted by 1.68×10^{5} J of heat?
5. A metal of mass 0.30 kg is heated to 100 °C and dropped into 0.20 kg of water at 20 °C. The final temperature is 30 °C. Ignoring heat losses, find the specific heat capacity of the metal.
6. A gas at a pressure of 3.0×10^{5} Pa and volume 0.40 m^3 expands at constant temperature to 1.2 m^3. What is its final pressure?
7. A gas of volume 300 cm^3 at 27 °C is heated at constant pressure to 127 °C. Find the new volume.
8. The temperature of a gas in a rigid container is raised from 27 °C to 87 °C. By what factor does its pressure change?
9. A wall of area 10 m^2 and thickness 0.20 m has a thermal conductivity of 0.50 W/(m K). The temperature difference across it is 20 K. Find the rate of heat flow.

10. The thread of a mercury thermometer is 40 mm long at 0 °C and 240 mm long at 100 °C. What is the temperature when the thread is 140 mm long? Give your answer in kelvin.

Answers to Exercise 7.1

1. 5.5 mm. $\Delta L = \alpha L_0 \Delta\theta = 1.1 \times 10^{-5} \times 5.0 \times 100 = 5.5 \times 10^{-3}$ m.
2. 5.04×10^5 J. $Q = mc\Delta\theta = 3.0 \times 4{,}200 \times 40 = 504{,}000$ J.
3. 420 s (7 minutes). $Q = 5.0 \times 4{,}200 \times 60 = 1.26 \times 10^6$ J; $t = Q/P = 1.26 \times 10^6 / 3{,}000 = 420$ s.
4. 0.50 kg. $m = Q/L = 1.68 \times 10^5 / 3.36 \times 10^5 = 0.50$ kg.
5. 400 J/(kg K). Heat lost = heat gained: $0.30 \times c \times 70 = 0.20 \times 4{,}200 \times 10$, so $21c = 8{,}400$ and $c = 400$.
6. 1.0×10^5 Pa. Boyle's law: $P_2 = P_1V_1/V_2 = 3.0 \times 10^5 \times 0.40/1.2 = 1.0 \times 10^5$ Pa.
7. 400 cm³. $V_2 = V_1T_2/T_1 = 300 \times 400/300 = 400$ cm³ (temperatures 300 K and 400 K).
8. 1.2 (the pressure increases by 20%). $P_2/P_1 = T_2/T_1 = 360/300 = 1.2$.
9. 500 W. $Q/t = kA\Delta\theta/L = 0.50 \times 10 \times 20/0.20 = 100/0.20 = 500$ W.
10. 323 K. $\theta = [(140 - 40)/(240 - 40)] \times 100 = 50$ °C, and $T = 50 + 273 = 323$ K.

Chapter summary

- Heat is energy in transit because of a temperature difference; temperature measures hotness, the average kinetic energy of the particles. $T(K) = \theta(°C) + 273$, and a thermometer is calibrated with the ice point and steam point.
- Solids expand when heated: $\Delta L = \alpha L_0 \Delta\theta$, with area expansivity about 2α and volume expansivity about 3α. Water has its maximum density at 4 °C.
- $Q = mc\Delta\theta$ for a change of temperature, and $Q = mL$ for a change of state. The heat lost by hot bodies equals the heat gained by cold ones in a lagged system.
- Evaporation occurs at any temperature at the surface and causes cooling; boiling occurs at a fixed temperature that rises with pressure.
- Boyle's law, Charles's law and the pressure law combine in $P_1V_1/T_1 = P_2V_2/T_2$ (T in kelvin); for an ideal gas, $PV = nRT$.
- Gas pressure arises from molecular collisions with the walls; Brownian motion is evidence of the motion of molecules.
- Conduction: $Q/t = kA\Delta\theta/L$; convection is the movement of heated fluid; radiation needs no medium, and dull black surfaces absorb and emit best.

Chapter 8: Waves and Sound

Almost everything we know about the world around us reaches us by means of waves. The voice of a friend, the music from a radio in a Lagos danfo bus, the light of the morning sun, the signal that carries a phone call from Kano to Port Harcourt, and the tremor of an earthquake far beneath the ground are all examples of energy travelling from one place to another in the form of a wave. In this chapter we study the general properties shared by every kind of wave, and then we apply those ideas to sound, which is the wave phenomenon that most directly shapes our daily experience. The same mathematical language will serve again in the next chapter, where we treat light.

The central idea is simple but important: a wave transfers *energy* from one point to another without any bulk transfer of matter. A cork floating on the River Niger bobs up and down as ripples pass, but it does not travel along with the ripples. Once you have grasped this, the rest of the chapter follows naturally.

8.1 The nature of waves

A **wave** is a disturbance that transmits energy through a medium or through space, as a result of repeated vibrations of the particles of the medium or of changing electric and magnetic fields. When the source of the disturbance vibrates regularly, a continuous train of waves, called a **progressive wave**, is produced.

Waves are classified in two ways. The first classification depends on the need for a material medium. **Mechanical waves**, such as sound, water waves and waves on a stretched string, need a material medium because they are passed on by the interaction between neighbouring particles. **Electromagnetic waves**, such as radio waves, light and X-rays, consist of oscillating electric and magnetic fields and can travel through a vacuum; this is why sunlight reaches the Earth across empty space.

The second classification depends on the direction of vibration of the particles relative to the direction in which the wave travels.

- In a **transverse wave** the particles vibrate at right angles to the direction of travel of the wave. Examples are waves on a rope, ripples on water, and all electromagnetic waves. A transverse wave consists of alternate *crests* (the highest points) and *troughs* (the lowest points). Transverse waves can be *polarised*, that is, confined to vibrate in a single plane.
- In a **longitudinal wave** the particles vibrate parallel to the direction of travel of the wave. Sound in air is the standard example. A longitudinal wave consists of alternate regions of high pressure, called *compressions*, and regions of low pressure, called *rarefactions*. Longitudinal waves cannot be polarised.

Key point. Sound is a longitudinal mechanical wave and therefore cannot travel through a vacuum. Light is a transverse electromagnetic wave and can. Polarisation is the test that distinguishes transverse waves from longitudinal ones.

8.2 Quantities that describe a wave

Several quantities are used to describe a wave quantitatively.

- The **amplitude** (a) is the maximum displacement of a vibrating particle from its mean (equilibrium) position. It is measured in metres and is a measure of the energy carried by the wave; the energy is proportional to the square of the amplitude.
- The **wavelength** (λ) is the distance between two successive points that are in the same state of vibration, for example crest to crest or compression to compression. It is measured in metres.
- The **period** (T) is the time taken for one complete oscillation of a particle, measured in seconds.
- The **frequency** (f) is the number of complete oscillations (or waves) passing a point in one second. It is measured in hertz (Hz), and $f = 1/T$.
- The **wave speed** (v) is the distance travelled by the wave in one second.

In one period T the wave advances by exactly one wavelength, so $v = \lambda/T$. Since $1/T = f$, we obtain the fundamental wave equation

$v = f\lambda$.

The equation holds for every kind of wave. An important consequence is that when a wave passes from one medium into another, its frequency stays the same (the frequency is fixed by the source), while its speed and wavelength both change in proportion.

Two particles are said to be **in phase** if they are in the same state of vibration at the same time, so that the separation between them is a whole number of wavelengths. They are in **antiphase** if one is at a crest while the other is at a trough, which corresponds to a separation of an odd number of half-wavelengths. The **phase difference** between two points separated by a distance x along the wave is $(x/\lambda) \times 360°$, or $(2\pi x/\lambda)$ in radians.

WORKED EXAMPLE 8.1

A tuning fork of frequency 250 Hz produces sound waves in air of wavelength 1.32 m. Calculate the speed of sound in air and the period of the vibration.

Solution: $v = f\lambda = 250 \times 1.32 = 330$ m/s. The period is $T = 1/f = 1/250 = 0.004$ s $= 4.0$ ms.

WORKED EXAMPLE 8.2

Two points on a progressive wave are 0.15 m apart along the direction of travel and the wavelength is 0.60 m. Find the phase difference between them.

Solution: Phase difference = $(x/\lambda) \times 360° = (0.15/0.60) \times 360° = 0.25 \times 360° = 90°$, which is $\pi/2$ radians. The two points are therefore a quarter of a cycle out of step.

8.3 Properties of waves

Waves in every part of the spectrum show the same characteristic behaviours: reflection, refraction, diffraction and interference. These behaviours are demonstrated most conveniently with water waves in a ripple tank.

Reflection

When a wave strikes a barrier it bounces back into the original medium. The laws are the same as for light: the angle of incidence equals the angle of reflection, and the incident ray, reflected ray and normal lie in one plane. The wave speed, frequency and wavelength are unchanged by reflection. Echoes are the best-known example in sound.

Refraction

Refraction is the change in the direction of travel of a wave when it passes from one medium into another in which its speed is different (or from one depth of water into another). The frequency is unchanged, so the wavelength alters in proportion to the speed. In a ripple tank, waves travelling from deep water into shallow water slow down and their wavelength decreases; if they arrive at an angle, they bend towards the normal.

WORKED EXAMPLE 8.3

Ripples of frequency 10 Hz travel across the deep part of a ripple tank with wavelength 3.0 cm. They then enter a shallow region where the wavelength is 2.0 cm. Find the speed of the ripples in each region.

Solution: The frequency is unchanged, 10 Hz. In deep water $v_1 = f\lambda_1 = 10 \times 0.030 = 0.30$ m/s. In shallow water $v_2 = f\lambda_2 = 10 \times 0.020 = 0.20$ m/s. The ripples slow down by a factor of 2/3, in the same ratio as the wavelengths.

Diffraction

Diffraction is the spreading of a wave into the region behind an obstacle or after passing through a gap. The effect is most pronounced when the size of the gap or obstacle is comparable with the wavelength. For this reason you can hear a person talking around the corner of a building (sound wavelengths are of the order of a metre) but you cannot see them

(light wavelengths are about half a micrometre). When the gap is much wider than the wavelength, the wave passes through almost undisturbed, with only slight bending at the edges.

Superposition and interference

The **principle of superposition** states that when two or more waves meet at a point, the resultant displacement is the algebraic sum of the displacements that each wave would produce on its own. **Interference** is the effect produced by the superposition of two waves of the same type. A steady, observable interference pattern requires **coherent** sources: sources that have the same frequency and a constant phase difference.

- **Constructive interference** occurs where the waves arrive in phase, so the resultant amplitude is the sum of the individual amplitudes. The condition is that the path difference is a whole number of wavelengths: path difference = $n\lambda$, where $n = 0, 1, 2, \ldots$
- **Destructive interference** occurs where the waves arrive in antiphase, so the resultant amplitude is the difference of the amplitudes (zero if they are equal). The condition is that the path difference is an odd number of half-wavelengths: path difference = $(n + \tfrac{1}{2})\lambda$.

WORKED EXAMPLE 8.4

Two loudspeakers A and B, connected to the same signal generator, emit sound of wavelength 0.40 m. A microphone is placed 3.0 m from A and 3.6 m from B. Will it detect a loud sound or a quiet one?

Solution: The path difference is $3.6 - 3.0 = 0.60$ m. In wavelengths this is $0.60/0.40 = 1.5$, an odd number of half-wavelengths ($3 \times \lambda/2$). The waves arrive in antiphase and interfere destructively, so the microphone detects a minimum, a quiet sound.

8.4 Stationary waves

When a progressive wave is reflected at a boundary and returns along its own path, it superposes with the oncoming wave. If the two have the same frequency and amplitude, the result is a **stationary (standing) wave**. Unlike a progressive wave, a stationary wave does not transfer energy along the medium. It has fixed points of zero amplitude, called **nodes**, and points of maximum amplitude, called **antinodes**. The distance between two adjacent nodes, or two adjacent antinodes, is $\lambda/2$. All particles between two adjacent nodes vibrate in phase with each other, but in antiphase with the particles in the next loop.

Feature	Progressive wave	Stationary wave
Energy	Transmitted along the wave	Not transmitted; stored in the loops
Amplitude	The same for all particles	Varies from zero at nodes to maximum at antinodes
Phase	Changes continuously along the wave	Same within a loop; opposite in adjacent loops
Wave profile	Moves forward	Does not move

Vibrating strings

A string of length L fixed at both ends, as in a guitar, a talking-drum cord or a sonometer wire, must have a node at each end. The simplest pattern, the **fundamental** or first harmonic, has one loop, so $L = \lambda/2$ and $\lambda = 2L$. The frequency is $f_1 = v/(2L)$. The other possible patterns, called harmonics, have frequencies that are whole-number multiples of the fundamental: $f_2 = 2f_1$, $f_3 = 3f_1$, and so on. The speed of a transverse wave on a string is given by

$$v = \sqrt{(T/\mu)},$$

where T is the tension in the string (in newtons) and μ is its mass per unit length (in kg/m). Hence the fundamental frequency is $f_1 = (1/2L)\sqrt{(T/\mu)}$. It follows that the pitch of a string is raised by shortening it, by increasing the tension, or by using a lighter string.

WORKED EXAMPLE 8.5

A sonometer wire of length 0.50 m and mass per unit length 0.010 kg/m is stretched with a tension of 64 N. Calculate the speed of waves on the wire and the frequency of its fundamental and second harmonic.

Solution: $v = \sqrt{(T/\mu)} = \sqrt{(64/0.010)} = \sqrt{6{,}400} = 80$ m/s. The fundamental frequency is $f_1 = v/(2L) = 80/(2 \times 0.50) = 80$ Hz. The second harmonic is $f_2 = 2f_1 = 160$ Hz.

8.5 Sound waves and their speed

Sound is produced by a vibrating body and travels as a longitudinal wave of compressions and rarefactions. It needs a material medium: a bell ringing inside a jar from which air is gradually pumped out becomes fainter and fainter until it can scarcely be heard. Sound travels fastest in solids, more slowly in liquids and slowest in gases, because the particles of a solid are closer together and the restoring forces between them are larger. Approximate values are 340 m/s in air at room temperature, about 1,500 m/s in sea water and about 5,000 m/s in steel.

In air the speed of sound is affected by several factors:

- **Temperature.** The speed is proportional to the square root of the absolute (kelvin) temperature: $v \propto \sqrt{T}$. It is about 331 m/s at 0 °C and rises by about 0.6 m/s for every degree Celsius.

- **Density.** Sound travels faster in a less dense gas at the same temperature, so it is faster in hydrogen than in air.
- **Humidity.** Moist air is slightly less dense than dry air, so sound travels slightly faster in humid conditions.
- **Pressure.** A change in the pressure of a gas at constant temperature has no effect on the speed of sound, because density changes in the same proportion.
- **Wind.** Wind carries the sound along with it, so the speed relative to the ground increases or decreases by the wind speed.

WORKED EXAMPLE 8.6

The speed of sound in air at 0 °C (273 K) is 331 m/s. Calculate its speed at 27 °C.

Solution: Since $v \propto \sqrt{T}$, $v_2/v_1 = \sqrt{(T_2/T_1)}$. With $T_2 = 27 + 273 = 300$ K, $v_2 = 331 \times \sqrt{(300/273)} = 331 \times \sqrt{1.0989} = 331 \times 1.0483 = 347$ m/s (to three significant figures).

Echoes, reverberation and sonar

An **echo** is the sound heard after reflection from a distant, hard surface such as a cliff, a wall or the side of a building. For the ear to distinguish an echo from the original sound, the delay must be at least about 0.1 s, which in air means a reflecting surface at least 17 m away. If the surface is closer, the reflected sound merges with the original and prolongs it; this is **reverberation**, and it is controlled in concert halls and studios by

using soft, absorbent materials. The distance to the reflecting surface is d = vt/2, where t is the time between the sound being sent and the echo returning, because the sound travels to the surface and back.

Ships use this principle in **sonar**: a pulse of ultrasonic sound (above 20,000 Hz) is sent down, and the time for the echo from the sea bed or a shoal of fish to return gives the depth. Hospitals use ultrasound for scanning unborn babies, and it is also used for cleaning delicate instruments and for detecting flaws inside metal castings.

WORKED EXAMPLE 8.7

A boy standing in front of a high wall in Ibadan claps his hands and hears the echo 0.50 s later. If the speed of sound is 340 m/s, how far is he from the wall? A ship's sonar, in turn, receives the echo from the sea bed 0.80 s after sending a pulse. If sound travels at 1,500 m/s in sea water, how deep is the water?

Solution: For the wall, d = vt/2 = 340 × 0.50/2 = 85 m. For the sea bed, depth = vt/2 = 1,500 × 0.80/2 = 600 m.

8.6 Characteristics of musical sound

Musical notes are distinguished from noise by regular, periodic vibrations. Three characteristics identify a note.

- **Pitch** is the sensation by which we judge a note to be high or low. It depends on the frequency: the higher the frequency, the higher the

pitch. A healthy young person hears frequencies from about 20 Hz to 20,000 Hz. Sounds below 20 Hz are called infrasound, and sounds above 20 kHz ultrasound.

- **Loudness** is the sensation of the strength of a sound. It depends on the amplitude of the wave (and so on its intensity) and also on the sensitivity of the ear.
- **Quality** (or timbre) is the characteristic that enables us to tell the difference between two notes of the same pitch and loudness played on different instruments, for example a flute and a trumpet. It depends on the number and relative strengths of the overtones that accompany the fundamental.

The **intensity** of a sound is the energy passing per second through unit area at right angles to the direction of travel; its unit is W/m^2. Because the ear responds to an enormous range of intensities, loudness is measured on a logarithmic scale. The **intensity level** in decibels (dB) is $\beta = 10 \log_{10}(I/I_0)$, where $I_0 = 1.0 \times 10^{-12}\ W/m^2$ is the threshold of hearing. Each increase of 10 dB corresponds to a tenfold increase in intensity. Normal conversation is about 60 dB and a jet engine nearby is about 120 dB, which is painful and can damage hearing.

WORKED EXAMPLE 8.8

The intensity of sound from a generator at a certain point is 1.0×10^{-6} W/m². What is the intensity level in decibels?

Solution: $\beta = 10 \log_{10}(I/I_0) = 10 \log_{10}(10^{-6}/10^{-12}) = 10 \log_{10}(10^{6}) = 10 \times 6 = 60$ dB.

8.7 Resonance and vibrating air columns

Resonance occurs when a system is made to vibrate at its own natural frequency by a periodic force of the same frequency, producing a large-amplitude vibration. Pushing a child on a swing at the right moments is a mechanical example. In sound, a vibrating tuning fork held over a tube of air of suitable length makes the air column resonate and the sound becomes markedly louder.

The air in a pipe can vibrate in stationary-wave patterns. At a closed end there is a node (the air cannot move); at an open end there is an antinode.

- **Closed pipe** (one end closed). The fundamental has a node at the closed end and an antinode at the open end, so $L = \lambda/4$ and $f_1 = v/(4L)$. Only odd harmonics are present: f_1, $3f_1$, $5f_1$, …
- **Open pipe** (both ends open). The fundamental has an antinode at each end, so $L = \lambda/2$ and $f_1 = v/(2L)$. All harmonics are present: f_1, $2f_1$, $3f_1$, …

Because the antinode at an open end lies slightly beyond the end of the pipe, the effective length is increased by an **end correction** e, which is about 0.6 times the radius of the tube. In a resonance-tube experiment, a tuning fork of known frequency f is held over a tube partly filled with water, and the water level is lowered until the first resonance is heard at air-column length l_1, and then the second at l_2. Then $l_1 + e = \lambda/4$ and $l_2 + e = 3\lambda/4$. Subtracting eliminates the end correction: $l_2 - l_1 = \lambda/2$, so $\lambda = 2(l_2 - l_1)$ and the speed of sound is $v = f\lambda$.

WORKED EXAMPLE 8.9

A closed pipe is 0.25 m long. Taking the speed of sound as 340 m/s, find its fundamental frequency and the frequency of the next harmonic that can be produced. Then find the fundamental frequency of an open pipe of length 0.50 m.

Solution: For the closed pipe, $f_1 = v/(4L) = 340/(4 \times 0.25) = 340/1.0 = 340$ Hz. The next harmonic is the third: $3 \times 340 = 1{,}020$ Hz. For the open pipe, $f_1 = v/(2L) = 340/(2 \times 0.50) = 340$ Hz, and its next harmonic is 680 Hz.

WORKED EXAMPLE 8.10

In a resonance-tube experiment with a tuning fork of frequency 440 Hz, the first resonance is obtained at an air-column length of 0.185 m and the second at 0.570 m. Calculate the speed of sound and the end correction.

Solution: $\lambda = 2(l_2 - l_1) = 2(0.570 - 0.185) = 2 \times 0.385 = 0.770$ m. The speed is $v = f\lambda = 440 \times 0.770 = 338.8$ m/s, or about 339 m/s. For the end correction, $l_1 + e = \lambda/4 = 0.1925$ m, so $e = 0.1925 - 0.185 = 0.0075$ m = 0.75 cm.

8.8 Beats and the Doppler effect

When two notes of nearly equal frequency are sounded together, the listener hears a note of steadily rising and falling loudness. Each loud pulse is called a **beat**. The waves alternately arrive in phase and in antiphase, and the number of beats heard per second equals the difference between the two frequencies: beat frequency = $|f_1 - f_2|$. Musicians use beats to tune instruments: when the beats disappear, the two notes have the same frequency.

The **Doppler effect** is the apparent change in the frequency of a wave when there is relative motion between the source and the observer. The pitch of an ambulance siren is higher as it approaches you and drops as it moves away. For a source of frequency f moving at speed u_s towards a stationary observer, with the speed of sound v, the observed frequency is

$f' = fv/(v - u_s)$ for an approaching source, and $f' = fv/(v + u_s)$ for a receding source.

The effect is also used in radar speed guns and in medical devices that measure the speed of blood flow. Astronomers apply it to light, whose shift towards the red end of the spectrum reveals that distant galaxies are receding.

WORKED EXAMPLE 8.11

An ambulance siren emits a note of frequency 500 Hz while travelling at 20 m/s along a road. Calculate the frequency heard by a stationary pedestrian when the ambulance approaches and when it recedes. (Speed of sound = 340 m/s.) Also state the number of beats per second heard when notes of 256 Hz and 260 Hz are sounded together.

Solution: Approaching: $f' = 500 \times 340/(340 - 20) = 170{,}000/320 = 531$ Hz (531.25 Hz). Receding: $f' = 500 \times 340/(340 + 20) = 170{,}000/360 = 472$ Hz (472.2 Hz). Beats: $260 - 256 = 4$ beats per second.

8.9 Applications and common errors

A frequent source of error in examination questions is the confusion between wave speed and particle speed. The speed $v = f\lambda$ is the speed at which the pattern advances; the individual particles merely oscillate about fixed positions and their speed is constantly changing. A second

common error is forgetting that the frequency of a wave does not change on refraction, so that in a change of medium only v and λ change. A third is to forget the doubling of distance in echo problems: always divide vt by two. Finally, remember that for pipes, a closed pipe supports only the odd harmonics, which is why a closed pipe and an open pipe of the same fundamental frequency sound different in quality.

Real-life applications abound. Loudspeaker enclosures and the shape of a church or mosque hall are designed to control reflection and reverberation; noise-cancelling headphones use destructive interference by generating a wave in antiphase with the unwanted noise; and the strings of a guitar are tuned by adjusting tension while the player alters pitch by pressing the strings against the frets to shorten the vibrating length.

Exercise 8.1

1. A wave has a frequency of 50 Hz and a wavelength of 6.8 m. Calculate its speed.
2. Radio waves travel at 3.0×10^8 m/s. What is the wavelength of a station broadcasting at 100 MHz?
3. Two points on a wave are 0.10 m apart and the wavelength is 0.40 m. What is their phase difference in degrees?
4. A lightning flash is seen and the thunder is heard 3.0 s later. How far away was the flash? (Speed of sound = 340 m/s.)
5. A ship sends an ultrasonic pulse and receives the echo from the sea bed 0.40 s later. If the speed of sound in sea water is 1,500

m/s, calculate the depth of the sea.

6. A string 0.80 m long is fixed at both ends. If the speed of waves on it is 160 m/s, find the fundamental frequency and the frequency of the third harmonic.
7. A closed pipe has a fundamental frequency of 100 Hz. What are the frequencies of its next two harmonics?
8. Calculate the length of an open pipe whose fundamental frequency is 425 Hz, if the speed of sound is 340 m/s.
9. A source of sound of frequency 400 Hz moves at 17 m/s towards a stationary observer. Find the observed frequency. (Speed of sound = 340 m/s.)
10. Two tuning forks of frequencies 512 Hz and 518 Hz are sounded together. How many beats are heard in 10 s?
11. Calculate the intensity level of a sound of intensity 1.0×10^{-8} W/m².
12. Waves of wavelength 5.0 cm and speed 0.40 m/s pass from deep to shallow water, where the speed is 0.30 m/s. Calculate the frequency and the new wavelength.

Answers to Exercise 8.1

1. 340 m/s. $v = f\lambda = 50 \times 6.8 = 340$.
2. 3.0 m. $\lambda = v/f = 3.0 \times 10^{8}/(100 \times 10^{6}) = 3.0$.
3. 90°. $(0.10/0.40) \times 360° = 90°$.

4. 1,020 m. $d = vt = 340 \times 3.0 = 1{,}020$ (the light arrives almost instantly).
5. 300 m. $d = vt/2 = 1{,}500 \times 0.40/2 = 300$.
6. 100 Hz and 300 Hz. $f_1 = v/2L = 160/1.6 = 100$ Hz; $f_3 = 3 \times 100 = 300$ Hz.
7. 300 Hz and 500 Hz. Only odd harmonics occur in a closed pipe: 3×100 and 5×100.
8. 0.40 m. $L = v/(2f_1) = 340/850 = 0.40$.
9. 421 Hz. $f' = 400 \times 340/(340 - 17) = 136{,}000/323 = 421$ Hz (421.05).
10. 60 beats. The beat frequency is 6 Hz, so $6 \times 10 = 60$.
11. 40 dB. $\beta = 10 \log_{10}(10^{-8}/10^{-12}) = 10 \times 4 = 40$.
12. $f = 8.0$ Hz; new wavelength 3.75 cm. $f = v/\lambda = 0.40/0.050 = 8.0$ Hz; $\lambda' = 0.30/8.0 = 0.0375$ m.

Exercise 8.2

1. State two differences between transverse and longitudinal waves, and give one example of each.
2. Explain why sound can be heard round a corner but light cannot.
3. In a resonance-tube experiment with a fork of 512 Hz, resonance occurs at air-column lengths of 0.160 m and 0.490 m. Calculate the speed of sound and the end correction.

4. The speed of sound in air at 0 °C is 331 m/s. Find the speed at 100 °C (373 K), correct to the nearest whole number.
5. A wire of mass per unit length 0.004 kg/m under a tension of 36 N vibrates in its fundamental mode with a node at each end 0.75 m apart. Find the fundamental frequency.
6. Explain how a stationary wave is formed and state two ways in which it differs from a progressive wave.
7. Two coherent sources emit waves of wavelength 0.20 m. At a point P, the distances from the sources are 2.5 m and 2.9 m. State whether the interference at P is constructive or destructive and give a reason.

Answers to Exercise 8.2

1. In transverse waves the particles vibrate perpendicular to the direction of travel (they can be polarised and form crests and troughs); in longitudinal waves they vibrate parallel to it (they form compressions and rarefactions and cannot be polarised). Examples: water ripples (transverse); sound in air (longitudinal).
2. Sound has a wavelength of about 0.3 m to 3 m, comparable with the size of doorways and corners, so it diffracts strongly; light has a wavelength of about 5×10^{-7} m, far smaller than any such opening, so its diffraction is negligible.
3. Speed 337.9 m/s (about 338 m/s); end correction 1.0 cm. $\lambda = 2(0.490 - 0.160) = 0.660$ m; $v = 512 \times 0.660 = 337.9$ m/s; $l_1 + e =$

$\lambda/4 = 0.165$, so $e = 0.005$ m = 0.5 cm. (Check: $0.165 - 0.160 = 0.005$ m, so $e = 0.5$ cm.)

4. 387 m/s. $v = 331 \times \sqrt{(373/273)} = 331 \times \sqrt{1.3663} = 331 \times 1.1689 = 386.9 \approx 387$.
5. 60 Hz. $v = \sqrt{(36/0.004)} = \sqrt{9{,}000} = 94.9$ m/s; $f_1 = v/2L = 94.9/1.5 = 63.2$ Hz (about 63 Hz).
6. A stationary wave forms when a progressive wave and its reflection, of equal frequency and amplitude, travel in opposite directions and superpose. Unlike a progressive wave, it has fixed nodes and antinodes, it does not transmit energy, and its amplitude varies from point to point.
7. Destructive. Path difference = 0.4 m = 2λ, which is a whole number of wavelengths, so the interference would be constructive. Correction: $0.4/0.2 = 2$, a whole number, so the interference at P is constructive.

Chapter summary

- A wave transfers energy without transferring matter. Mechanical waves need a medium; electromagnetic waves do not.
- In transverse waves, particles vibrate at right angles to the direction of travel; in longitudinal waves, parallel to it. Sound is longitudinal.
- The fundamental wave equation is $v = f\lambda$, with $f = 1/T$; frequency stays constant when a wave changes medium.

- Reflection, refraction, diffraction and interference are the characteristic properties of waves. Constructive interference needs a path difference of $n\lambda$; destructive needs $(n + ½)\lambda$.
- Stationary waves have nodes and antinodes $\lambda/2$ apart. A string fixed at both ends has $f_1 = (1/2L)\sqrt{(T/\mu)}$ and all harmonics.
- Sound travels fastest in solids and slowest in gases; in air $v \propto \sqrt{T}$. Echo distance is $d = vt/2$.
- Pitch depends on frequency, loudness on amplitude and quality on overtones. Intensity level is $\beta = 10 \log_{10}(I/I_0)$ decibels.
- A closed pipe has $f_1 = v/4L$ and odd harmonics only; an open pipe has $f_1 = v/2L$ and all harmonics. In a resonance tube $\lambda = 2(l_2 - l_1)$.
- The beat frequency equals the difference between the two frequencies. The Doppler shift gives $f' = fv/(v \mp u_s)$ for a source approaching or receding.

Chapter 9: Light, Reflection, Refraction, Lenses and the Electromagnetic Spectrum

Almost everything we know about the world reaches us as light, and the images in mirrors, cameras and spectacles depend on its behaviour. In this chapter we study how light travels, reflection, refraction, lenses, dispersion, simple optical instruments and the electromagnetic spectrum of which visible light is a small part.

Throughout this chapter we take the speed of light in a vacuum (and, to a very good approximation, in air) as $c = 3.0 \times 10^8$ m/s.

9.1 The nature and propagation of light

Light is a form of energy that is transmitted as a transverse electromagnetic wave and that stimulates the sense of sight. It needs no material medium, which is why sunlight reaches the Earth across empty space. A body that produces its own light, such as the Sun, a candle flame or an electric lamp, is **luminous**; a body such as the Moon or this page, which is seen only because it reflects light from a luminous source, is **non-luminous**.

In a uniform medium light travels in straight lines, a fact called the **rectilinear propagation of light**. We represent the path of light by a straight line with an arrow, called a **ray**, and a collection of rays is a

beam. Sharp shadows behind opaque objects are the everyday evidence. A shadow has a dark central region, the **umbra**, and, when the source is large, a lighter fringe called the **penumbra**, from which part of the source is visible.

Eclipses are shadows on a cosmic scale. In a **solar eclipse** the Moon passes between the Sun and the Earth; an observer in the umbra sees a total eclipse and one in the penumbra a partial one. A **pinhole camera** forms an inverted, real image on a screen, and by similar triangles $h'/h = v/u$, where h and h' are the object and image heights and u and v are the distances of object and screen from the hole.

9.2 Reflection at plane surfaces

When light meets a boundary some of it is thrown back; this is **reflection**. The line drawn at right angles to the surface at the point of incidence is the **normal**. The angle of incidence i is measured between the incident ray and the normal, and the angle of reflection r between the reflected ray and the normal. The two laws of reflection are:

> **Laws of reflection.** (1) The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane. (2) The angle of incidence is equal to the angle of reflection ($i = r$). Angles are always measured from the normal, never from the mirror surface.

A smooth surface produces **regular reflection**; a rough surface produces **diffuse reflection**, with rays scattered in all directions.

The image in a plane mirror is **virtual** (rays only appear to come from it and it cannot be caught on a screen), **erect**, the **same size** as the object, and as far behind the mirror as the object is in front. It is also **laterally inverted**, so that your right hand appears to be the left hand of the image.

Two results are especially useful. First, if a mirror is rotated through an angle θ while the incident ray stays fixed, the reflected ray turns through 2θ in the same direction. Second, when two plane mirrors are inclined at an angle θ to each other, and $360°/\theta$ is a whole number, the number of images of an object placed between them is $n = (360°/\theta) - 1$. If the mirrors are parallel, $\theta = 0$ and an infinite number of images is formed in principle, as you see in a barber's shop with mirrors on opposite walls.

WORKED EXAMPLE 9.1

A ray of light strikes a plane mirror, and the mirror is turned through 15° about an axis in its surface while the incident ray is held fixed. Through what angle does the reflected ray turn? Two other mirrors are placed at 60° to each other with a small object between them. How many images are formed?

Solution: The reflected ray turns through twice the angle of rotation of the mirror, so the turn is $2 \times 15° = 30°$. For the inclined mirrors, $n = 360°/60° - 1 = 6 - 1 = 5$ images.

9.3 Reflection at curved mirrors

A **spherical mirror** is part of a hollow sphere. If the inside is the reflecting surface it is **concave** (converging); if the outside reflects it is **convex** (diverging). The important terms are as follows. The **pole** P is the centre of the mirror surface. The **centre of curvature** C is the centre of the sphere of which the mirror is a part, and the distance PC is the **radius of curvature** R. The **principal axis** is the line through P and C. Rays parallel and close to the principal axis converge to (concave) or appear to diverge from (convex) a point on the axis called the **principal focus** F, and the distance PF is the **focal length** f. For small apertures,

$f = R/2$.

To locate an image by drawing, use two of three rays: a ray parallel to the axis reflects through (or as if from) F; a ray through F reflects parallel to the axis; and a ray through C returns along its own path. A concave mirror gives a real, inverted, diminished image when the object is beyond C; a real, inverted image of the same size at C when the object is at C; a real, inverted, magnified image beyond C when the object is between C and F; and a virtual, erect, magnified image when the object is inside F, which is why it is used as a shaving mirror. A convex mirror always gives a virtual, erect, diminished image and a wide field of view, so it is used as a vehicle wing mirror.

For calculations we use the **mirror formula** with the "real-is-positive" convention: distances of real objects and real images are positive, distances of virtual images and virtual foci are negative. Thus a concave

mirror has a positive f and a convex mirror a negative f.

1/u + 1/v = 1/f, and the linear magnification is m = image height/object height = v/u (ignoring sign, which only tells us whether the image is erect or inverted).

WORKED EXAMPLE 9.2

A concave mirror has a radius of curvature of 30 cm. An object is placed 20 cm in front of it. Find the position and magnification of the image. Then find the image formed by a convex mirror of focal length 12 cm when the object is 24 cm away.

Solution: For the concave mirror f = R/2 = 15 cm and u = 20 cm. 1/v = 1/f − 1/u = 1/15 − 1/20 = (4 − 3)/60 = 1/60, so v = 60 cm. The image is real, 60 cm in front of the mirror, and m = v/u = 60/20 = 3, so it is inverted and three times as large. For the convex mirror f = −12 cm and u = 24 cm. 1/v = −1/12 − 1/24 = (−2 − 1)/24 = −3/24 = −1/8, so v = −8 cm. The image is virtual, 8 cm behind the mirror, and m = 8/24 = 1/3, so it is erect and one-third the size of the object.

9.4 Refraction of light

Refraction is the change in direction of a ray of light when it passes from one transparent medium into another of different optical density. It happens because the speed of light changes at the boundary. A ray entering a denser medium (glass or water from air) is bent towards the

normal, and a ray entering a less dense medium is bent away from it. A ray that meets the boundary along the normal is not deviated. The laws of refraction are: (1) the incident ray, the refracted ray and the normal lie in one plane; (2) **Snell's law**, which says that the ratio sin i/sin r is a constant for a given pair of media and a given colour of light.

The constant is the **refractive index** n of the second medium relative to the first. When the first medium is a vacuum (or air to a good approximation) we call it the absolute refractive index, and it equals the ratio of the speed of light in a vacuum to its speed in the medium:

$n = \sin i / \sin r = c/v$.

Typical values are 1.33 (4/3) for water, 1.5 for ordinary glass and 2.4 for diamond. If light passes from medium 1 (index n_1) into medium 2 (index n_2), then $n_1 \sin \theta_1 = n_2 \sin \theta_2$.

Real and apparent depth

A swimming pool or a river looks shallower than it is, because rays from the bottom bend away from the normal as they leave the water and so appear to come from a point nearer the surface. For viewing nearly vertically from above,

n = real depth/apparent depth.

Total internal reflection

When light passes from a denser to a less dense medium, the refracted ray bends away from the normal. As the angle of incidence increases there comes a value, called the **critical angle** C, at which the refracted ray just grazes along the boundary (angle of refraction 90°). For any larger angle of incidence there is no refracted ray and all the light is reflected back into the dense medium: **total internal reflection**. Since n = sin 90°/sin C,

sin C = 1/n.

> **Conditions for total internal reflection.** The light must travel from the optically denser to the less dense medium, and the angle of incidence must exceed the critical angle. Many candidates forget the first condition.

Total internal reflection is used in right-angled glass prisms (for which C ≈ 42° is less than 45°) in periscopes and binoculars, in the sparkle of cut diamonds, in road reflectors and in **optical fibres**, which carry telephone and internet signals as pulses of light.

WORKED EXAMPLE 9.3

(a) The refractive index of glass is 1.5. Calculate the speed of light in glass and the critical angle for the glass–air boundary. (b) A ray in air strikes a glass block at 30° to the normal. Find the angle of refraction. (c) A coin lies at the bottom of a pond 3.0 m deep (n = 4/3). How deep does it appear when viewed from directly above?

Solution: (a) $v = c/n = 3.0 \times 10^8 / 1.5 = 2.0 \times 10^8$ m/s. $\sin C = 1/1.5 = 2/3 = 0.667$, so $C = 41.8°$. (b) $\sin r = \sin 30° / 1.5 = 0.5/1.5 = 1/3 = 0.3333$, so $r = 19.5°$. The ray bends towards the normal, as expected for entry into a denser medium. (c) Apparent depth = real depth/n = $3.0/(4/3) = 3.0 \times 3/4 = 2.25$ m.

9.5 Lenses

A **lens** is a piece of transparent material, usually glass, bounded by two curved surfaces (or one curved and one plane) that refracts light to form images. A **convex (converging) lens** is thicker at the centre than at the edges and brings a parallel beam to a real focus; a **concave (diverging) lens** is thinner at the centre and spreads a parallel beam so that it appears to come from a virtual focus. The **optical centre** is the point through which a ray passes undeviated, the **principal focus** is the point to which rays parallel to the principal axis converge (or from which they appear to diverge), and the **focal length** f is its distance from the optical centre. A lens has a principal focus on each side.

Ray diagrams again use two of three rays: a ray parallel to the axis goes through the focus; a ray through the optical centre is undeviated; a ray through the focus emerges parallel to the axis. A convex lens gives a real, inverted, diminished image for an object beyond 2f (camera), a real, inverted, magnified image beyond 2f for an object between f and 2f (projector), and a virtual, erect, magnified image for an object inside f (magnifying glass). A concave lens always forms a virtual, erect, diminished image.

The same formulas apply, with the real-is-positive convention: $1/u + 1/v = 1/f$, and $m = v/u$. A convex lens has a positive f and a concave lens a negative f. The **power** of a lens is the reciprocal of its focal length in metres, $P = 1/f$, and is measured in **dioptres** (D). A convex lens has positive power and a concave lens negative power. When thin lenses are placed in contact their powers add: $P = P_1 + P_2$, which is equivalent to $1/f = 1/f_1 + 1/f_2$.

WORKED EXAMPLE 9.4

An object is placed 15 cm from a convex lens of focal length 10 cm. Find the image distance, the magnification and the power of the lens. Then find the focal length of a combination of a convex lens of f = 20 cm in contact with a concave lens of f = −30 cm.

Solution: $1/v = 1/f - 1/u = 1/10 - 1/15 = (3 - 2)/30 = 1/30$, so v = 30 cm. The image is real and inverted, with $m = 30/15 = 2$. The power is $P = 1/0.10 = +10$ D. For the combination, $1/f = 1/20 - 1/30 = (3 - 2)/60 = 1/60$, so f = 60 cm, a weak converging combination of power $1/0.6 \approx +1.67$ D.

9.6 Dispersion of light and colour

When a narrow beam of white light passes through a glass prism it spreads out into a band of colours called a **spectrum**. This is **dispersion**. In the order of increasing deviation the colours are red, orange, yellow, green, blue, indigo and violet. Dispersion occurs because the refractive index of glass is slightly different for each colour: it is smallest for red light, which therefore travels fastest in glass and is deviated least, and largest for violet light, which travels slowest and is deviated most. In a vacuum all colours travel at the same speed, 3.0×10^8 m/s.

The colour of an object depends on the light it reflects: a red rose in white light reflects red and absorbs the rest, so in green light it looks black. The **primary colours** of light are red, green and blue; red and green make yellow, and all three together make white.

9.7 Optical instruments

The **human eye** is a natural optical instrument. Light is refracted mainly at the curved surface of the cornea and then by the lens, which is made to change its focal length by the ciliary muscles (accommodation) so that a sharp real inverted image falls on the retina, from which the optic nerve carries signals to the brain. The **near point** of a normal eye is about 25 cm and its **far point** is at infinity.

A **short-sighted** person (myopia) sees near objects clearly but not distant ones, because the image forms in front of the retina; a diverging lens corrects it. A **long-sighted** person (hypermetropia) cannot see near objects clearly, because the image would form behind the retina; a converging lens corrects it.

WORKED EXAMPLE 9.5

A short-sighted girl has a far point of 80 cm. What power of spectacle lens will let her see distant objects clearly? A long-sighted man has a near point of 100 cm. What lens lets him read a book held 25 cm away?

Solution: For the girl the lens must form a virtual image of an object at infinity at her far point, so v = −80 cm and f = −80 cm = −0.80 m. P = 1/f = −1.25 D (a concave lens). For the man, we need a virtual image at 100 cm for an object at 25 cm: u = 25 cm, v = −100 cm. 1/f = 1/25 + 1/(−100) = 4/100 − 1/100 = 3/100, so f = 33.3 cm = 0.333 m and P = +3.0 D (a convex lens).

A **camera** forms a real, inverted, diminished image on a sensor; a **projector** places a slide just outside the focal length of a convex lens to form a magnified real image on a screen. A **simple magnifying glass** is a convex lens of short focal length with the object inside its focal length. If the image is at the near point D = 25 cm, the angular magnification is M = 1 + D/f; if it is at infinity (normal adjustment) M = D/f.

A **compound microscope** uses a short-focus objective to form a magnified real image that the eyepiece then magnifies again. An **astronomical refracting telescope** has an objective of long focal length f_o, which forms a real image of a distant object at its focus, and an eyepiece of short focal length f_e used as a magnifier. In normal adjustment the lenses are separated by $f_o + f_e$ and the angular magnification is $M = f_o/f_e$.

WORKED EXAMPLE 9.6

A magnifying glass has focal length 5.0 cm. Find its magnification when the image is at infinity and at the near point (25 cm). A telescope has an objective of focal length 100 cm and an eyepiece of focal length 4.0 cm. Find its magnification and its length in normal adjustment.

Solution: At infinity M = D/f = 25/5 = 5. At the near point M = 1 + 25/5 = 6. For the telescope, $M = f_o/f_e = 100/4 = 25$ and the length is 100 + 4 = 104 cm.

9.8 The electromagnetic spectrum

Visible light is one band of a continuous family of **electromagnetic waves**. All are transverse waves consisting of oscillating electric and magnetic fields at right angles to each other and to the direction of travel. They need no medium, all travel at $c = 3.0 \times 10^8$ m/s in a vacuum, and are not deflected by electric or magnetic fields. They differ in frequency f and wavelength λ, linked by $c = f\lambda$. In order of increasing frequency (decreasing wavelength) the bands are:

- **Radio waves** (above about 1 mm) are used for broadcasting and communications.
- **Microwaves** (about 1 mm to 30 cm) are used in radar, mobile links and microwave ovens.
- **Infrared** (about 700 nm to 1 mm) is emitted by hot bodies and used in remote controls and night-vision cameras.
- **Visible light** spans roughly 400 nm (violet) to 700 nm (red).
- **Ultraviolet** (about 10 nm to 400 nm) causes sun tan, can cause skin cancer, and is used to sterilise water.
- **X-rays** (about 0.01 nm to 10 nm) pass through soft tissue but not bone, and are used in medical imaging.
- **Gamma rays** (shorter than about 0.01 nm) come from radioactive nuclei and are used to sterilise equipment and treat cancer.

The higher the frequency, the greater the photon energy and the more harmful the radiation to living tissue. Radio, microwave and infrared radiation are generally less dangerous than ultraviolet, X-rays and

gamma rays, which are **ionising**.

WORKED EXAMPLE 9.7

A radio station in Ibadan broadcasts on 90 MHz. Calculate the wavelength of its waves.

Solution: $\lambda = c/f = 3.0 \times 10^8 / (9.0 \times 10^7) = 3.33$ m.

Exercise 9.1

1. A ray of light strikes a plane mirror at an angle of incidence of 35°. What is the angle between the incident and the reflected ray?
2. Two plane mirrors are inclined at 45° to each other. How many images of an object between them are formed?
3. A concave mirror has a radius of curvature of 40 cm. An object is placed 60 cm from it. Find the image distance and the magnification, and describe the image.
4. A convex mirror has a focal length of magnitude 20 cm. An object is placed 20 cm in front of it. Find the image distance and the magnification.
5. A coin lies at the bottom of a tank of water ($n = 4/3$) of depth 1.2 m. How deep does it appear to an observer looking vertically downwards?
6. Calculate the critical angle for diamond, whose refractive index is 2.4.

7. A convex lens of focal length 8.0 cm forms a real image 24 cm from the lens. Where is the object, and what is the magnification?
8. A thin convex lens of power +2.5 D is placed in contact with a concave lens of power −1.5 D. Find the power and focal length of the combination.
9. An astronomical telescope has an objective of focal length 120 cm and an eyepiece of focal length 5.0 cm. Find the magnification and the distance between the lenses in normal adjustment.
10. A radio station transmits at 600 kHz. What is the wavelength of the waves?

Answers to Exercise 9.1

1. 70°. The angle of reflection equals 35°, so the angle between the two rays is 35° + 35° = 70°.
2. 7 images. n = 360°/45° − 1 = 8 − 1 = 7.
3. f = 20 cm; 1/v = 1/20 − 1/60 = 2/60 = 1/30, so v = 30 cm; m = 30/60 = 0.5. The image is real, inverted and diminished.
4. f = −20 cm; 1/v = −1/20 − 1/20 = −1/10, so v = −10 cm (virtual, behind the mirror); m = 10/20 = 0.5, erect and diminished.
5. 0.90 m. Apparent depth = 1.2/(4/3) = 1.2 × 3/4 = 0.90 m.
6. sin C = 1/2.4 = 0.4167, so C ≈ 24.6°.
7. 1/u = 1/8 − 1/24 = 2/24 = 1/12, so u = 12 cm; m = 24/12 = 2.

8. P = 2.5 + (−1.5) = +1.0 D, so f = 1/1.0 = 1.0 m (100 cm).
9. M = 120/5.0 = 24; separation = 120 + 5.0 = 125 cm.
10. $\lambda = c/f = 3.0 \times 10^8/(6.0 \times 10^5) = 500$ m.

Chapter summary

- Light travels in straight lines in a uniform medium; shadows, eclipses and the pinhole camera follow from this.
- Reflection obeys i = r. A plane-mirror image is virtual, erect, same size and as far behind the mirror as the object is in front; mirrors at θ give $(360°/\theta) - 1$ images, and a rotated mirror turns the reflected ray through twice its own rotation.
- For spherical mirrors f = R/2 and 1/u + 1/v = 1/f, with m = v/u (real-is-positive).
- Refraction: n = sin i/sin r = c/v; real depth/apparent depth = n; sin C = 1/n, and total internal reflection needs a dense-to-less-dense path at an angle above C.
- Lens formula as for mirrors; P = 1/f in dioptres, and powers in contact add. Short sight needs a concave lens, long sight a convex lens.
- Dispersion occurs because n is larger for violet than for red.
- Magnifier: M = D/f or 1 + D/f. Telescope in normal adjustment: $M = f_o/f_e$, length $f_o + f_e$.

- All electromagnetic waves travel at $c = 3.0 \times 10^8$ m/s and obey $c = f\lambda$.

Chapter 10: Electrostatics, Capacitors and Current Electricity

When a power cut ends in Ibadan and the fans turn again, a stream of charges is once more drifting through copper wires, carrying energy from a generating station to your home. Electricity is the physics of electric charge, both at rest and in motion. In this chapter we first study charges at rest (electrostatics): how bodies become charged, the forces between charges, electric fields and potential, and the capacitor, which stores charge. We then turn to charges in motion: current, resistance, Ohm's law, the behaviour of resistors in series and parallel, the e.m.f. and internal resistance of a cell, electrical energy and power, and the measurement of resistance. Magnetic effects of currents are left to the next chapter.

Use these constants where needed: Coulomb's constant $k = 9.0 \times 10^{9}$ N m²/C²; permittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m; charge of the electron $e = 1.6 \times 10^{-19}$ C.

10.1 Electric charge and Coulomb's law

Matter contains positive protons, negative electrons and neutral neutrons. A body is neutral when it has equal numbers of protons and electrons; it becomes **charged** when it gains or loses electrons. A body

with an excess of electrons is negatively charged, and one with a deficit is positively charged. When an ebonite rod is rubbed with fur, the rod gains electrons and the fur loses them; when a glass rod is rubbed with silk, the glass loses electrons. The **law of conservation of charge** states that charge can neither be created nor destroyed, only transferred. The SI unit of charge is the **coulomb** (C), and the smallest free charge is that of the electron, $e = 1.6 \times 10^{-19}$ C, so any charge is a whole-number multiple of e (charge is *quantised*). Like charges repel and unlike charges attract.

A **conductor** (a metal, graphite, salt solution) has charges free to move; an **insulator** (plastic, dry wood, rubber, glass) does not. A body can be charged by friction, by contact, or by **induction**. In charging by induction, a charged rod is brought near, but not touching, an insulated metal sphere. Free electrons in the sphere are repelled (by a negative rod) to the far side. If the far side is then earthed momentarily and the earth connection is removed before the rod is taken away, the sphere is left with a charge opposite to that of the rod. The **gold-leaf electroscope** detects and tests charge: a charged body brought near the cap makes the leaf diverge because the leaf and stem acquire like charges. Repulsion is the only sure test of the sign of a charge, since a neutral body is also attracted by a charged one. Charge also collects on the points of a conductor, where it leaks away into the air; this action of points explains why lightning conductors have sharp spikes.

Coulomb's law states that the force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them:

$F = kQ_1Q_2/r^2$, where $k = 1/(4\pi\varepsilon_0) = 9.0 \times 10^9$ N m²/C².

WORKED EXAMPLE 10.1

Two small spheres carrying charges of +2.0 μC and +3.0 μC are 0.30 m apart in air. Find the force between them. What happens to the force if the distance is doubled?

Solution: $Q_1 = 2.0 \times 10^{-6}$ C, $Q_2 = 3.0 \times 10^{-6}$ C. $F = (9.0 \times 10^9 \times 2.0 \times 10^{-6} \times 3.0 \times 10^{-6})/(0.30)^2 = (9.0 \times 10^9 \times 6.0 \times 10^{-12})/0.090 = 0.054/0.090 = 0.60$ N, repulsive. Doubling r divides F by $2^2 = 4$, so F becomes 0.15 N.

10.2 Electric field and potential

An **electric field** is a region in which a charge experiences a force. The **electric field strength** E at a point is the force per unit positive charge placed there, E = F/q, in newtons per coulomb (N/C) or volts per metre (V/m). Field lines show the direction of the force on a positive charge: they start on positive charges, end on negative ones, never cross, and are closer together where the field is stronger. For a point charge Q, the field at distance r is $E = kQ/r^2$. Between two parallel plates the field is uniform, and its strength is E = V/d where V is the potential difference between them and d is their separation.

The **electric potential** at a point is the work done in bringing a unit positive charge from infinity to that point. For a point charge, V = kQ/r, in volts (V = J/C). The **potential difference** between two points is the

work done per unit charge in moving charge between them: $V = W/Q$, so a charge Q moved through a pd V gains or loses energy $W = QV$. An electron accelerated through 1 volt gains 1.6×10^{-19} J, a quantity called one **electron-volt** (eV). Field strength is a vector, but potential is a scalar.

WORKED EXAMPLE 10.2

A point charge of 4.0 nC (4.0×10^{-9} C) is placed in air. Find the field strength and the potential at a point 0.20 m from it. What force acts on a 2.0×10^{-6} C charge placed there?

Solution: $E = kQ/r^2 = (9.0 \times 10^{9} \times 4.0 \times 10^{-9})/(0.20)^2 = 36/0.040 = 900$ N/C. $V = kQ/r = 36/0.20 = 180$ V. $F = qE = 2.0 \times 10^{-6} \times 900 = 1.8 \times 10^{-3}$ N.

10.3 Capacitors

A **capacitor** is a device that stores electric charge and energy; in its simplest form it is two parallel metal plates separated by an insulator (the **dielectric**), such as air, paper or mica. When a battery is connected to it, one plate gains charge +Q and the other −Q. The **capacitance** C is the charge stored per unit potential difference:

$C = Q/V$, measured in **farads** (F). In practice the microfarad (1 μF = 10^{-6} F) and picofarad (1 pF = 10^{-12} F) are used.

For a parallel-plate capacitor the capacitance is $C = \varepsilon_0 A/d$ in air, where A is the overlap area of the plates and d their separation. A dielectric of relative permittivity ε_r increases the capacitance by the factor ε_r. Capacitance is therefore increased by large plate area, small separation and a good dielectric.

For capacitors in **parallel** the pd across each is the same and the total capacitance is the sum, $C = C_1 + C_2 + C_3$. For capacitors in **series** each carries the same charge and the reciprocals add: $1/C = 1/C_1 + 1/C_2 + 1/C_3$. The energy stored in a charged capacitor is

$W = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = Q^2/(2C)$.

Capacitors are used to smooth rectified current in power supplies, to tune radio receivers, to store energy in camera flashes and to block direct current in amplifiers.

WORKED EXAMPLE 10.3

A 6 µF capacitor and a 3 µF capacitor are connected in series across a 12 V supply. Find the total capacitance, the charge on each capacitor and the energy stored. What would the capacitance be if they were connected in parallel?

Solution: In series, $1/C = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2$, so $C = 2$ µF. The charge is $Q = CV = 2 \times 10^{-6} \times 12 = 2.4 \times 10^{-5}$ C $= 24$ µC on each capacitor. Energy $W = \tfrac{1}{2}CV^2 = \tfrac{1}{2} \times 2 \times 10^{-6} \times 144 = 1.44 \times 10^{-4}$ J. In parallel, $C = 6 + 3 = 9$ µF.

10.4 Electric current, Ohm's law and resistance

Electric current is the rate of flow of charge: $I = Q/t$, in amperes (1 A = 1 C/s). In a metal the carriers are free electrons, which drift slowly in a direction opposite to the **conventional current**, which is taken as the direction in which positive charges would move, from the positive terminal of the cell round the circuit to the negative terminal. The potential difference across a component is the energy converted per coulomb, $V = W/Q$, and it is measured with a voltmeter connected in parallel; current is measured by an ammeter connected in series.

Ohm's law states that the current through a metallic conductor is directly proportional to the potential difference across its ends, provided the temperature and other physical conditions remain constant. The **resistance** $R = V/I$ is measured in ohms (Ω), where 1 Ω = 1 V/A. A graph of V against I for an *ohmic* conductor is a straight line through the origin whose gradient is R. A filament lamp and a diode are non-ohmic: the resistance of the lamp filament rises as it heats.

The resistance of a uniform wire depends on its length l, its cross-sectional area A and the material, which is described by the **resistivity** ρ (unit: Ω m):

$$R = \rho l / A.$$

Doubling the length doubles the resistance; doubling the diameter divides the resistance by four. Copper has $\rho \approx 1.7 \times 10^{-8}$ Ω m, which is why it is used for wiring, while nichrome, $\rho \approx 1.1 \times 10^{-6}$ Ω m, is used

for heating elements. The resistance of a metal increases with temperature, whereas that of a semiconductor or a thermistor decreases.

WORKED EXAMPLE 10.4

A nichrome wire 5.0 m long has a cross-sectional area of 5.5×10^{-7} m². If the resistivity of nichrome is 1.1×10^{-6} Ω m, find the resistance of the wire and the current when it is connected across a 20 V supply.

Solution: $R = \rho l/A = (1.1 \times 10^{-6} \times 5.0)/(5.5 \times 10^{-7}) = (5.5 \times 10^{-6})/(5.5 \times 10^{-7}) = 10\ \Omega$. $I = V/R = 20/10 = 2.0$ A.

10.5 Resistors in series and parallel

When resistors are joined end to end (**series**) the same current flows through each and the pd across the combination is the sum of the individual pds, so the total resistance is $R = R_1 + R_2 + R_3$. When they are joined across the same two points (**parallel**) the pd across each is the same and the currents add, so

$1/R = 1/R_1 + 1/R_2 + 1/R_3$.

For two resistors in parallel this reduces to $R = R_1R_2/(R_1 + R_2)$, and the combined resistance is always less than the smallest resistor. Houses are wired in parallel so that every appliance receives the full mains voltage and works independently. In a series combination, the largest resistor receives the largest share of the voltage, and in a parallel combination

the smallest resistor carries the largest current. These rules, together with the two principles that current is conserved at a junction (the total current entering a point equals the total leaving it) and that the sum of the e.m.f.s in a closed loop equals the sum of the pds, are enough to solve nearly every UTME circuit problem.

10.6 E.m.f. and internal resistance

The **electromotive force** (e.m.f.) E of a cell is the energy supplied by the cell to each coulomb of charge that passes through it, that is, the total energy per unit charge converted from chemical energy to electrical energy. It equals the terminal pd on open circuit. Because the chemical materials in the cell resist the passage of current, the cell has an **internal resistance** r. When a current I flows, a pd Ir is "lost" inside the cell, so the pd V across the terminals is less than E:

$E = V + Ir = I(R + r)$, so $V = E - Ir$.

Here R is the external (load) resistance. A cell whose internal resistance is large, such as a nearly flat battery, gives a very low terminal voltage when it supplies a large current; this is why a weak torch battery can read 1.5 V yet fail to light a bulb. A graph of V against I is a straight line of gradient $-r$ and intercept E.

WORKED EXAMPLE 10.5

A battery of e.m.f. 14 V and internal resistance 1.0 Ω is connected to a 4.0 Ω resistor in series with a parallel combination of a 6.0 Ω and a 3.0 Ω resistor. Find the current from the battery, its terminal pd, and the current in each parallel branch.

Solution: Parallel part: $R = (6 \times 3)/(6 + 3) = 18/9 = 2.0\ \Omega$. External resistance = 4.0 + 2.0 = 6.0 Ω. $I = E/(R + r) = 14/(6.0 + 1.0) = 2.0$ A. Terminal pd $V = E - Ir = 14 - 2.0 \times 1.0 = 12$ V. The pd across the 4.0 Ω resistor = $2.0 \times 4.0 = 8.0$ V, so the pd across the parallel pair = 12 − 8.0 = 4.0 V. The current in the 6.0 Ω branch = 4.0/6.0 = 0.67 A and in the 3.0 Ω branch = 4.0/3.0 = 1.33 A; these add to 2.0 A, as they must.

10.7 Electrical energy and power

When a charge Q moves through a pd V the energy transferred is $W = QV = IVt$. **Electrical power** is the rate of transfer of energy, measured in watts (W):

$$P = IV = I^2R = V^2/R.$$

The energy transferred as heat in a resistor in time t is therefore $H = I^2Rt$, a relation known as **Joule's law**. Electric companies measure energy in **kilowatt-hours** (kWh), where 1 kWh = 1,000 W × 3,600 s = 3.6×10^6 J. The cost of electricity is the number of kWh used multiplied by the price per kWh.

A **fuse** is a thin wire of low melting point placed in the live wire of an appliance. If the current exceeds the fuse rating the wire melts and breaks the circuit. The fuse chosen should have a rating just above the normal working current. Appliances with metal cases are also connected to the earth wire, so that a fault current flows safely to earth and blows the fuse instead of electrocuting the user.

WORKED EXAMPLE 10.6

(a) A lamp is marked 240 V, 60 W. Find its resistance and its working current. (b) A 750 W pressing iron is used for 4 hours a day for 30 days. If electricity costs ₦200 per kWh, find the cost. (c) What fuse should be used for a 2.0 kW heater on a 240 V supply, if fuses of 3 A, 5 A, 10 A and 13 A are available?

Solution: (a) $R = V^2/P = 240^2/60 = 57{,}600/60 = 960\ \Omega$. $I = P/V = 60/240 = 0.25$ A. (b) Energy = 0.75 kW × 4 h × 30 = 90 kWh. Cost = 90 × ₦200 = ₦18,000. (c) $I = P/V = 2{,}000/240 = 8.3$ A, so the 10 A fuse is the smallest that will not blow in normal use.

10.8 Measurement of resistance: the Wheatstone bridge

An unknown resistance can be found with an ammeter and voltmeter ($R = V/I$) or, more accurately, with a **Wheatstone bridge**. Four resistors P, Q, R and S form a diamond. A cell is connected across one pair of

opposite corners and a sensitive galvanometer across the other pair. When the galvanometer shows no deflection the bridge is **balanced**, the two middle points are at the same potential, and

$P/Q = R/S$.

If P, Q and R are known, S can be calculated. The **metre bridge** is a practical form in which P and Q are the two parts of a uniform wire, one metre long, divided by a sliding contact, so that the resistance ratio equals the ratio of the lengths. If a known resistance R is in one gap and an unknown S in the other, and balance is found at length l from the end next to R, then $R/S = l/(100 - l)$. The bridge is most accurate when the balance point lies near the middle of the wire. The **potentiometer** compares or measures e.m.f.s with no current drawn at balance, so it gives the true e.m.f. of a cell, which a voltmeter cannot do because the voltmeter draws some current.

WORKED EXAMPLE 10.7

In a Wheatstone bridge the arms P = 30 Ω and Q = 60 Ω, and the adjustable arm R = 20 Ω balances the bridge when the unknown is S. Find S. In a metre bridge a 6.0 Ω resistor in the left gap balances at 40 cm. Find the unknown resistance in the right gap.

Solution: $P/Q = R/S$ gives $S = RQ/P = (20 \times 60)/30 = 40\ \Omega$. For the metre bridge, $R/S = l/(100 - l) = 40/60$, so $S = 6.0 \times 60/40 = 9.0\ \Omega$.

Exercise 10.1

1. Two point charges of 4.0 μC and 5.0 μC are 0.20 m apart in air. Calculate the force between them.
2. Find the electric field strength at a distance of 0.50 m from a point charge of 5.0 μC.
3. Two parallel plates 2.0 cm apart have a pd of 400 V across them. Find the field strength between them and the force on an electron placed there.
4. Find the total capacitance when capacitors of 4 μF and 12 μF are connected (a) in series and (b) in parallel.
5. A 5.0 μF capacitor is charged to 200 V. Find the charge on it and the energy stored.
6. A charge of 120 C passes a point in a circuit in 2.0 minutes. What is the current?
7. A copper wire of length 10 m and cross-sectional area 2.0×10^{-6} m² has resistivity 1.7×10^{-8} Ω m. Find its resistance.
8. Find the effective resistance of 12 Ω, 6.0 Ω and 4.0 Ω resistors connected in parallel.
9. A cell of e.m.f. 6.0 V and internal resistance 0.50 Ω is connected to a 5.5 Ω resistor. Find the current, the terminal pd and the "lost volts".
10. A 1.5 kW heater is used for 4.0 hours a day for 30 days. At ₦200 per kWh, what is the cost of the electricity?

11. The arms of a Wheatstone bridge are P = 30 Ω, Q = 60 Ω and R = 20 Ω. Find S for balance.

Answers to Exercise 10.1

1. 4.5 N. $F = (9.0 \times 10^{9} \times 4.0 \times 10^{-6} \times 5.0 \times 10^{-6})/0.04 = 0.18/0.04 = 4.5$ N.
2. 1.8×10^{5} N/C. $E = kQ/r^{2} = (9.0 \times 10^{9} \times 5.0 \times 10^{-6})/0.25 = 45{,}000/0.25 = 1.8 \times 10^{5}$ N/C.
3. $E = V/d = 400/0.020 = 2.0 \times 10^{4}$ V/m; $F = eE = 1.6 \times 10^{-19} \times 2.0 \times 10^{4} = 3.2 \times 10^{-15}$ N.
4. (a) $1/C = 1/4 + 1/12 = 4/12 = 1/3$, so C = 3 µF. (b) 4 + 12 = 16 µF.
5. $Q = CV = 5.0 \times 10^{-6} \times 200 = 1.0 \times 10^{-3}$ C; $W = \tfrac{1}{2}CV^{2} = \tfrac{1}{2} \times 5.0 \times 10^{-6} \times 40{,}000 = 0.10$ J.
6. 1.0 A. t = 120 s, so I = 120/120 = 1.0 A.
7. 0.085 Ω. $R = (1.7 \times 10^{-8} \times 10)/(2.0 \times 10^{-6}) = 1.7 \times 10^{-7}/2.0 \times 10^{-6} = 0.085$ Ω.
8. $1/R = 1/12 + 2/12 + 3/12 = 6/12 = 1/2$, so R = 2.0 Ω.
9. $I = E/(R + r) = 6.0/6.0 = 1.0$ A; $V = IR = 5.5$ V; lost volts $= Ir = 0.50$ V (6.0 − 5.5).
10. Energy = 1.5 × 4.0 × 30 = 180 kWh; cost = 180 × 200 = ₦36,000.
11. $S = RQ/P = 20 \times 60/30 = 40$ Ω.

Chapter summary

- Charge is conserved and quantised ($e = 1.6 \times 10^{-19}$ C); like charges repel, unlike attract; Coulomb's law is $F = kQ_1Q_2/r^2$.
- $E = F/q$; for a point charge $E = kQ/r^2$ and $V = kQ/r$; between plates $E = V/d$. Work done in moving charge is $W = QV$.
- Capacitance $C = Q/V$; parallel plates $C = \varepsilon_0 A/d$; parallel capacitors add, series capacitors add as reciprocals; energy $W = \tfrac{1}{2}CV^2$.
- $I = Q/t$; $V = IR$ for ohmic conductors; $R = \rho l/A$.
- Series resistors add; parallel resistors add as reciprocals. Current is conserved at junctions and e.m.f.s equal the sum of pds round a loop.
- $E = V + Ir = I(R + r)$; the terminal pd falls as the current drawn rises.
- $P = IV = I^2R = V^2/R$; 1 kWh = 3.6×10^6 J; choose a fuse just above the working current.
- A balanced Wheatstone bridge has $P/Q = R/S$; the metre bridge uses $R/S = l/(100 - l)$.